



Scienza delle Costruzioni

Esercizi svolti

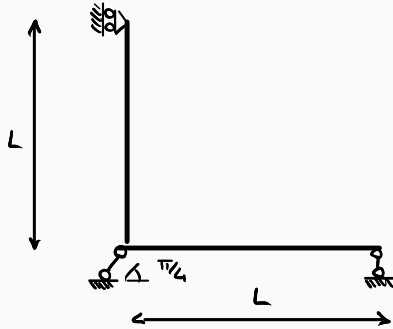
Prof. Alessio Gizzi
Dott. Lorenzo Zoboli

Problema cinematico

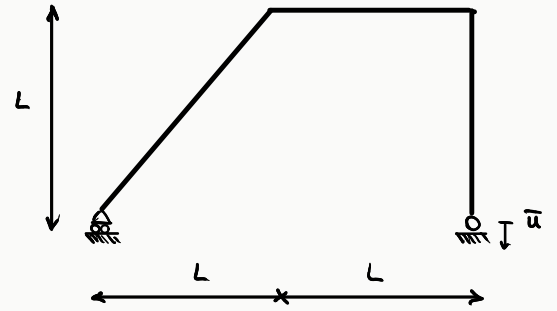
Per le seguenti strutture:

1. Studiare il problema cinematico per via algebrica, dimostrando l'esistenza della soluzione, laddove possibile, e calcolando il grado di labilità ℓ .
2. Studiare il problema cinematico attraverso i centri dello spostamento.
3. Verificare che $\mathbf{A}^t = \mathbf{B}$ scrivendo le equazioni del problema statico nello stesso ordine e scegliendo gli stessi poli adottati per il problema cinematico

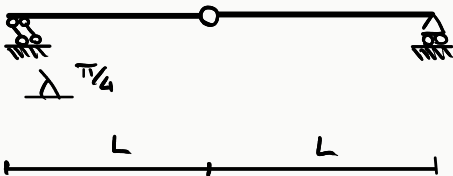
A.1



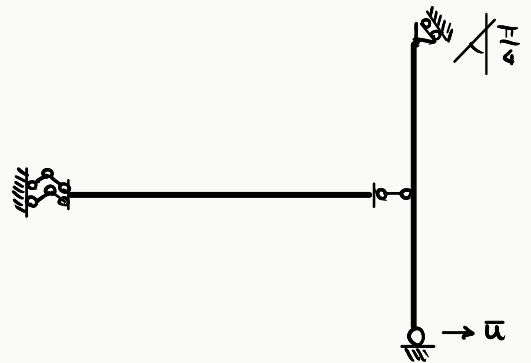
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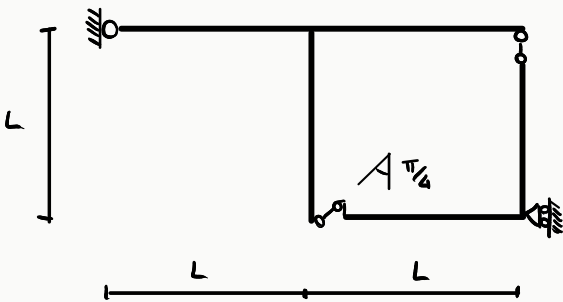
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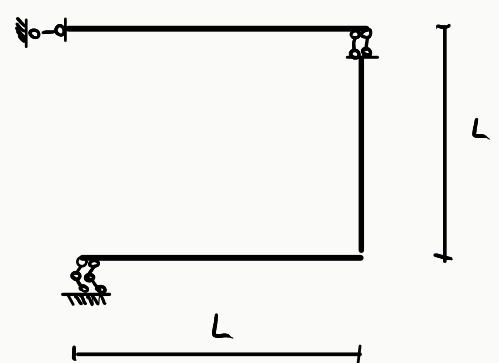
A.4



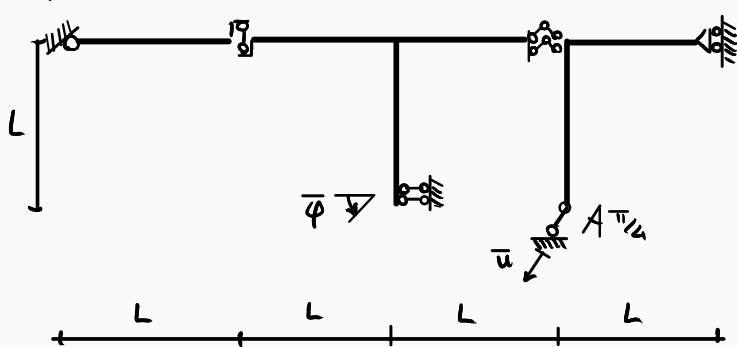
A.5



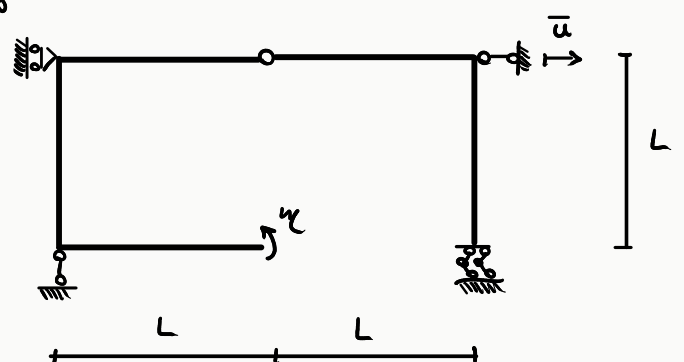
A.6



A.7



A.8

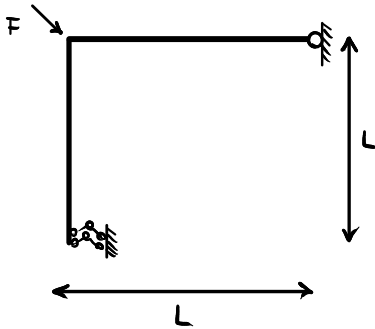


Problema statico

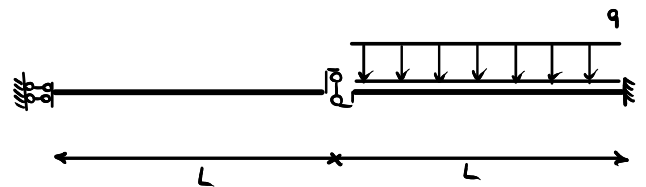
Per le seguenti strutture:

1. Studiare il problema statico per via algebrica, dimostrando l'esistenza della soluzione, laddove possibile, e calcolando il grado di iperstaticità i .
2. Calcolare il grado di iperstaticità usando la formula di Grassmann $3t - s = \ell - i$, calcolando ℓ mediante i centri dello spostamento.

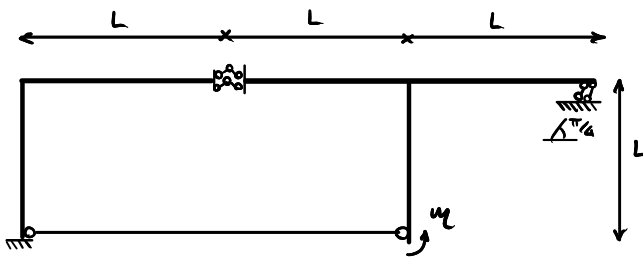
B.1



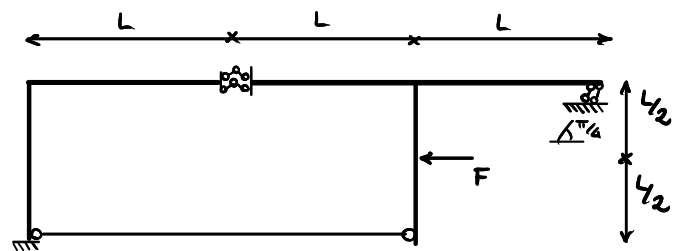
B.2



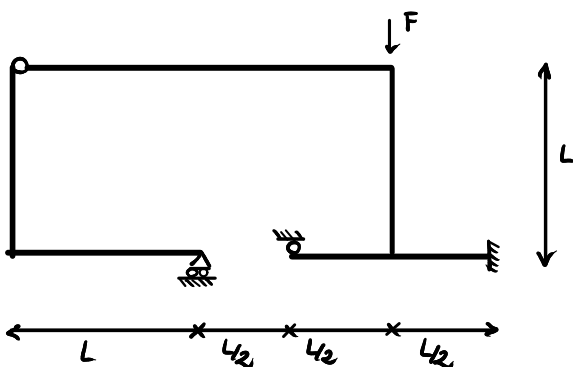
B.3



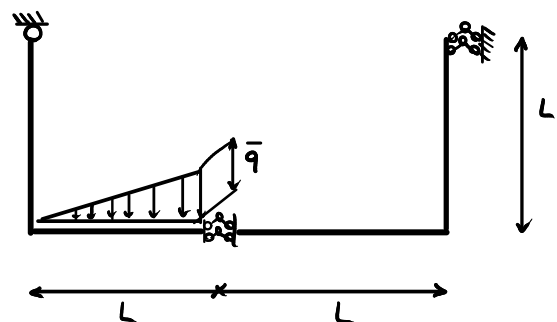
B.4



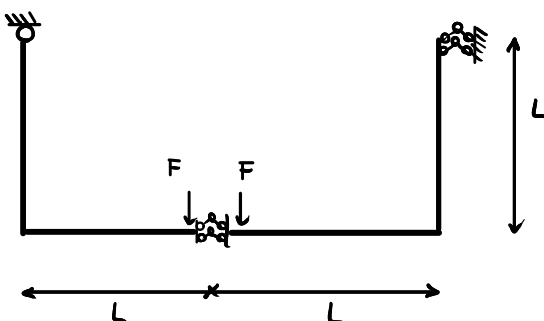
B.5



B.6



B.7

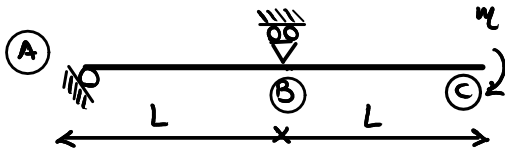


Esercizi svolti

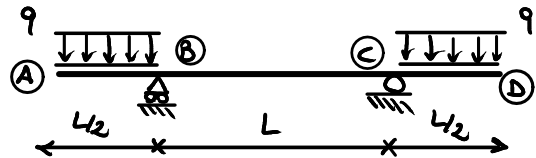
Caratteristiche della sollecitazione

Per le seguenti strutture determinare il grado di iperstaticità, le reazioni vincolari, le caratteristiche della sollecitazione

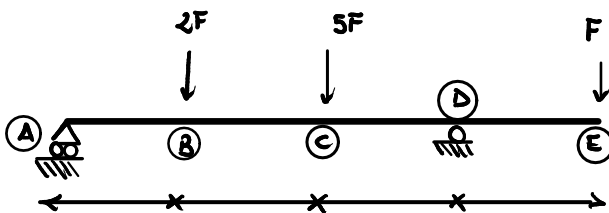
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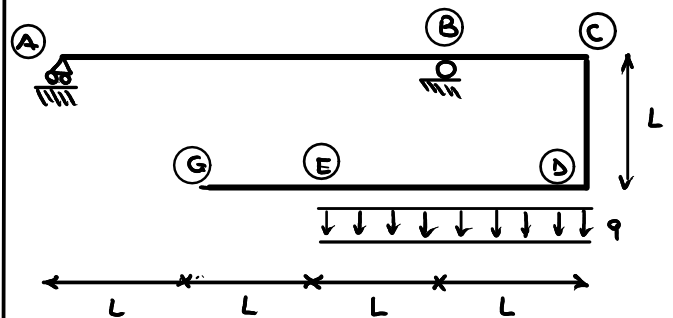
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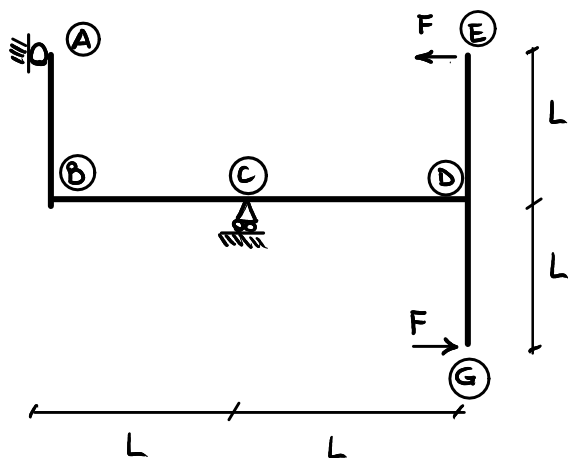
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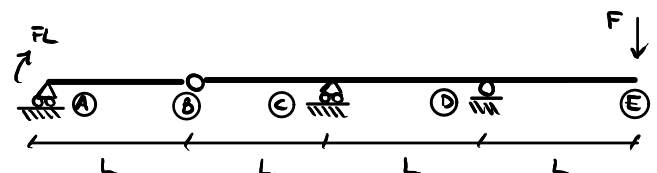
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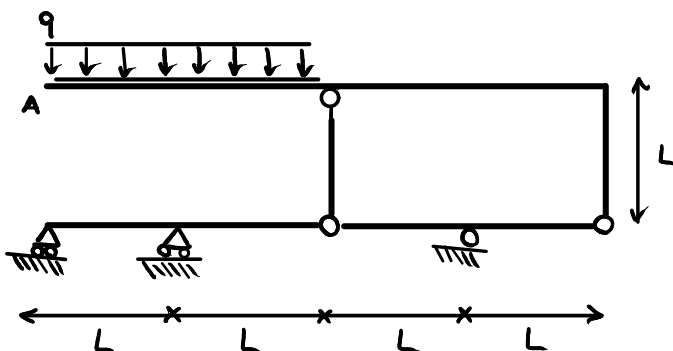
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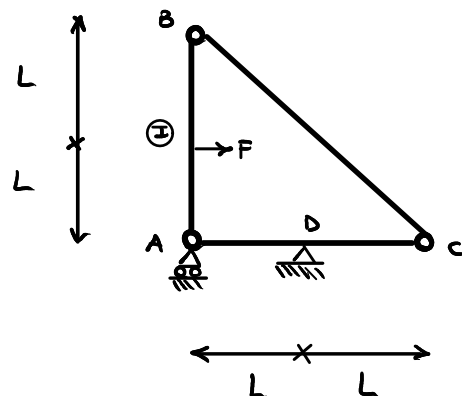
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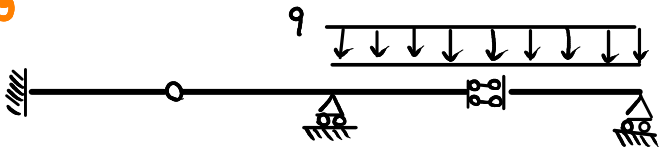
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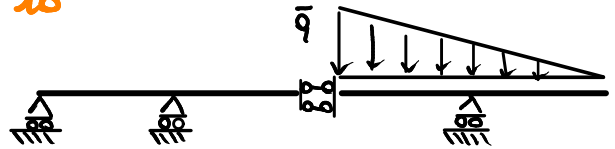
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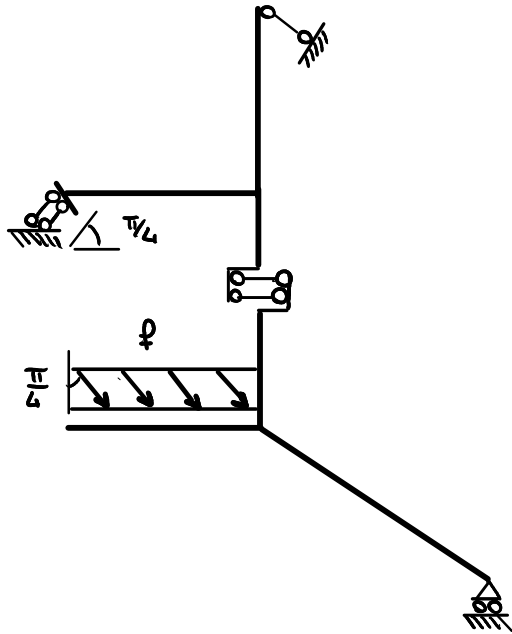
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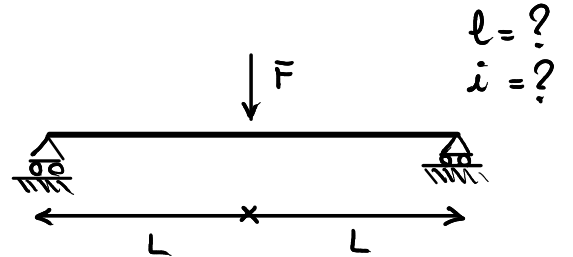
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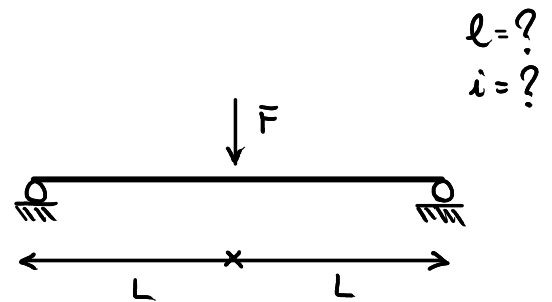
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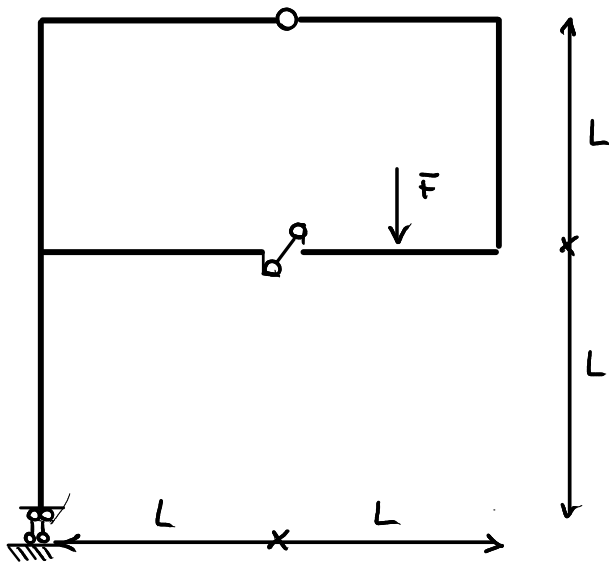
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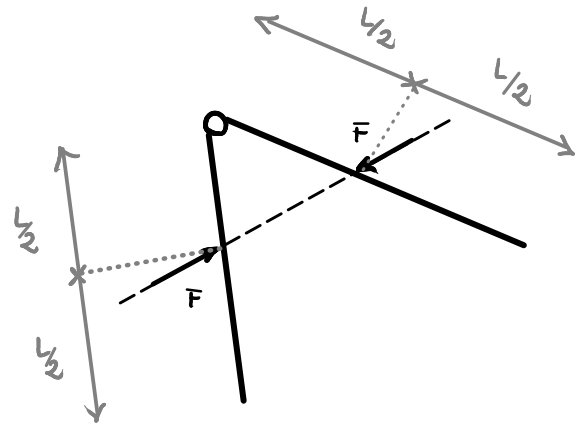
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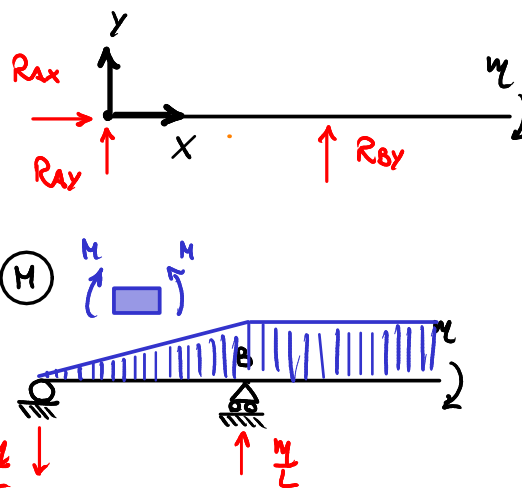
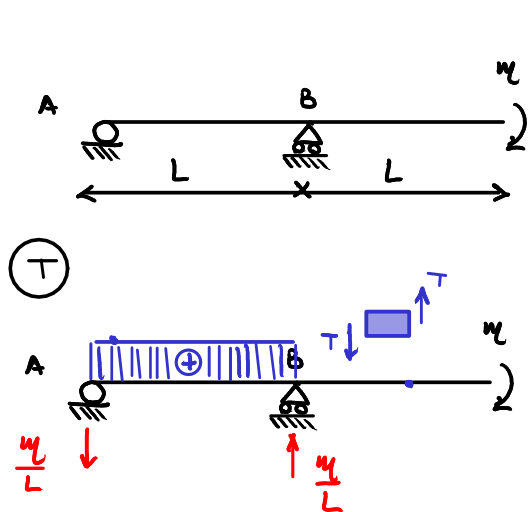
14



15



Esercizio 1



EQUILIBRIO A FORZA

$$\begin{aligned} \parallel X: R_{ax} &= 0 \\ \parallel Y: R_{ay} + R_{by} &= 0 \end{aligned}$$

EQUILIBRIO A MOMENTO

$$\parallel Z: M_z(A) = -R_{by}L - wL^2/2 = 0$$

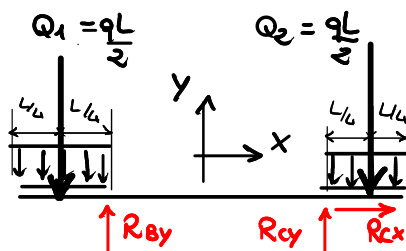
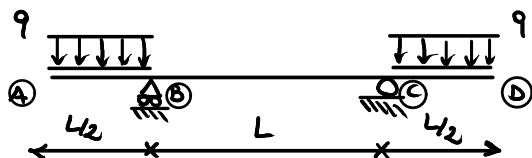
$$R_{by} = -\frac{wL}{2}, R_{ay} = \frac{wL}{2}, R_{ax} = 0$$

$$T(z) = \begin{cases} wL/2 & z \in AB \\ 0 & z \in BC \end{cases}$$

$$M(z) = \begin{cases} -\frac{w}{2}z & z \in AB \\ -wL & z \in BC \end{cases}$$

$$N(z) = 0 \quad \forall z$$

Esercizio 2



DISTRIBUZIONE DI CARICO
CONSTANTE $q(z) = q$

Staticamente equivalente a una forza Q pari all' "area" del rettangolo della distribuzione (qui $q \cdot L/2$), e posta nel suo baricentro

EQUILIBRIO

$$\parallel X: R_{cx} = 0$$

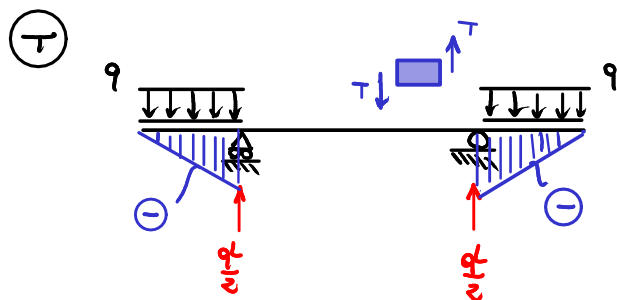
$$\parallel Y: R_{by} + R_{cy} + Q_1 + Q_2 = 0$$

$$\parallel Z: M_z(B) = +Q_1(L + \frac{L}{4}) + R_{ay}L - Q_2\frac{L}{4} = 0$$

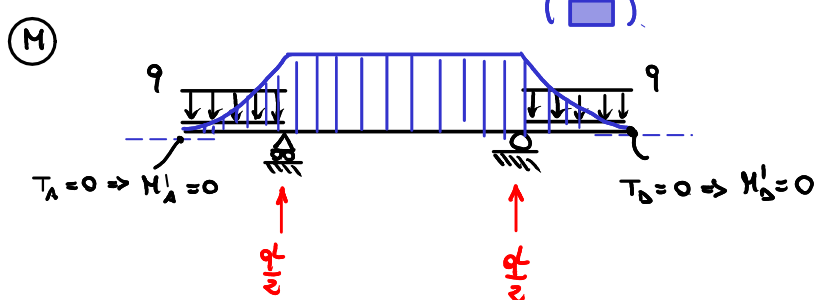
$$R_{cx} = 0$$

$$R_{by} = -\frac{qL}{2}$$

$$R_{cy} = -\frac{qL}{2}$$

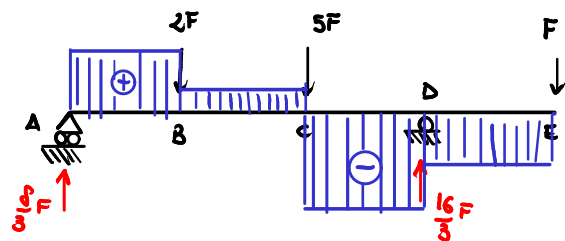
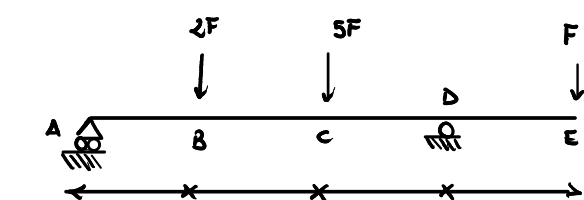


$$T(z) = \begin{cases} -qz & z \in AB \\ 0 & z \in BC \\ -q(L-z) & z \in CD \end{cases}$$

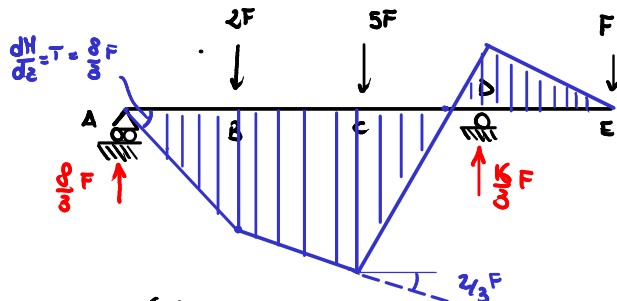
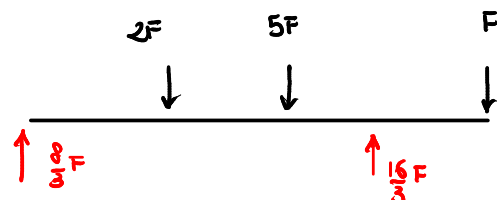


$$M(z) = \begin{cases} -qz \cdot \frac{z}{2} & z \in AB \\ -\frac{qL^2}{2} & z \in BC \\ -q\frac{(L-z)^2}{2} & z \in CD \end{cases}$$

Esercizio 3

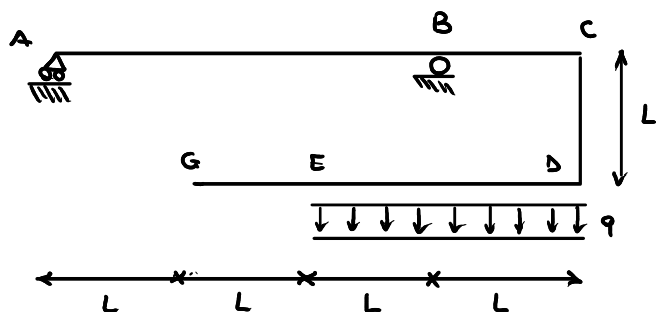


$$T(z) = \begin{cases} \frac{8}{3} F & z \in AB \\ \frac{8}{3} F - 2F = \frac{2}{3} F & z \in BC \\ \frac{8}{3} F - 2F - 5F = -\frac{13}{3} F & z \in CD \\ \frac{8}{3} F - 2F - 5F + \frac{16}{3} F = F & z \in DE \end{cases}$$

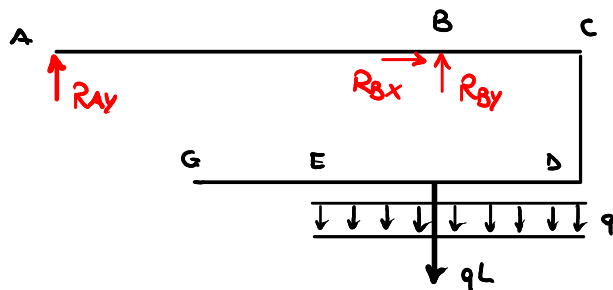


$$H(z) = \begin{cases} \frac{8}{3} F z & z \in AB \\ \frac{8}{3} F(L+z) - 2Fz = \frac{8}{3} FL + \frac{2}{3} Fz & z \in BC \\ \frac{8}{3} F(2L+z) - 2F(L+z) - 5Fz = \frac{10}{3} FL - \frac{13}{3} Fz & z \in CD \\ \frac{8}{3} F(3L+z) - 2F(2L+z) - 5F(L+z) + \frac{16}{3} Fz = F(L-z) & z \in DE \end{cases}$$

Esercizio 4



EQUAZIONI DI EQUILIBRIO



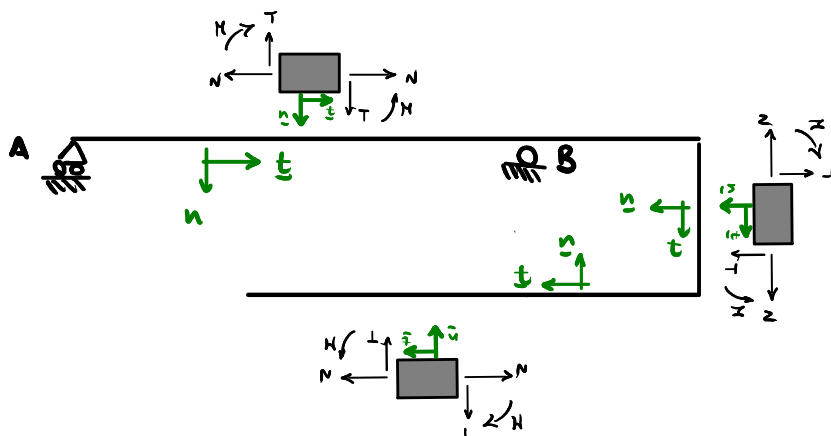
$$R_{bx} = 0$$

$$R_{by} + R_{ay} + 2qL = 0$$

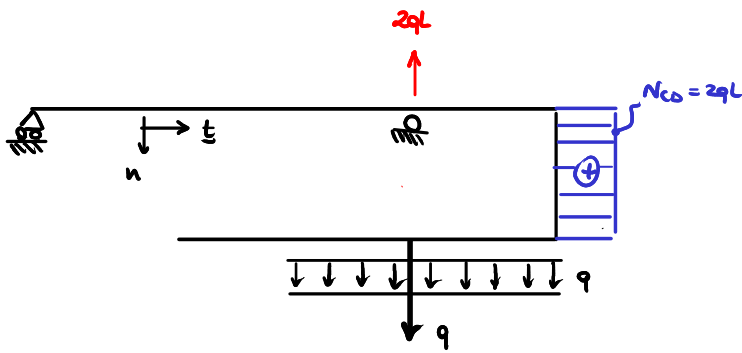
$$M_z(B) = R_{ay} 3L = 0$$

$$\begin{aligned} R_{ay} &= 0 \\ R_{bx} &= 0 \\ R_{by} &= -2qL \end{aligned}$$

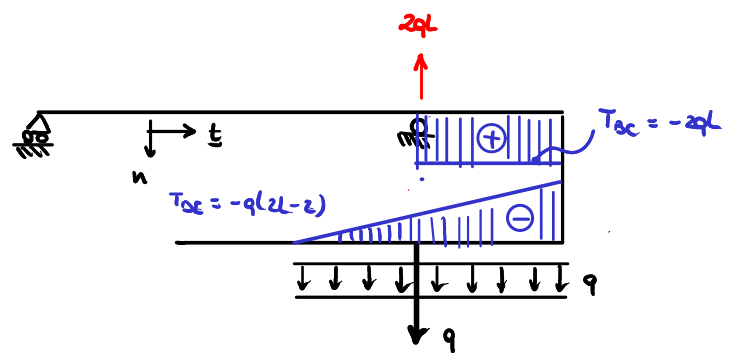
VERSI DI PERCORRENZA



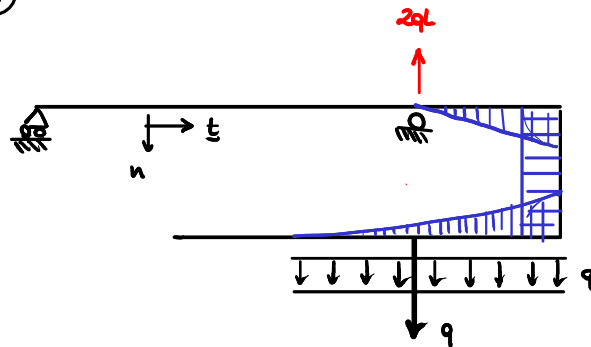
(N)



(T)



(M)

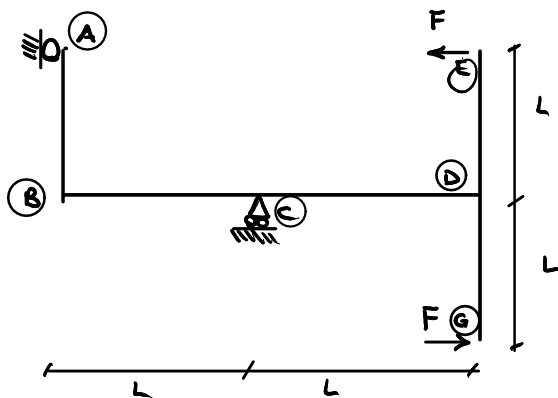


$$M_{BC} = 2qL^2$$

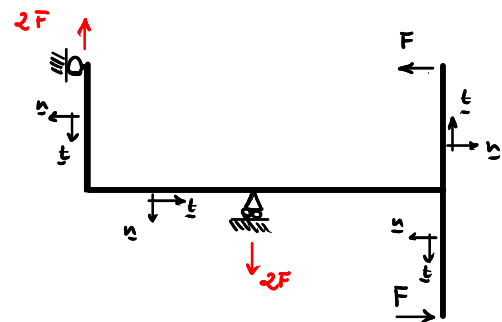
$$M_{CD} = 2qL^2$$

$$M_{DE} = \frac{q(2L-z)^2}{2}$$

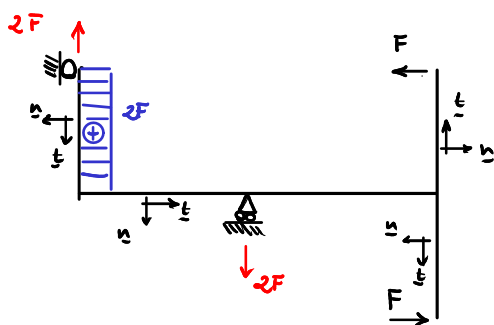
Esercizio 5



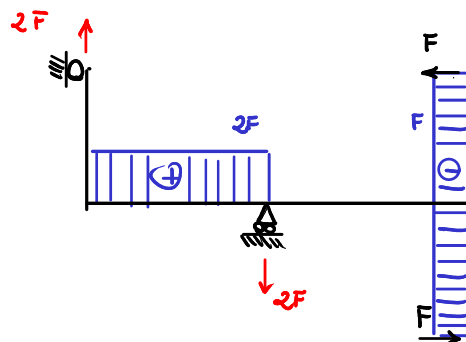
REAZIONI VINCOLARI ; VERSO DI PERCORRENZA DELLA LINEA D'ASSE



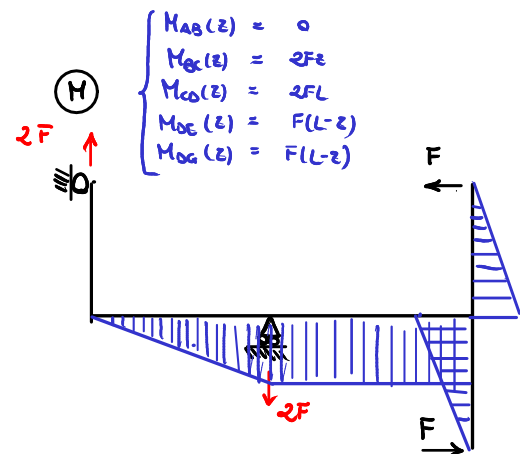
(N)



(T)

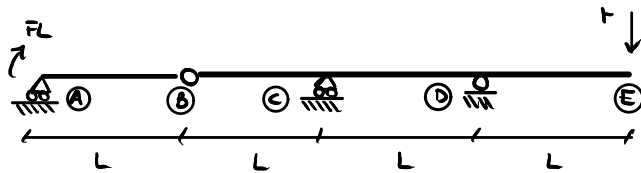


(M)



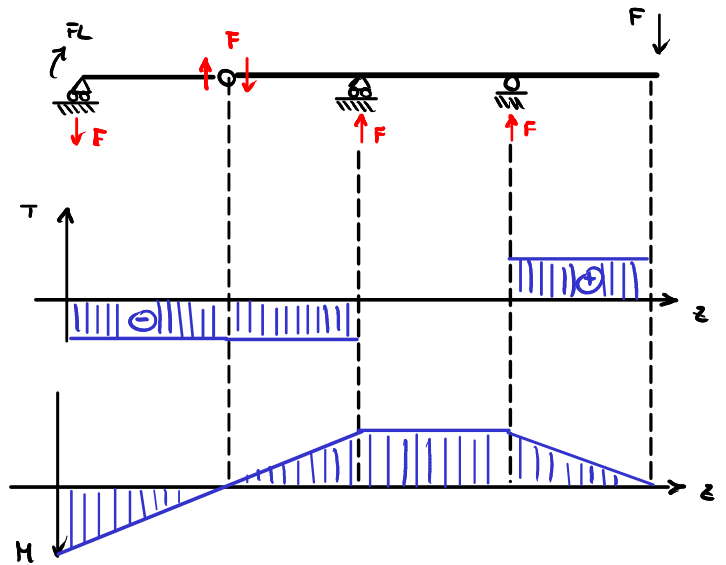
$$\begin{cases} M_{AB}(z) = 0 \\ M_{BC}(z) = 2Fz \\ M_{CD}(z) = 2FL \\ M_{DE}(z) = F(L-z) \\ M_{EG}(z) = F(L-z) \end{cases}$$

Esercizio 6

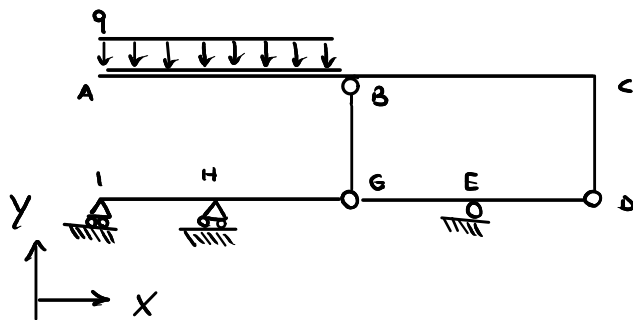


$$T(z) = \begin{cases} -F & z \in AB \\ -F & z \in BC \\ 0 & z \in CD \\ F & z \in DE \end{cases}$$

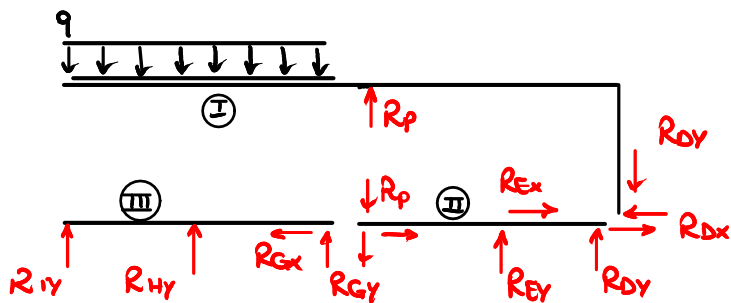
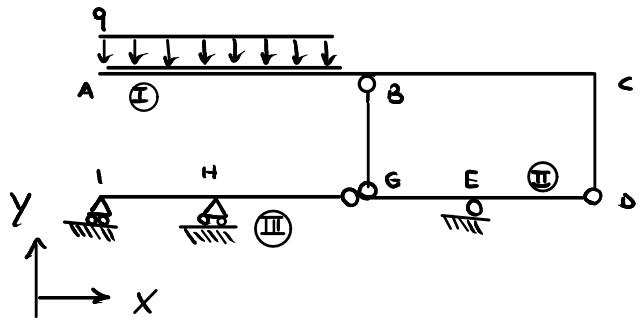
$$M(z) = \begin{cases} F(L-z) & z \in AB \\ -Fz & z \in BC \\ -FL & z \in CD \\ -F(L-z) & z \in DE \end{cases}$$



Esercizio 7



Soluzione della statica 1: BG come pendolo che connette B e G⁺

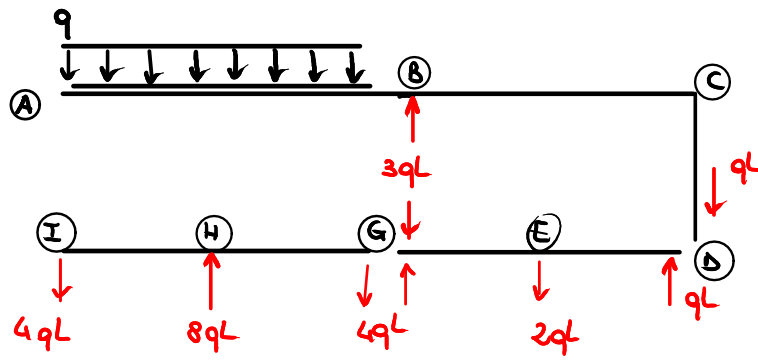


$$\textcircled{I} \begin{cases} -R_{Dx} = 0 \\ -2qL + R_p - R_{Dy} = 0 \\ M_z(B) = 2qL \cdot L - R_{Dy} \cdot 2L - R_{Dx} L = 0 \end{cases}$$

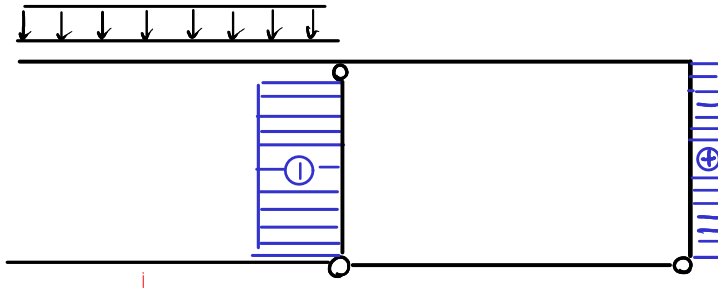
$$\textcircled{II} \begin{cases} R_{Gx} + R_{Ex} + R_{Dx} = 0 \\ -R_{Gy} + R_{Ey} + R_{Dy} - R_p = 0 \\ M_z(D) = -R_{Ey}L + R_p 2L - R_{Gy} 2L = 0 \end{cases}$$

$$\textcircled{III} \begin{cases} -R_{Gx} = 0 \\ R_{Iy} + R_{Hy} + R_{Gy} = 0 \\ M_z(H) = -R_{Iy}L + R_{Gy}L = 0 \end{cases}$$

$$R_p = 3qL, R_{Dy} = qL, R_{Ey} = -2qL, R_{Gy} = -4qL, R_{Hy} = 8qL, R_{Iy} = -4qL$$



(N)



$$N_{AB} = 0$$

$$N_{EG} = 0$$

$$N_{BC} = 0$$

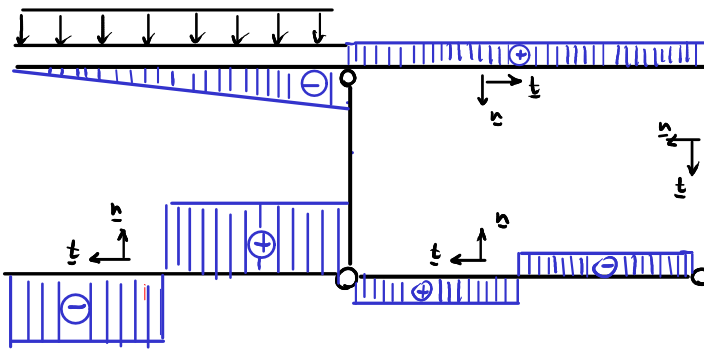
$$N_{GH} = 0$$

$$N_{CD} = qL$$

$$N_{HI} = 0$$

$$N_{DE} = 0$$

(T)



$$T_{AB} = -qz$$

$$T_{EG} = qL$$

$$T_{BC} = qL$$

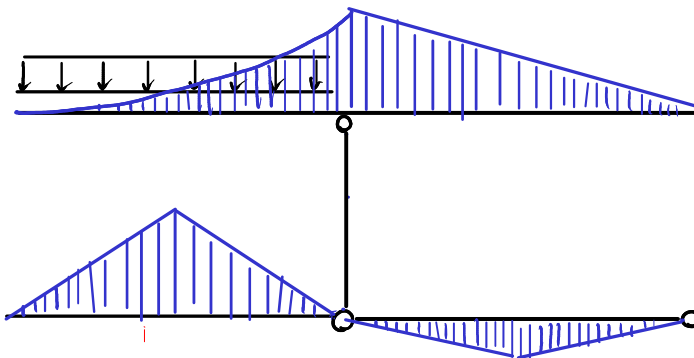
$$T_{GH} = 4qL$$

$$T_{CD} = 0$$

$$T_{HI} = -4qL$$

$$T_{DE} = -qL$$

(M)



$$M_{AB} = -\frac{qL^2}{2}$$

$$M_{EG} = -qL(L-z)$$

$$M_{BC} = -qL(2L-z)$$

$$M_{GH} = 4qLz$$

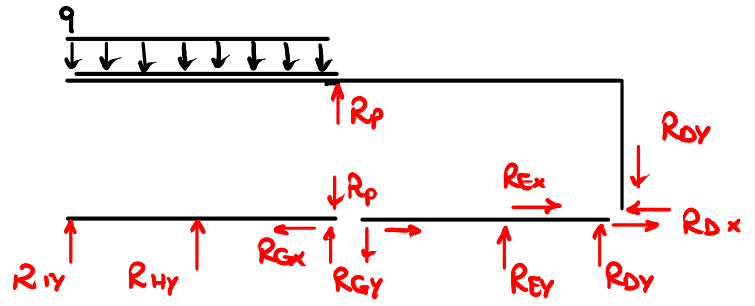
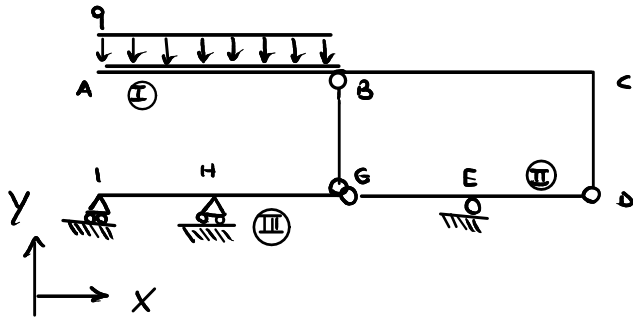
$$M_{CD} = 0$$

$$M_{HI} = 4qL(L-z)$$

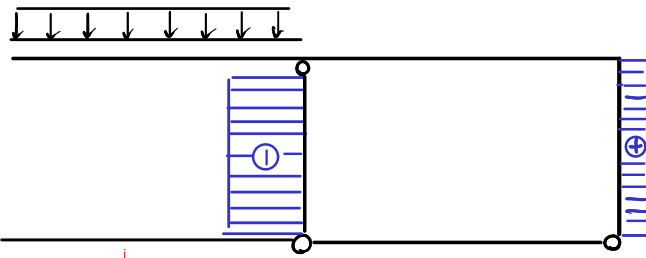
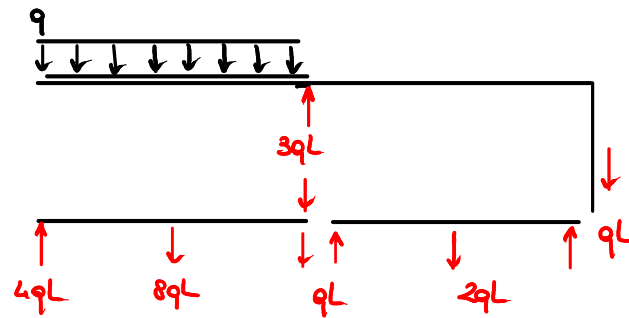
$$M_{DE} = -qLz$$

Si osservi che la presenza di una cerniera interna (G) rende indifferente la scelta di porre uno degli estremi del pendolo interno BG in G^+ o G^- .

Soluzione della statica 2: BG come pendolo che connette B e G



$$\begin{cases} -R_{Dx} = 0 \\ -2qL + R_p - R_{Dy} = 0 \\ M_z(B) = 2qL \cdot L - R_{Dy} \cdot 2L - R_{Dx} L = 0 \end{cases} \quad \begin{cases} R_{Gx} + R_{Ex} + R_{Dx} = 0 \\ -R_{Gy} + R_{Ey} + R_{Dy} = 0 \\ M_z(D) = -R_{Ey} L + R_{Gy} 2L = 0 \end{cases} \quad \begin{cases} -R_{Gx} = 0 \\ R_{1y} + R_{Hy} + R_{Gy} - R_p = 0 \\ M_z(H) = -R_{1y} L + R_{Gy} L - R_p L = 0 \end{cases}$$



$$N_{AB} = 0$$

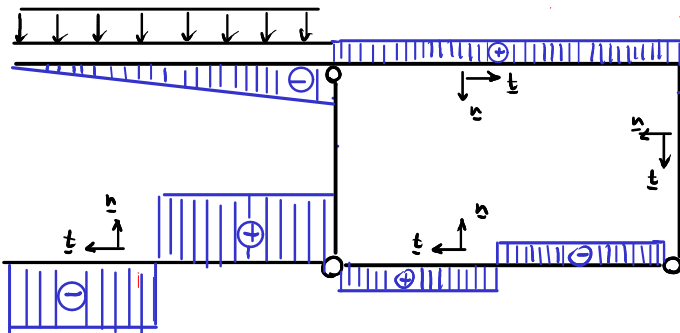
$$N_{EG} = 0$$

$$N_{BC} = 0$$

$$N_{GH} = 0$$

$$N_{CD} = qL$$

$$N_{HI} = 0$$



$$T_{AB} = -qz$$

$$T_{EG} = qL$$

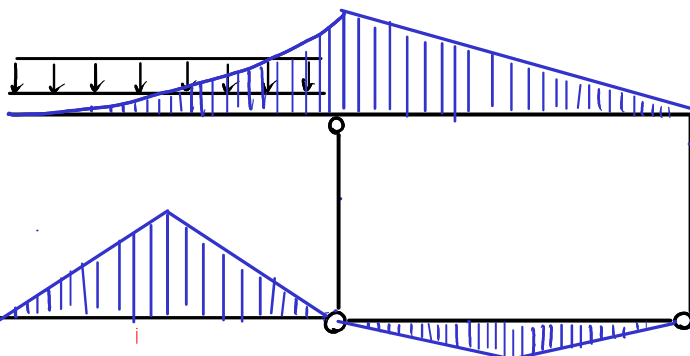
$$T_{BC} = qL$$

$$T_{GH} = 4qL$$

$$T_{CD} = 0$$

$$T_{HI} = -4qL$$

$$T_{DE} = -qL$$



$$M_{AB} = -\frac{qz^2}{2}$$

$$M_{EG} = -qL(L-z)$$

$$M_{BC} = -qL(2L-z)$$

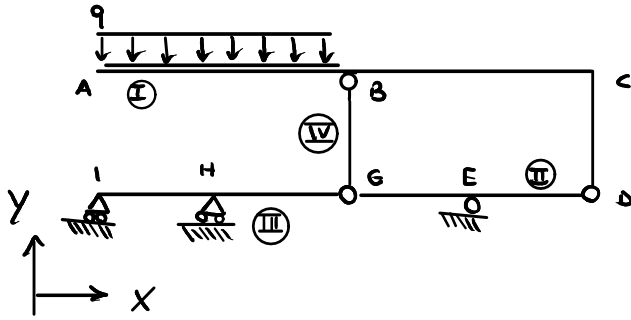
$$M_{GH} = 4qLz$$

$$M_{CD} = 0$$

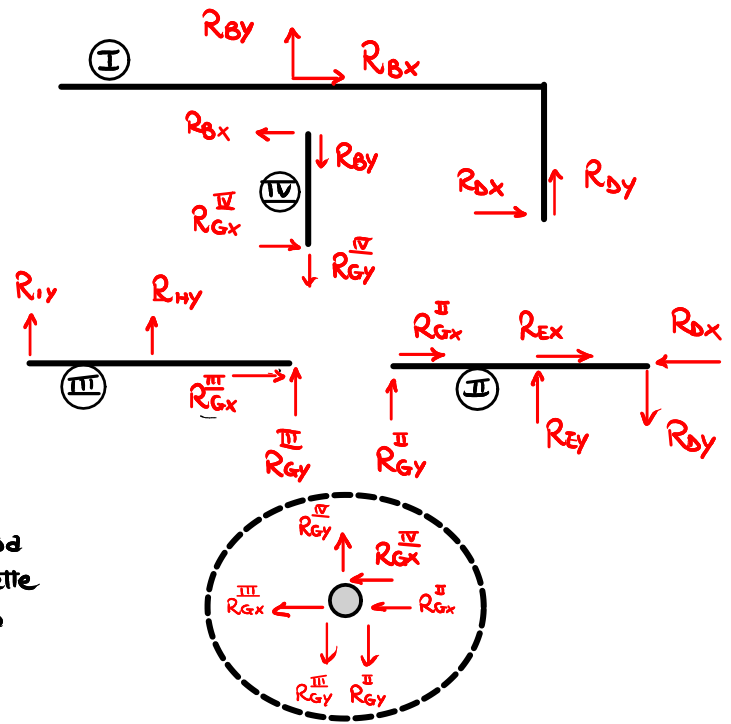
$$M_{HI} = 4qL(L-z)$$

$$M_{DE} = -qLz$$

Soluzione della statica 3: BG come corpo a sé



In questa scelta la cerniera G viene vista come cerniera a tre vie, o cerniera tripla: essa esprimerà su ciascuno dei corpi II, III, IV che connette delle reazioni vincolari in generale diverse tra loro ($R_{Gx}^I \neq R_{Gx}^{II} \neq R_{Gx}^{III}$ e $R_{Gy}^I \neq R_{Gy}^{II} \neq R_{Gy}^{III}$)



EQUILIBRIO DEI CORPI TRAVE

$$\textcircled{I} \quad R_{Bx} + R_{Dx} = 0$$

$$R_{By} + R_{Dy} - 2qL = 0$$

$$M_z(B) = 2qL^2 + R_{Dx}L + R_{Dy}2L = 0$$

$$\textcircled{II} \quad R_{Gx}^{II} + R_{Ex} - R_{Dx} = 0$$

$$R_{Gy}^{II} + R_{Ey} - R_{Dy} = 0$$

$$M_z(D) = -R_{Gy}^{II}2L - R_{Ey}L = 0$$

$$\textcircled{IV} \quad -R_{Bx} + R_{Gx}^{IV} = 0$$

$$-R_{By} - R_{Gy}^{IV} = 0$$

$$M_z(G) = R_{Bx}L = 0$$

$$\textcircled{III} \quad R_{Gx}^{III} = 0$$

$$R_{Iy} + R_{Hy} + R_{Gy}^{III} = 0$$

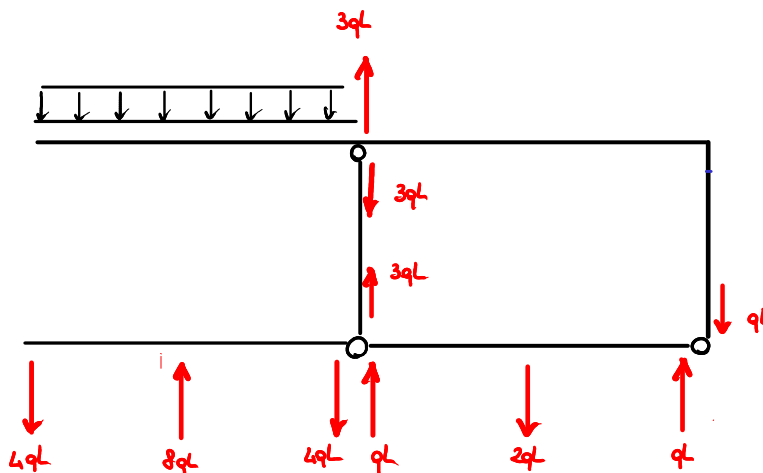
$$M_z(G) = -R_{Iy}2L - R_{Hy}L = 0$$

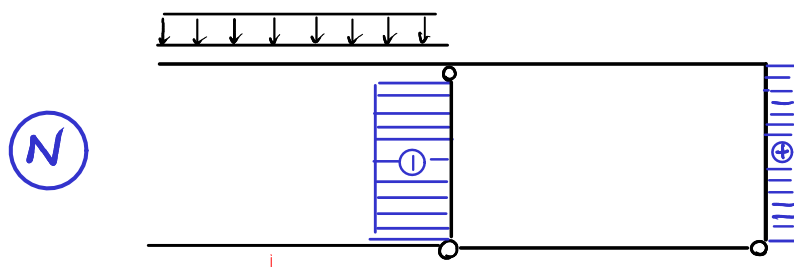
EQUILIBRIO DEL "CORPO" CERNIERA

$$\underline{R}_G^{II} + \underline{R}_G^{III} + \underline{R}_G^{IV} = \underline{0}$$

$$R_{Gx}^{II} + R_{Gx}^{III} + R_{Gx}^{IV} = 0$$

$$R_{Gy}^{II} + R_{Gy}^{III} - R_{Gy}^{IV} = 0$$





$$N_{AB} = 0$$

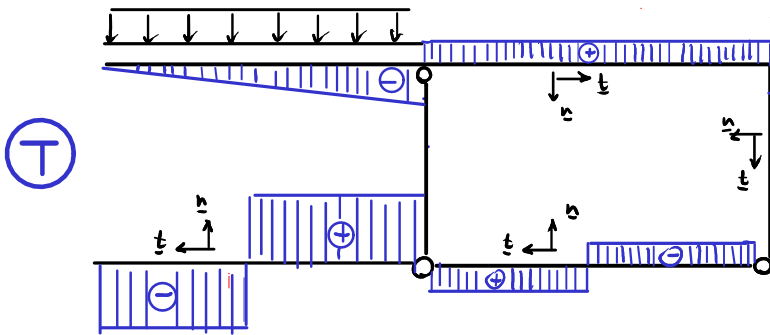
$$N_{EG} = 0$$

$$N_{BC} = 0$$

$$N_{GH} = 0$$

$$N_{CD} = qL$$

$$N_{HI} = 0$$



$$T_{AB} = -qz$$

$$T_{EG} = qL$$

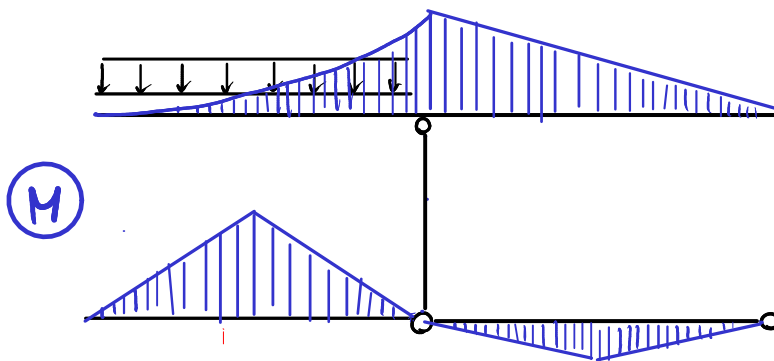
$$T_{BC} = qL$$

$$T_{GH} = 4qL$$

$$T_{CD} = 0$$

$$T_{HI} = -4qL$$

$$T_{DE} = -qL$$



$$M_{AB} = -\frac{qz^2}{2}$$

$$M_{EG} = -qL(L-z)$$

$$M_{BC} = -qL(2L-z)$$

$$M_{GH} = 4qLz$$

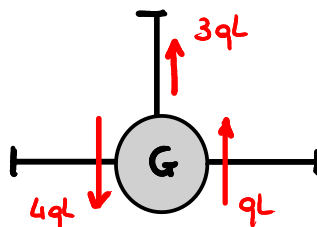
$$M_{CD} = 0$$

$$M_{HI} = 4qL(L-z)$$

$$M_{DE} = -qLz$$

Dai tre casi analizzati è evidente che quando una forza concentrata agisce su una cerniera interna (qui la forza era la reazione R_D), è indifferente vedere tale forza applicata poco prima, poco dopo o direttamente sulla cerniera. Infatti:

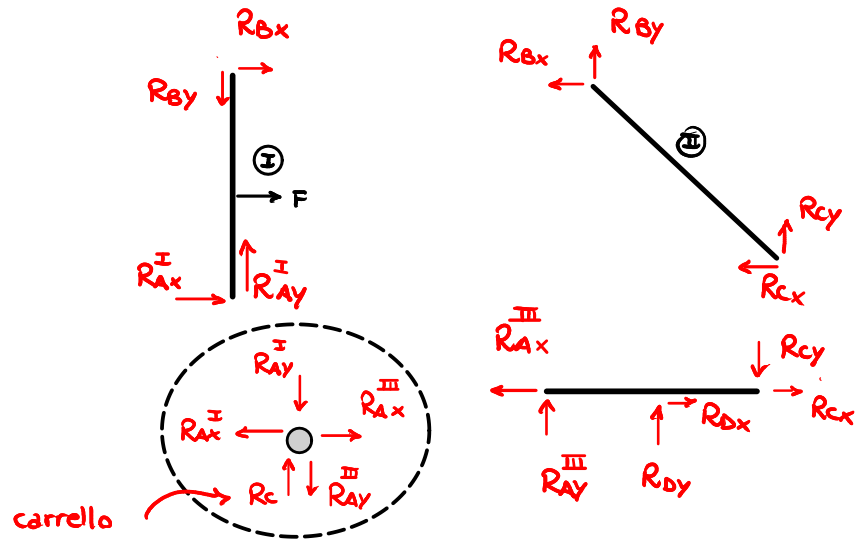
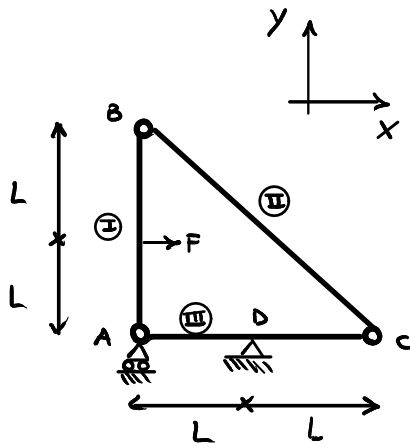
- a seconda di dove si vuole considerare applicato il carico concentrato sulla cerniera le reazioni di quest'ultima cambiano, ma la forza netta agente su ciascuna sezione connessa dalla cerniera è invariata. Ad esempio nei tre casi qui risolti:



- se la forza netta agente su ciascuna sezione è la stessa le caratteristiche della sollecitazione non cambiano.

Esercizio 8

Soluzione 1 (lunga): reazione del carrello sulla cerniera
BC trattato come corpo



EQUILIBRIO CORPO I	EQUILIBRIO CORPO II	EQUILIBRIO CORPO III
$R_{Bx} + F + R_{Ax}^I = 0$	$-R_{Bx} - R_{Cx} = 0$	$-R_{Ax}^III + R_{Bx} + R_{Cx} = 0$
$-R_{By} + R_{Ay}^I = 0$	$R_{By} + R_{Cy} = 0$	$R_{Ay}^III + R_{By} - R_{Cy} = 0$
$M_z(A) = -FL - R_{Bx}2L = 0$	$M_z(A) = R_{Bx}2L - R_{By}2L = 0$	$M_z(A) = R_{By}L - R_{Cy}2L = 0$

EQUILIBRIO CERNIERA

$$\begin{aligned} -R_{Ax}^I + R_{Ax}^III &= 0 \\ -R_{Ay}^I + R_C - R_{Ay}^III &= 0 \end{aligned}$$

$$R_{Ax}^I = -F/2 \quad R_{Ay}^I = -F/2$$

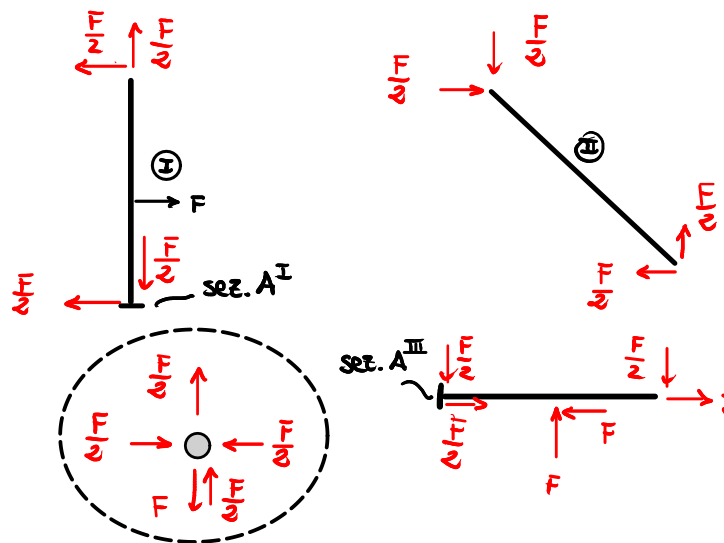
$$R_{Bx} = -F/2 \quad R_{By} = -F/2$$

$$R_{Cx} = F/2 \quad R_{Cy} = F/2$$

$$R_{Bx} = -F \quad R_{By} = F$$

$$R_{Ax}^III = -F/2 \quad R_{Ay}^III = -F/2$$

$$R_C = -F$$



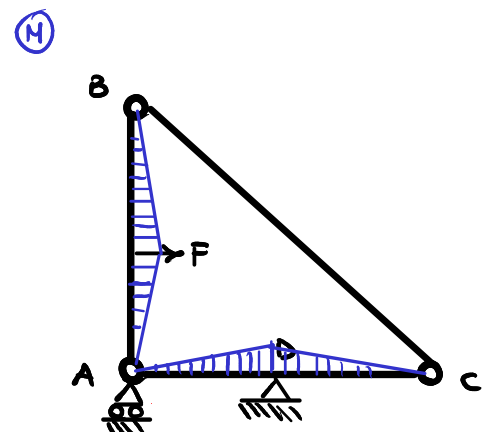
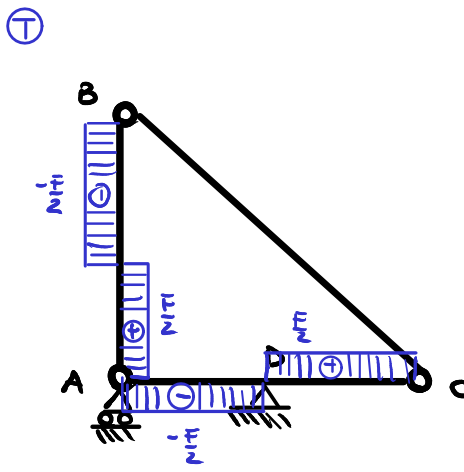
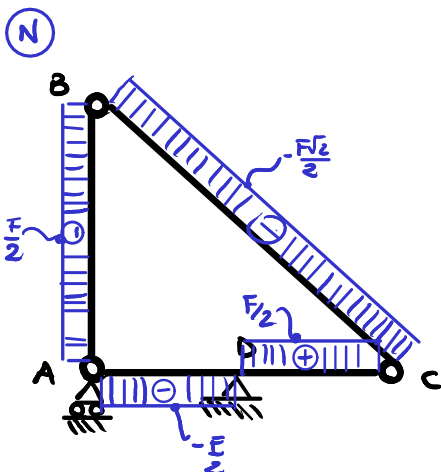
OSS

Forza netta sulla sezione A^I

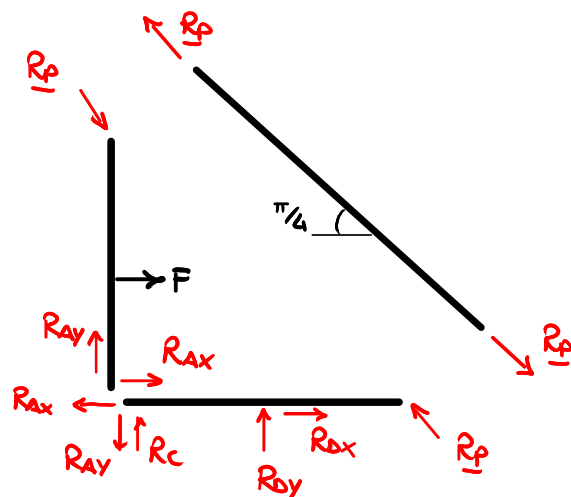
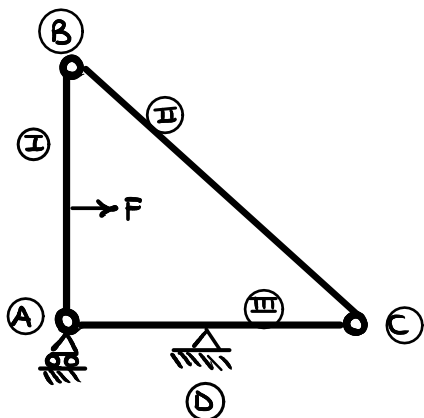
$$\underline{R}(A^I) = -\frac{F}{2}\underline{e}_x - \frac{F}{2}\underline{e}_y$$

Forza netta sulla sezione A^{III}

$$\underline{R}(A^{III}) = \frac{F}{2}\underline{e}_x - \frac{F}{2}\underline{e}_y$$



Soluzione 2 : reazione del carrello sul corpo (III), BC come pendolo



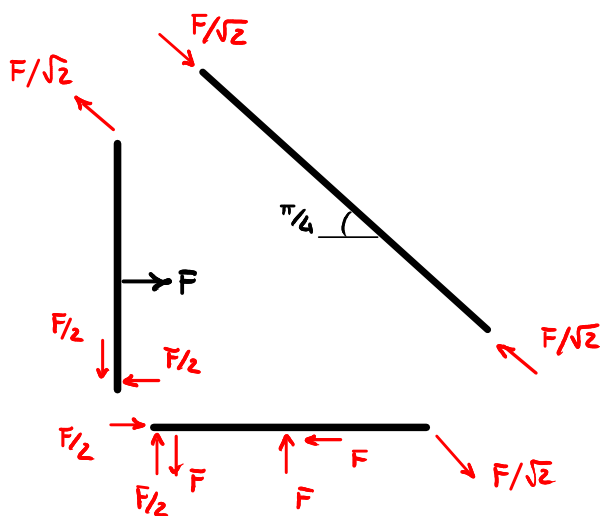
EQUILIBRIO CORPO I	EQUILIBRIO CORPO (III)
$R_{Ax} + F + R_p \frac{\sqrt{2}}{2} = 0$	$-R_{Ax} + R_{Cx} - R_p \frac{\sqrt{2}}{2} = 0$
$R_{Ay} - \frac{R_p}{\sqrt{2}} = 0$	$-R_{Ay} + R_{Cy} + R_p \frac{\sqrt{2}}{2} = 0$
$M_z(B) = FL + 2R_{Ax}L = 0$	$M_z(b) = R_p \frac{\sqrt{2}}{2}L - R_{Cx}L + R_{Ay}L = 0$

$$R_{Ax} = -F/2 \quad R_{Ay} = -F/2$$

$$R_p = -\frac{\sqrt{2}}{2} F$$

$$R_{Dx} = -F \quad R_{Dy} = F$$

$$R_c = -F$$



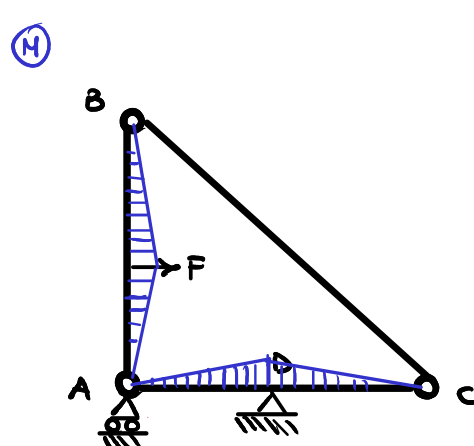
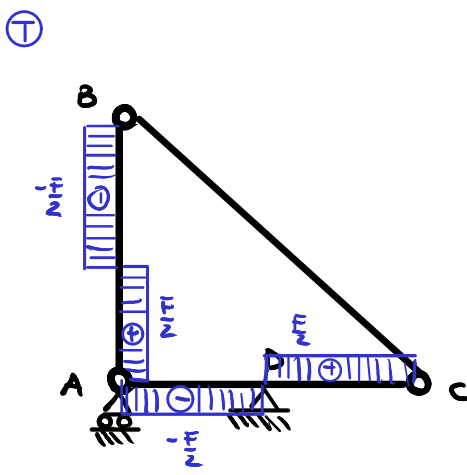
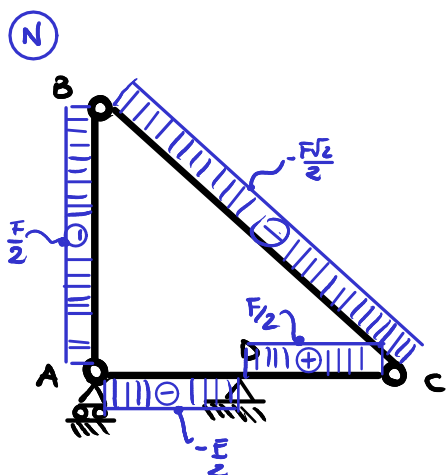
OSS

Forza netta sulla sezione A^I

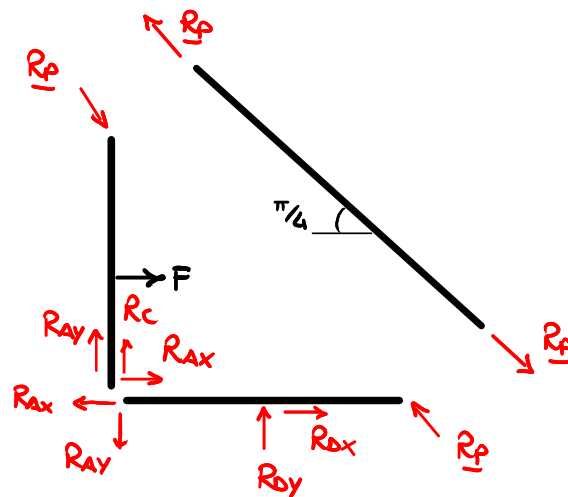
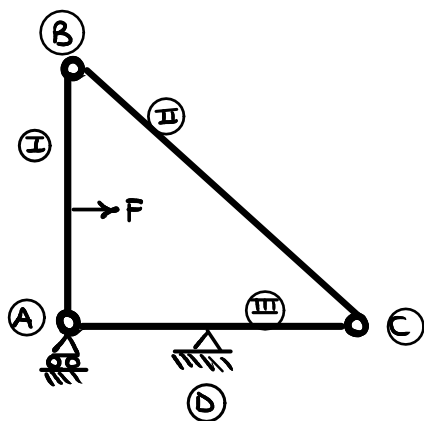
$$\underline{R}(A^I) = -\frac{F}{2} \underline{e}_x - \frac{F}{2} \underline{e}_y$$

Forza netta sulla sezione A^{III}

$$\underline{R}(A^{III}) = \frac{F}{2} \underline{e}_x - \frac{F}{2} \underline{e}_y$$



Soluzione 3 : reazione del carrello sul corpo (I), Bc come pendolo



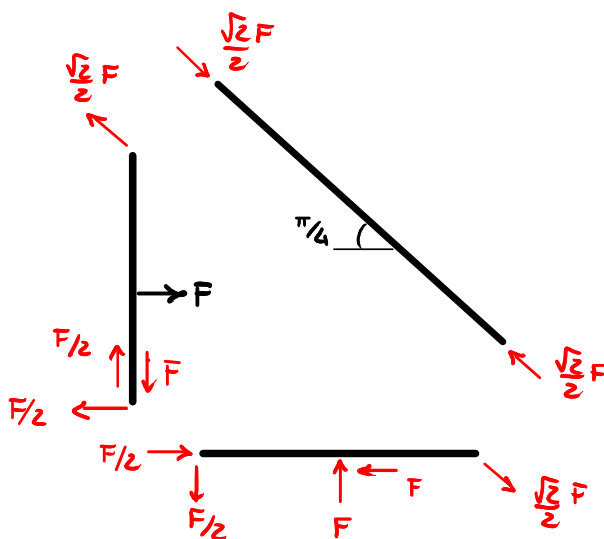
EQUILIBRIO CORPO I	EQUILIBRIO CORPO III
$R_{Ax} + F + R_P \frac{\sqrt{2}}{2} = 0$	$-R_{Ax} + R_{Cx} - R_P \frac{\sqrt{2}}{2} = 0$
$R_{Ay} - \frac{R_P}{\sqrt{2}} + R_C = 0$	$-R_{Ay} + R_{Cy} + R_P \frac{\sqrt{2}}{2} = 0$
$H_z(B) = FL + 2R_{Ax}L = 0$	$H_z(b) = R_P \frac{\sqrt{2}}{2}L + R_{Ay}L = 0$

$$R_{Ax} = -F/2 \quad R_{Ay} = F/2$$

$$R_P = -\frac{\sqrt{2}}{2}F$$

$$R_{Cx} = -F \quad R_{Cy} = F$$

$$R_C = -F$$



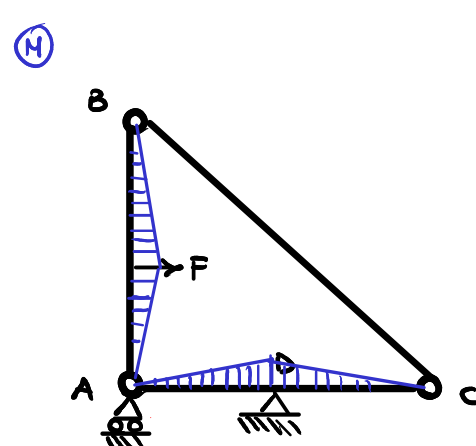
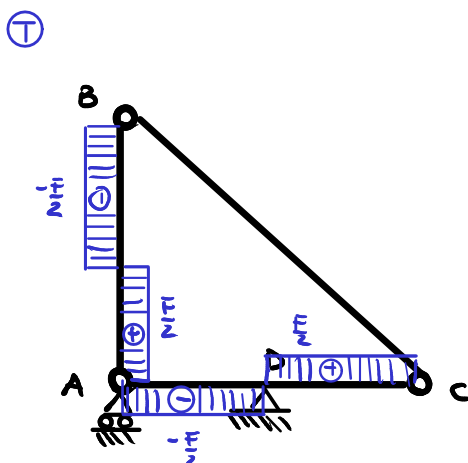
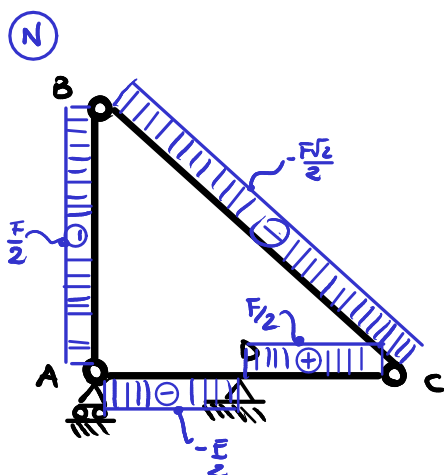
OSS

Forza netta sulla sezione A^I

$$\underline{R}(A^I) = -\frac{F}{2}\underline{e}_x - \frac{F}{2}\underline{e}_y$$

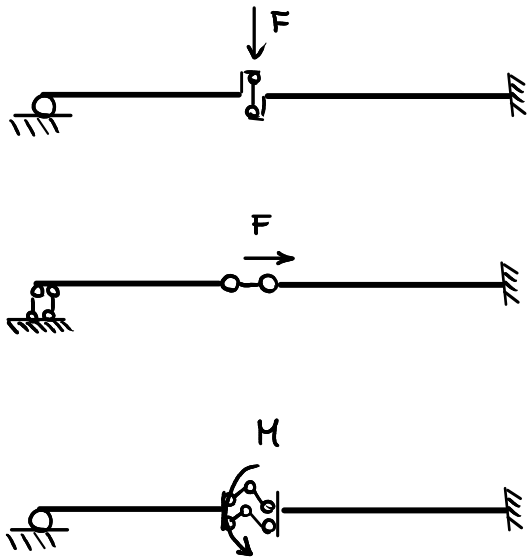
Forza netta sulla sezione A^{III}

$$\underline{R}(A^{III}) = \frac{F}{2}\underline{e}_x - \frac{F}{2}\underline{e}_y$$

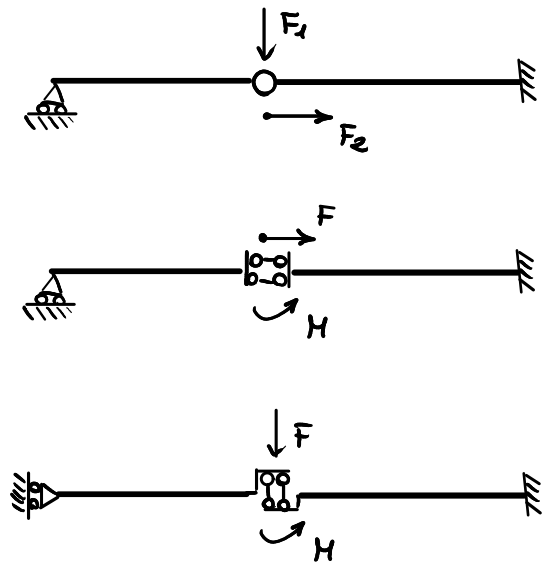


Osservazione - Ritenere indifferente il punto di applicazione di una forza / momento poco prima / poco dopo / direttamente su un vincolo interno è valido solo se l'ente concentrato ha la stessa direzione di quella del vincolo (se questo è semplice), o la stessa direzione di una delle direzioni del vincolo (se questo è doppio) -

VINCOLO INTERNO SEMPLICE



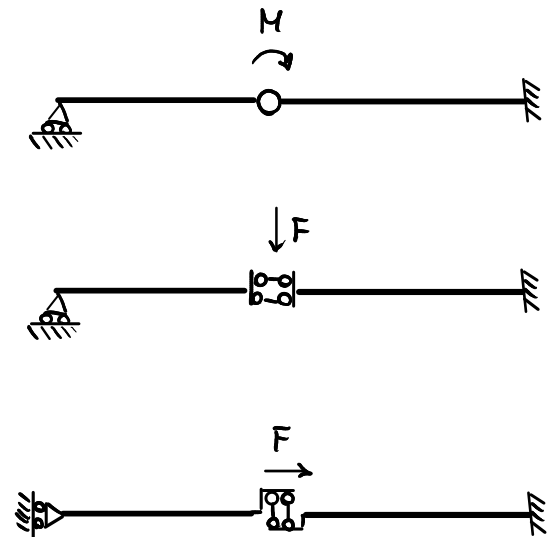
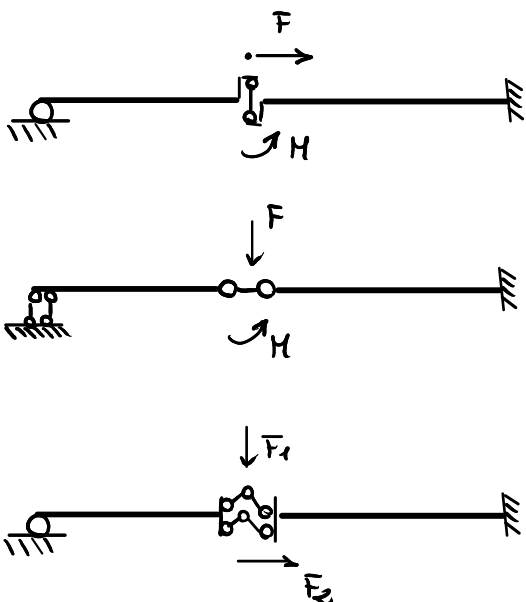
VINCOLO INTERNO DOPPIO



Negli esempi proposti qui sopra è indifferente vedere l'ente concentrato prima/dopo/sul vincolo B:

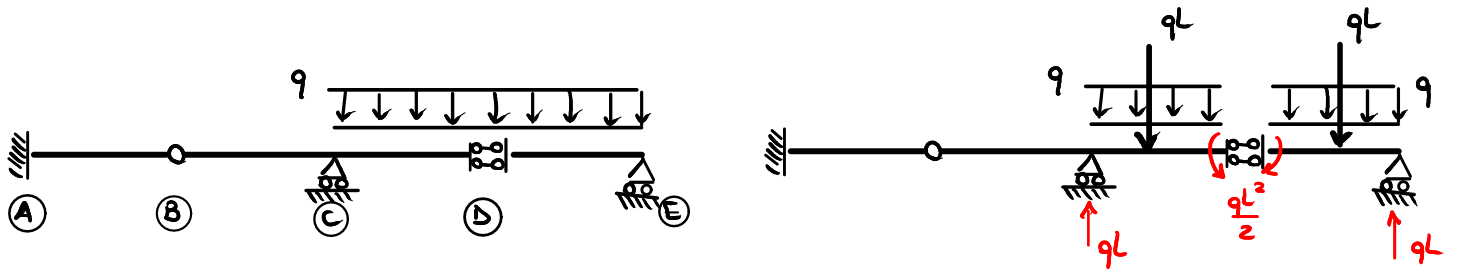
- le reazioni del vincolo interno saranno DIVERSE nei tre casi
- la forza netta agente sulle sezioni B^- , B^+ sarà invece la stessa nei tre casi, e quindi le caratteristiche della sollecitazione saranno uguali a prescindere da dove $\approx$ è scelto di vedere la forza applicata

Negli esempi che seguono, invece, bisogna sapere dove è applicato l'ente concentrato perché la soluzione della statica (reazioni e caratteristiche) è diversa a seconda che esso sia in B^- , B^+



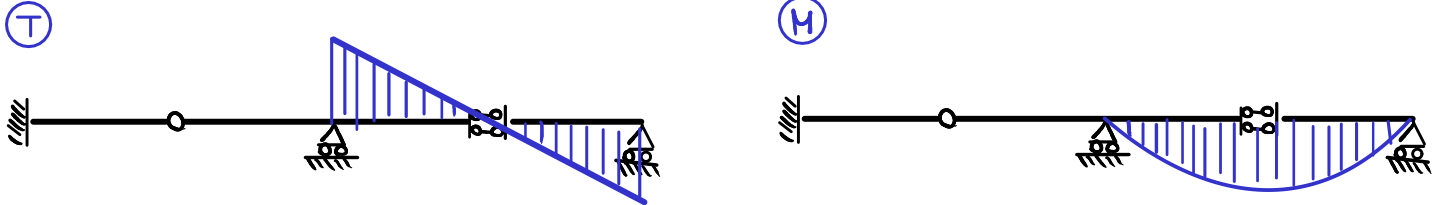
In casi come questi non $\approx$ può decidere arbitrariamente il punto di applicazione: esso deve essere fornito dal testo.

Esercizio 9

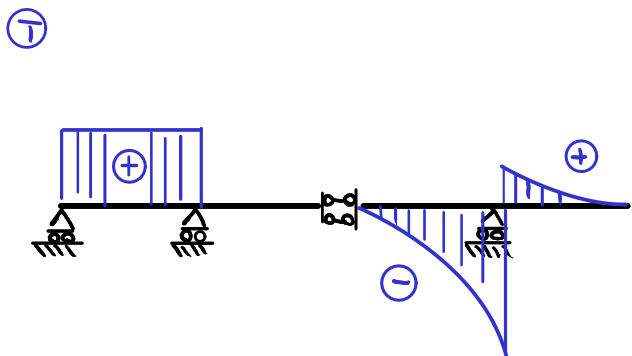
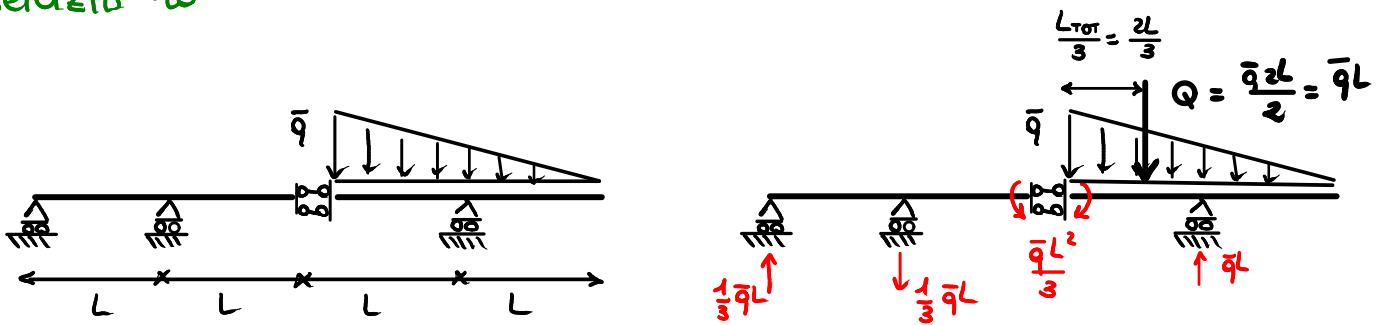


Si osservi che in questo caso non si può trattare il carico distribuito come un'unica forza risultante qL posta in D, per quanto discusso nell'esercizio precedente.

Conviene separare il carico in due distribuzioni, una su CB e l'altra su DE, considerando separatamente le risultanti



Esercizio 10

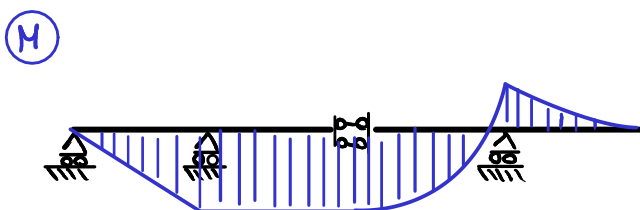


$$T_{AB} = \frac{\bar{q}L}{3}$$

$$T_{BC} = 0$$

$$T_{CD} = \frac{\bar{q}z^2}{4L} - \bar{q}z$$

$$T_{DE} = \bar{q} \frac{(L-z)^2}{4L}$$



$$M_{AB} = \frac{\bar{q}L}{3} z$$

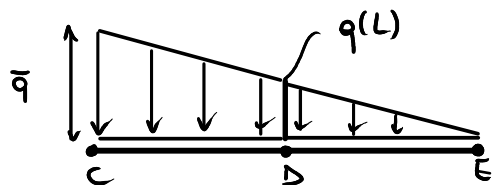
$$M_{BC} = \frac{\bar{q}L^2}{3}$$

$$M_{CD} = \frac{\bar{q}z^3}{12L} - \frac{\bar{q}z^2}{2} + \frac{\bar{q}L^2}{3}$$

$$M_{DE} = -\bar{q} \frac{(L-z)^3}{12L}$$

Derivazione di T e M sui tratti cd e DE

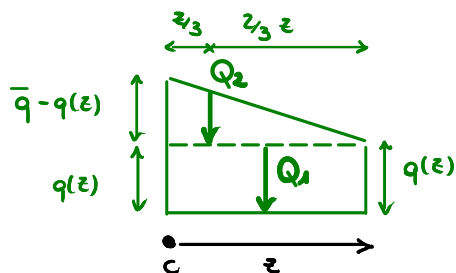
Sul tratto cd ($z=0$ in C e $z=L$ in D) il carico distribuito varia con una legge lineare che può essere ricavata ragionando su tutto il tratto CE .



$$\frac{q_{CE}(z) - \bar{q}}{0 - \bar{q}} = \frac{z - 0}{2L - 0} \Rightarrow q_{CE}(z) = \bar{q} \left(\frac{2L - z}{2L} \right) \equiv q_{\infty}(z)$$

In particolare $q(D) = q_{\infty}(L) = \frac{\bar{q}}{2}$

Perfano, preso un tratto intermedio su CD ed immaginando di guardare a precedere, la porzione di canco distribuito si può scomporre come segue



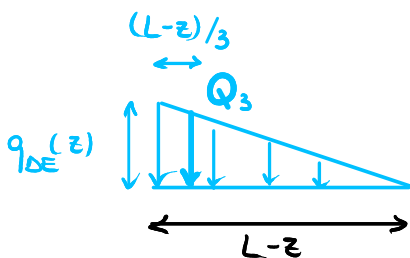
$$Q_A = q(z) z = \frac{\bar{q} z (2L - z)}{2L} \quad \text{risultante della pressione "rettangolare"}$$

$$Q_2 = \frac{(\bar{q} - q(z))z}{2} = \frac{\bar{q}z^2}{4L} \quad \text{risultante della porzione "triangolare"}$$

Consequentemente

$$T_{cd} = -(Q_1 + Q_2) = -q(z)z - \frac{(\bar{q} - q(z))z}{2} = \frac{\bar{q}z^2}{4L} - \bar{q}z$$

$$M_{CD} = \frac{\bar{q}L^2}{3} - \frac{2}{3}zQ_2 - \frac{z}{2}Q_1 = \frac{\bar{q}z^3}{12L} - \frac{\bar{q}z^2}{2} + \frac{\bar{q}L^2}{3}$$



Sul tratto DE avendo riparametrizzato il tratto DE a partire da D ($z=0$ in D, $z=L$ in E) non è possibile usare la funzione $q_{CE}(z)$ su questo tratto perché essa è stata costruita assumendo $z=0$ in A e $z=2L$ in E. Osservando però che in D l'intensità del campo distribuito è $q(D) = \bar{q}/2$, si può ricavare

$$\frac{q_{DE}(z) - \bar{q}/2}{0 - \bar{q}/2} = \frac{z - 0}{L - 0} \Rightarrow q_{DE}(z) = \frac{\bar{q}}{2} \left(\frac{L-z}{L} \right)$$

Guardando a seguire allora

$$T_{DE} = Q_3 = \frac{q_{DE}(z)(L-z)}{2} = \bar{q} \frac{(L-z)^2}{4L}$$

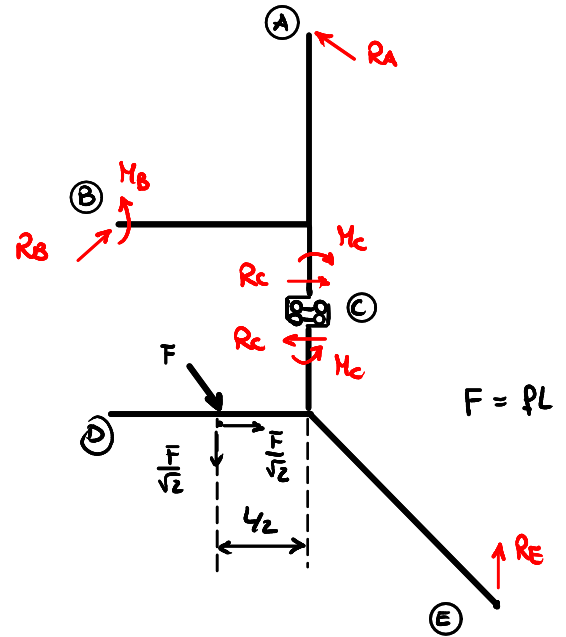
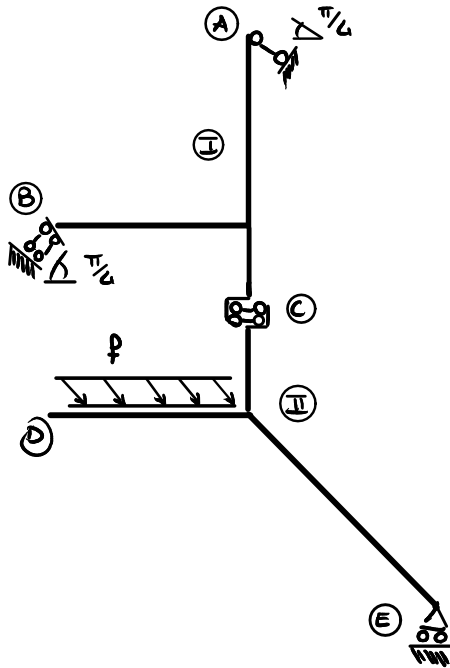
$$M_{DE} = Q_3 \left(\frac{L-z}{3} \right) = -9 \frac{(L-z)^3}{12L}$$

Verifica in D

$$\Delta T_D = T_D^+ - T_D^- = T_{D\in}(0) - T_{D\in}(L) = \bar{q}L \quad (\Delta T_D = -F_n)$$

$$\Delta M_D = M_D^+ - M_D^- = M_{DE}(0) - M_{CD}(L) = 0 \quad (\Delta M_D = M_{\text{ext}} = 0)$$

Esercizio 11



$$-R_A \frac{\sqrt{2}}{2} + R_B \frac{\sqrt{2}}{2} + R_C = 0$$

$$R_A \frac{\sqrt{2}}{2} + R_B \frac{\sqrt{2}}{2} = 0$$

$$M_Z(B) = M_B + R_A \frac{\sqrt{2}}{2} L + R_A \frac{\sqrt{2}}{2} L - M_C + R_C \frac{L}{2} = 0$$

$$-R_C + F \frac{\sqrt{2}}{2} = 0$$

$$-\frac{F \sqrt{2}}{2} + R_E = 0$$

$$M_Z(F) = R_C \frac{L}{2} + M_C + R_E \left(L + \frac{L}{2} \right) = 0$$

$$R_B = -F/2$$

$$R_A = F/2$$

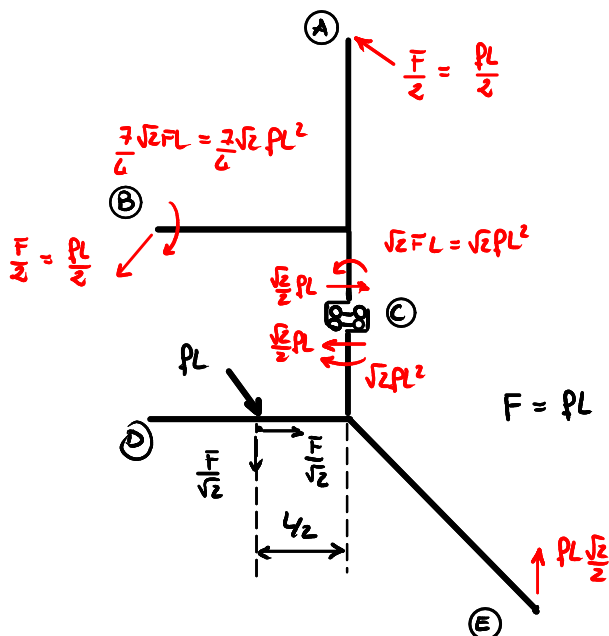
$$M_B = -\frac{7}{4} FL \sqrt{2}$$

$$R_C = F/\sqrt{2}$$

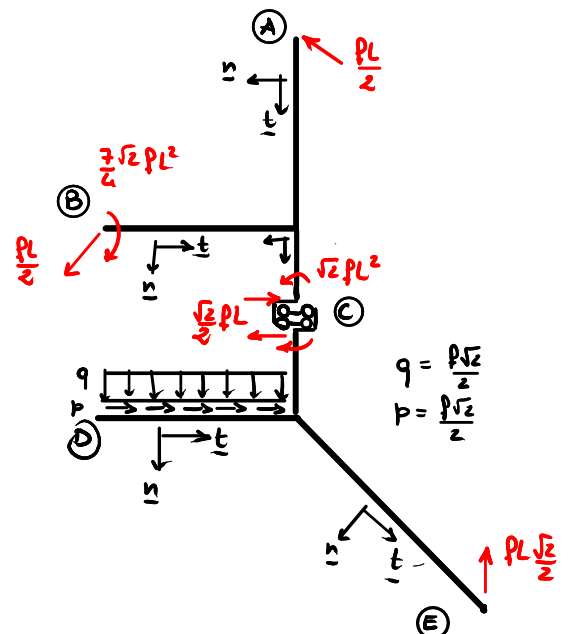
$$R_E = F/\sqrt{2}$$

$$M_C = -\sqrt{2} FL$$

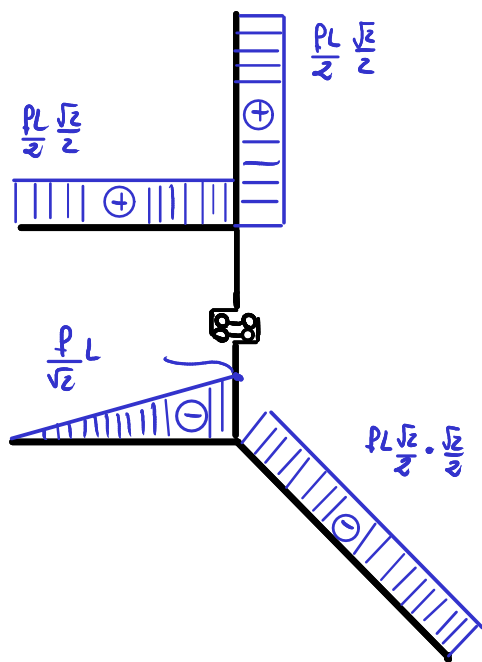
Diagramma di struttura libera finale



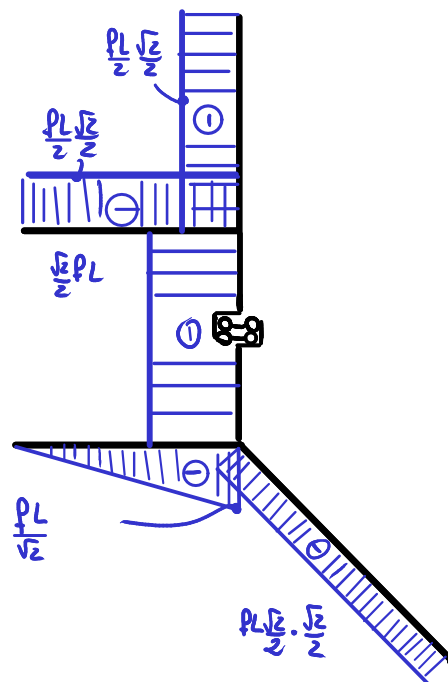
Per il tracciamento delle caratteristiche



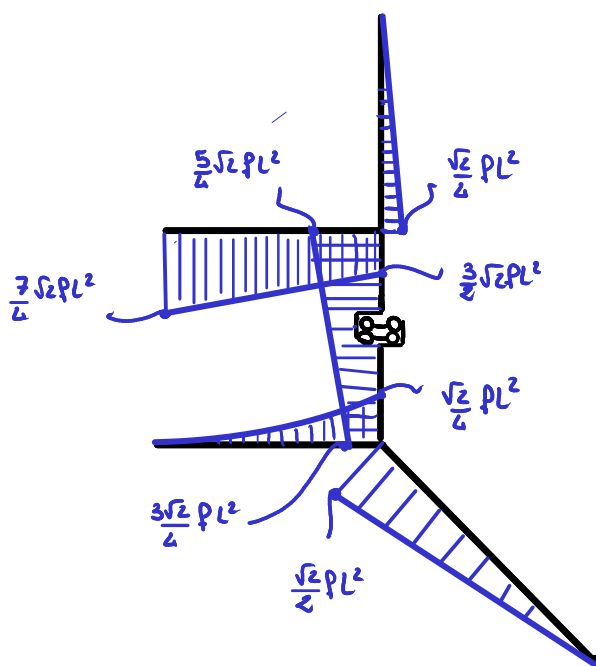
(N)



(T)



(M)



$$M_{AT_1} = \frac{PL}{2} \frac{\sqrt{2}}{2} z$$

$$M_{BT_1} = \frac{7\sqrt{2}}{4} PL^2 - \frac{\sqrt{2}}{4} PLz$$

$$M_{TC} = \frac{\sqrt{2}}{2} PL \left(\frac{L}{2} - z \right) + \sqrt{2} PL^2$$

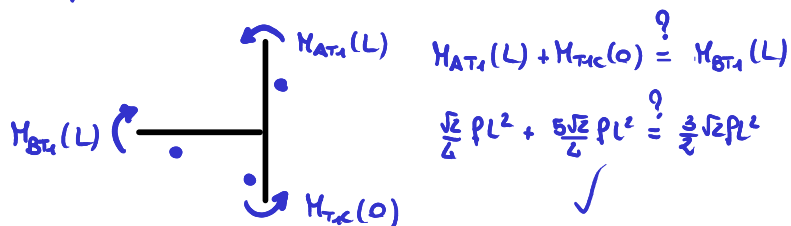
$$M_{CT_2} = \sqrt{2} PL^2 - \frac{\sqrt{2}}{2} PLz$$

$$M_{DT_2} = -\frac{\sqrt{2}}{2} PL \frac{z^2}{2}$$

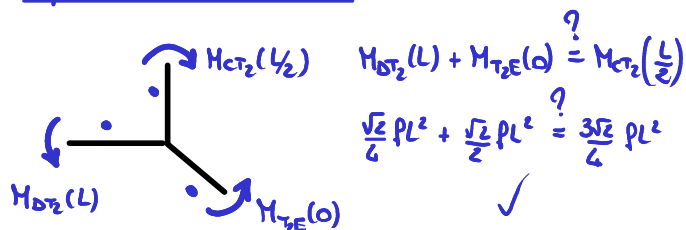
$$M_{TE} = -\frac{\sqrt{2}}{2} PL \cdot \frac{\sqrt{2}}{2} (\sqrt{2}L - z)$$

VERIFICA EQUILIBRIO A MOMENTO DEI NODI TRIPLI

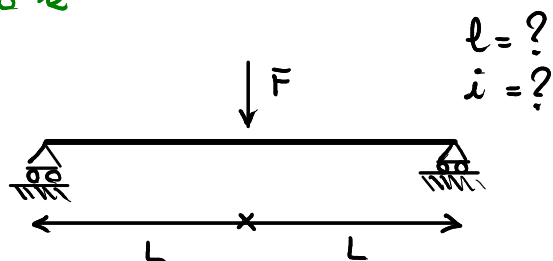
Equilibrio nodo triplo T₁



Equilibrio nodo triplo T₂



Esercizio 12



$$\begin{aligned} t &= 1 \\ s &= 2 \\ l &= 1 \quad (C = C_{\infty}) \end{aligned}$$

$$\left. \begin{aligned} & \\ & \end{aligned} \right\} i = 0$$

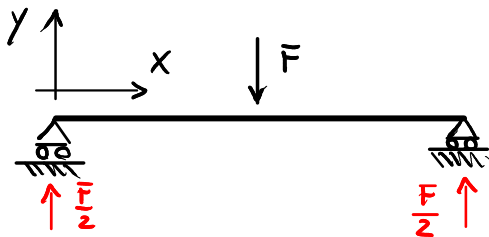
Questa struttura è una volta labile: il centro dello spostamento è un punto improprio di direzione verticale, ovvero la struttura può muoversi di moto rigido lungo la direzione orizzontale.

Tuttavia essa è staticamente determinata sotto l'azione della forza verticale F : le equazioni di equilibrio infatti sono sufficienti a determinare le reazioni vincolari.

$$\parallel X \quad 0 = 0$$

$$\parallel Y \quad R_{Ay} - F + R_{By} = 0$$

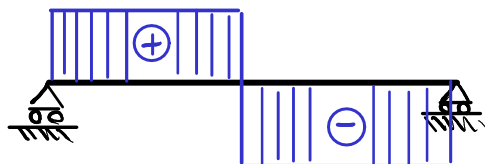
$$\parallel Z \quad M_z(A) = -FL + 2R_{By}L = 0$$



Questo è, cioè, un esempio di struttura labile ma in equilibrio per le particolari condizioni di carico. Se il carico esterno avesse una componente "lungo la direzione della labilità", ossia la direzione orizzontale, non ci sarebbe modo di avere equilibrio perché non ci sono vincoli in grado di reagire in quella direzione.

Si noti che, per quanto appena detto, la struttura risulterebbe staticamente determinata anche se il carico esterno fosse un momento concentrato posto in una sezione qualsiasi.

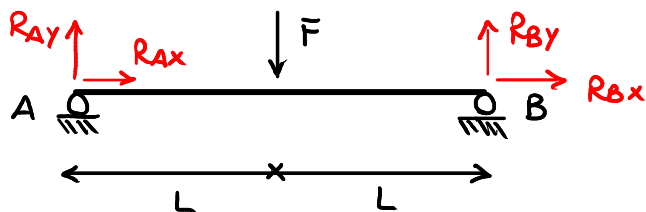
I diagrammi di T e M sono



Si osservi che le medesime considerazioni possono farsi anche per la struttura dell'esercizio 10

Le strutture labili, in generale, ammettono soluzione al problema di equilibrio solo se la direzione del carico esterno non ha componenti parallela alla direzione della labilità.

Esercizio 13



$$\ell = 0 \quad i = 1$$

Questa struttura è solo apparentemente simile a quella dell'esercizio precedente: la soluzione $R_{Ay} = R_{By} = F/2$ è solo parte di un insieme di $\infty^1 = \infty^1$ soluzioni dove R_{Ax} e R_{Bx} sono indeterminate:

$$\parallel X \quad R_{Ax} + R_{Bx} = 0$$

$$R_{Ax} = -R_{Bx}$$

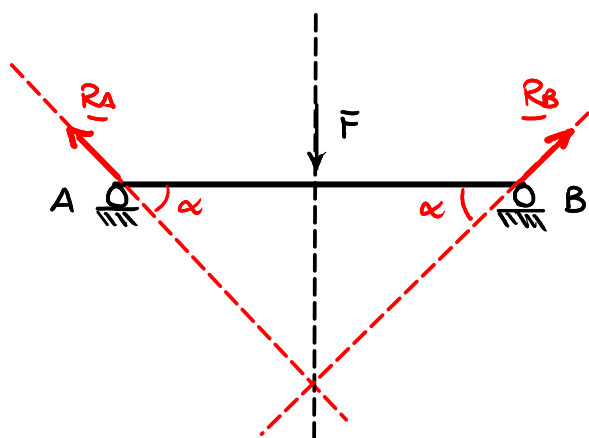
$\Rightarrow$ R_{Ax} qualsiasi, $R_{Bx} = -R_{Ax}$
(o equivalentemente R_{Bx} qualsiasi, $R_{Ax} = -R_{Bx}$)

$$\parallel Y \quad R_{Ay} + R_{By} - F = 0$$

$$R_{Ay} = R_{By} = \frac{F}{2}$$

$$\parallel Z \quad M_z(A) = -FL + R_{By}L = 0$$

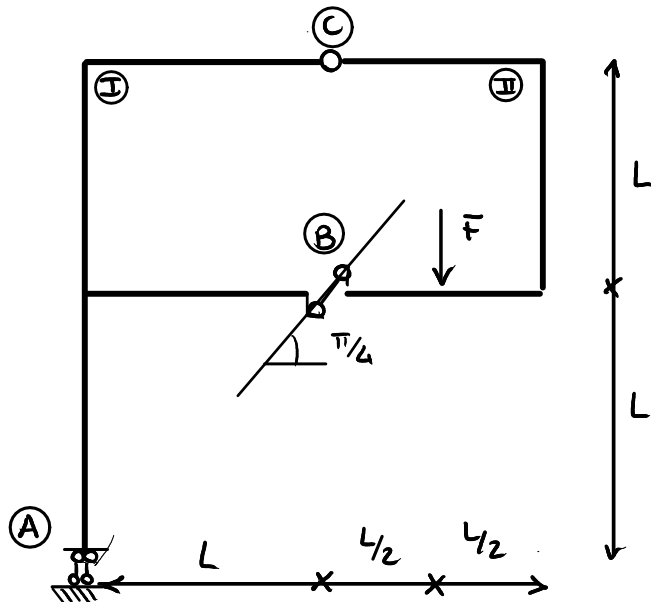
E cioè le reazioni $\underline{R}_A$, $\underline{R}_B$ hanno direzione incognita perché le rispettive componenti x restano incognite



Qualunque direzione a , b inclinata di α rispetto alla linea d'asse è valida

Le strutture iperstatiche, in generale, non ammettono soluzione unica al problema di equilibrio. La soluzione è unica se si considera il problema di equilibrio elastico

Esercizio 14

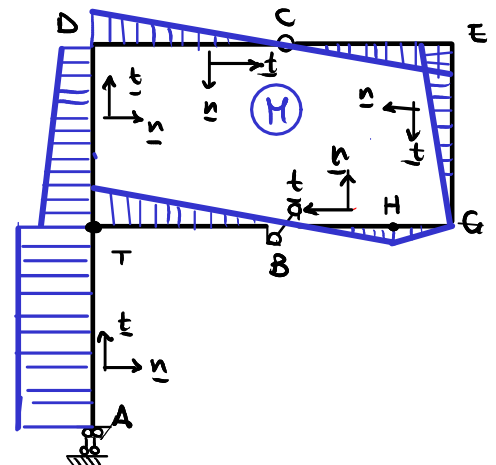
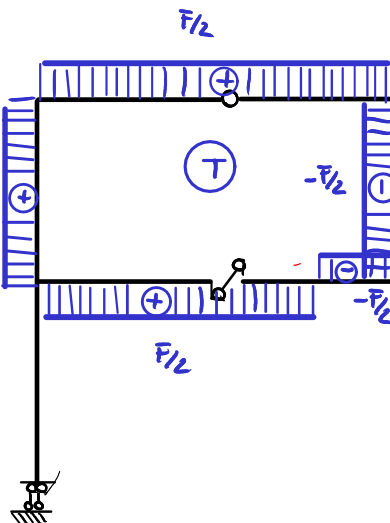
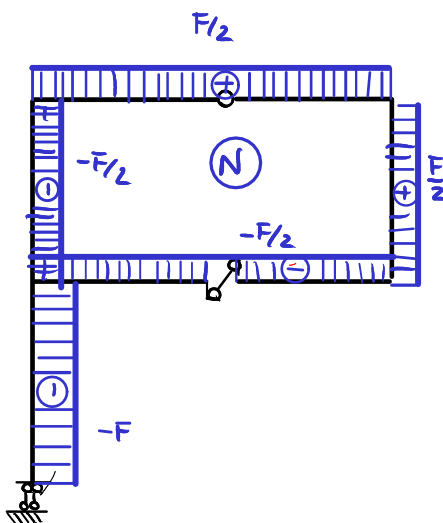
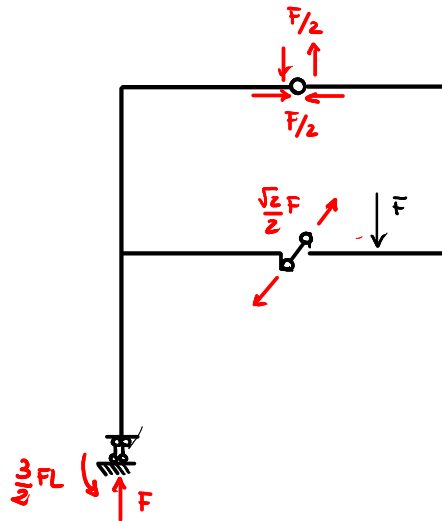


La struttura è una volta labile per vincoli esterni, dato che l'unico vincolo che la connette al telaio è un giunto.

Si osservi che la struttura non ha labilità interne, mancando l'allineamento tra le due condizioni dei vincoli interni (la direzione del pendolo B non passa per la cerniera C): non esistendo il centro dello spostamento relativo cioè lo spostamento relativo tra corpo I e corpo II è nullo, e i due corpi si comportano come se fossero uno solo.

Per tanto la struttura è assimilabile a un unico corpo vincolato esternamente con un giunto ad asse verticale ($C = \cos \uparrow$).

È facile riconoscere che questa è una struttura labile ma in equilibrio per la particolare condizione di carico.



$$M_{AT} = -\frac{3}{2}FL$$

$$M_{EG} = \frac{FL}{2} - \frac{Fz}{2}$$

$$M_{TB} = -\frac{F}{2}(L-z) - \frac{FL}{2}$$

$$M_{GH} = -\frac{F}{2}z$$

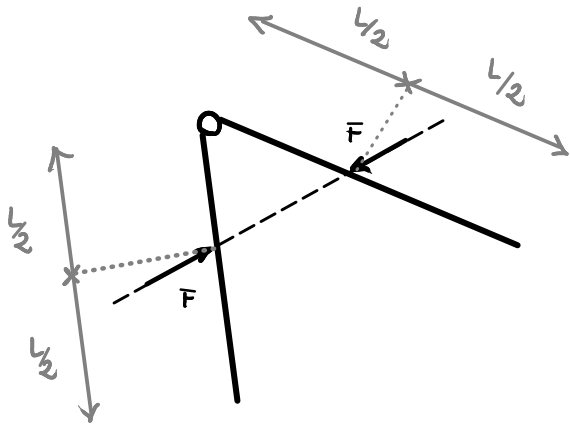
$$M_{BC} = -\frac{F}{2}(L-z)$$

$$M_{HB} = -\frac{F}{2}\left(\frac{L}{2} - z\right)$$

$$M_{CE} = +\frac{Fz}{2}$$

$$M_{BT} = \frac{Fz}{2}$$

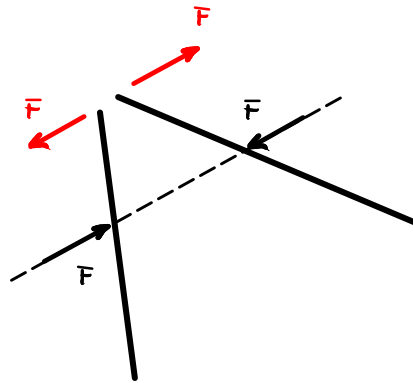
Esercizio 15



La struttura è una volta labile internamente (la cerniera interna lascia libere le rotazioni relative) e tre volte esternamente, dato che non vi sono vincoli esterni.

Le due forze esterne F costituiscono un sistema di forze autoequilibrato (sono uguali, opposte e colineari). Questo comporta che da un punto di vista globale l'equilibrio a forza e momento è soddisfatto.

Tuttavia la struttura non è in equilibrio perché le singole aste non lo sono: la cerniera interna può infatti esplicare delle reazioni che su ciascuna asta sono uguali e contrarie alla forza F che agisce su di essa, ma l'equilibrio a momento non è soddisfatto.



In altri termini il sistema di carico esterno attiva la labilità della struttura. E, si osservi, su una struttura labile con carichi nella direzione della labilità, non esistendo soluzione al problema di equilibrio, non ha senso definire le caratteristiche della sollecitazione.

Soluzione di strutture iperstatiche: metodo delle forze (con TLV)

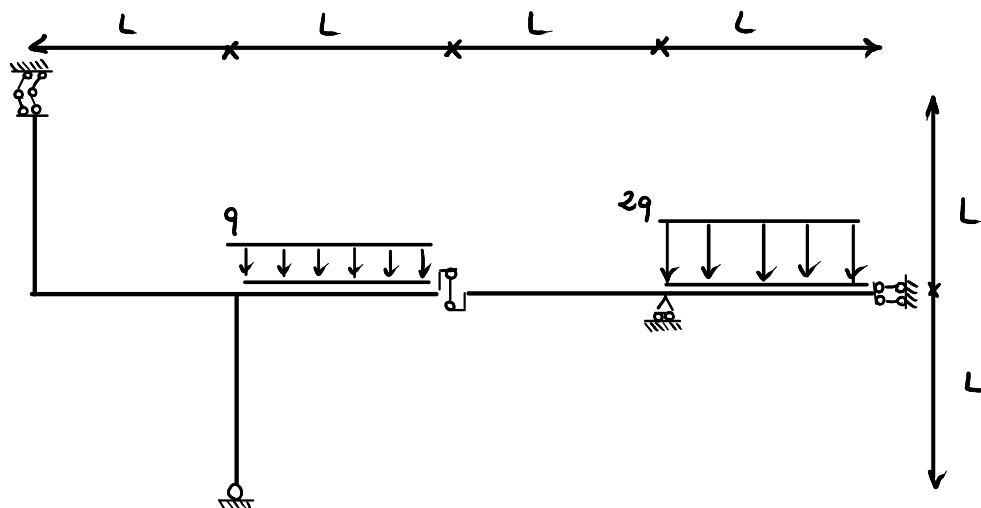
Per le seguenti strutture determinare:

- 1) il grado di labilità ℓ per via algebrica, attraverso le proprietà della matrice cinematica $[A]$, e attraverso la nozione di centri di rotazione, sfruttando i teoremi di allineamento
- 2) il grado di iperstaticità i per via algebrica, attraverso le proprietà della matrice di equilibrio $[B]$, e attraverso la formula di Grassmann $3t - s = \ell - i$
- 3) Risolvere la struttura (reazioni vincolari, caratteristiche della sollecitazione) usando il metodo delle forze.

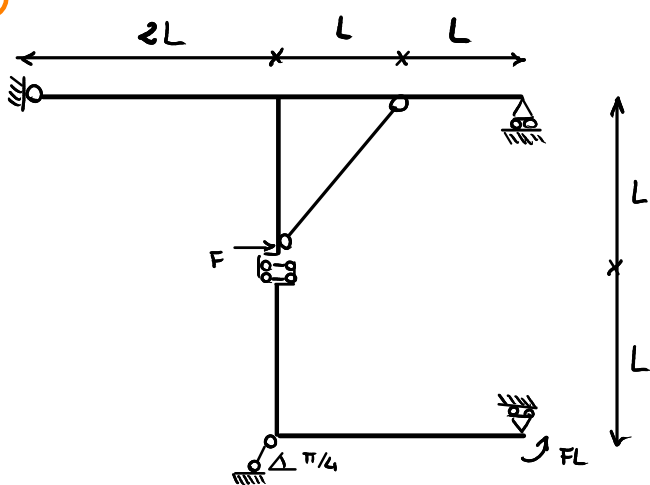
METODOLOGIA DI SOLUZIONE CON IL METODO DELLE FORZE

- 3.1) Noto il grado di iperstaticità i , individuare i vincoli semplici sovrabbondanti e rimuoverli. Si ottiene così una struttura dove $3t = s$.
N.B. La scelta di quali vincoli considerare sovrabbondanti è arbitraria. Più scelte sono possibili, ma non tutte sono corrette (vedi punto seguente)
- 3.2) Verificare che la struttura ottenuta non sia labile. Se lo è, la scelta fatta sui vincoli sovrabbondanti è sbagliata. Ripetere il punto 3.1. Se non lo è, la scelta fatta sui vincoli sovrabbondanti è una delle possibili scelte corrette. Procedere al punto seguente
- 3.3) Disegnare la struttura isostatica ottenuta, sostituendo al posto dei vincoli rimossi le loro reazioni vincolari incognite (fissare dei versi arbitrari) e scrivere a lato le condizioni cinematiche di vincolo soppressi (condizioni di congruenza).
Si ottiene così il sistema equivalente $S^{(eq)}$
- 3.4) Per sovrapposizione degli effetti $S^{(eq)} = S^{(0)} + x_1 S^{(1)} + x_2 S^{(2)} + \dots + x_i S^{(i)}$, dove
 $x_k, k=1, \dots, i$ sono le reazioni vincolari iperstatiche (di tipo forza e/o di tipo momento)
 $S^{(0)}$ è lo schema strutturale dove sulla struttura isostatica disegnata in $S^{(eq)}$ vanno applicati tutti e soli i carichi esterni assegnati
 $S^{(k)}$ è il k -esimo schema strutturale dove sulla struttura isostatica disegnata in $S^{(eq)}$ va applicata la k -esima incognita iperstatica x_k assumendone modulo unitario
Risolvere separatamente $S^{(0)}, \dots, S^{(i)}$ (reazioni + caratteristiche della sollecitazione)
- 3.5) Calcolo delle $x_1, \dots, x_i$ (soluzione del problema iperstatico) applicando il Teorema dei Lavori Virtuali i volte separatamente. Per ciascuna applicazione considerare come sistema delle forze il sistema k -esimo $S^{(k)}$ e come sistema degli spostamenti il sistema equivalente $S^{(eq)}$
Se $i=1$ (struttura una volta iperstatica) si ottiene una equazione in una incognita (x_1)
Se $i > 1$ (struttura i volte iperstatica) si ottiene un sistema di i equazioni in i incognite ($x_1, \dots, x_i$)

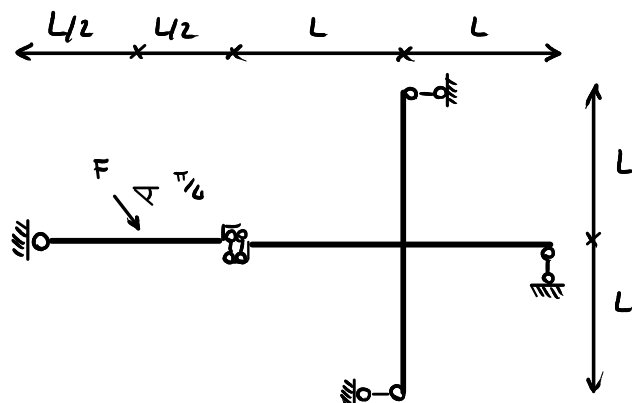
1



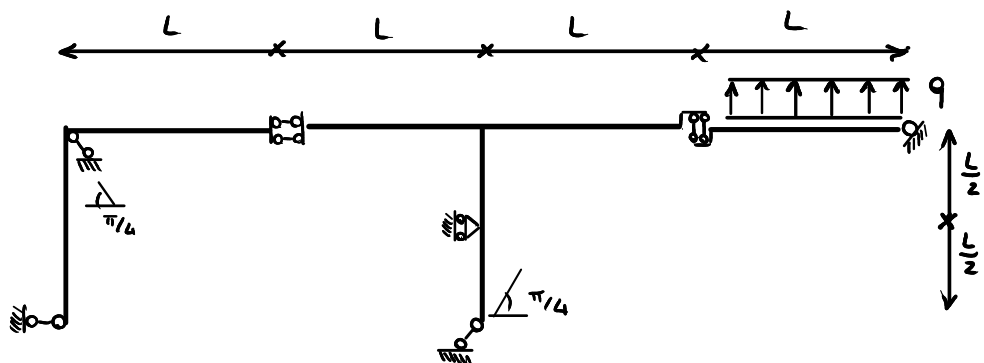
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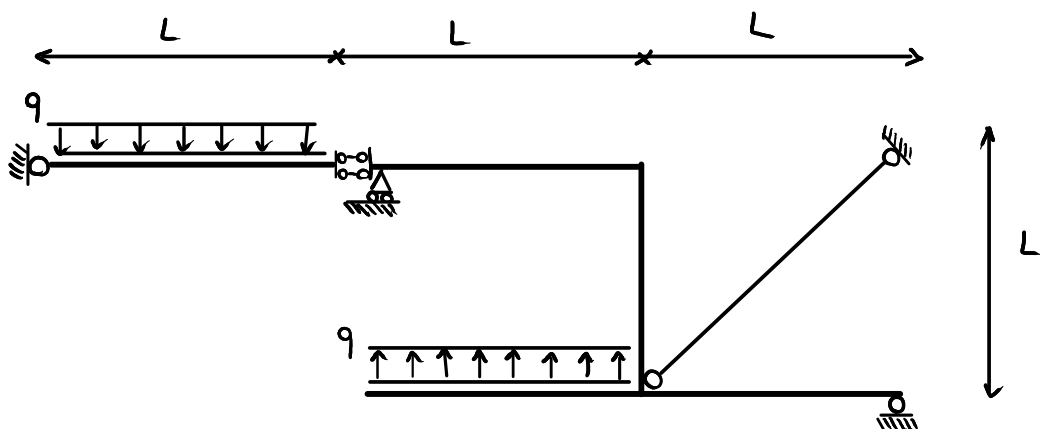
3



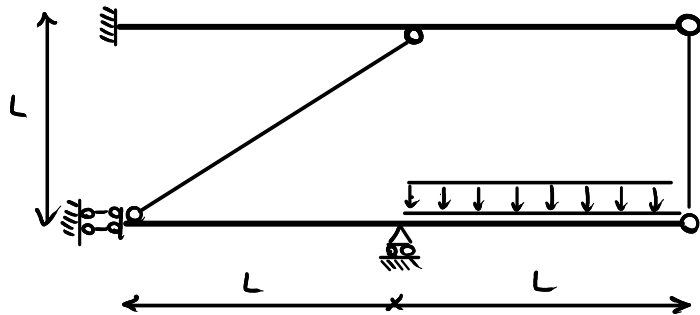
4



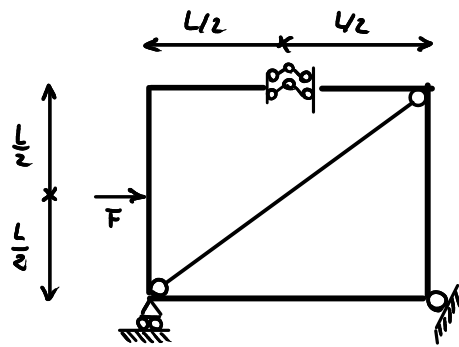
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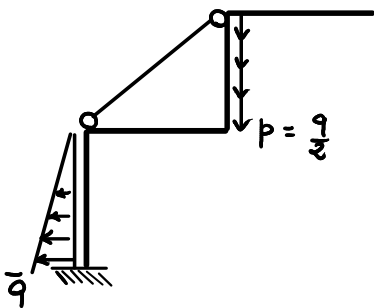
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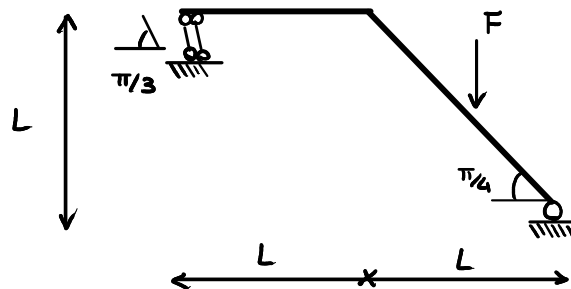
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8

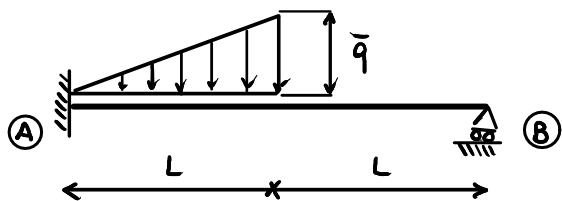


9



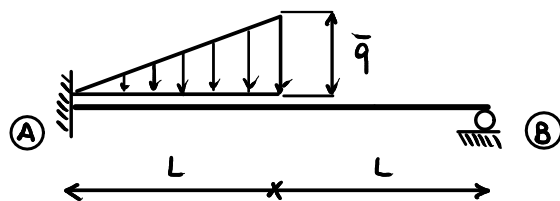
10

Trascurare la deformabilità assiale e quella tagliante



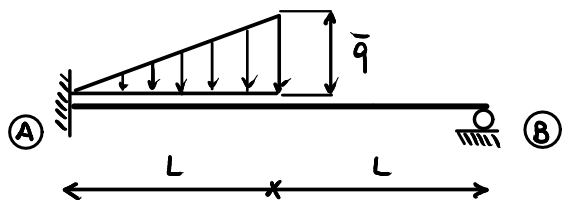
11

Trascurare la deformabilità assiale e quella tagliante



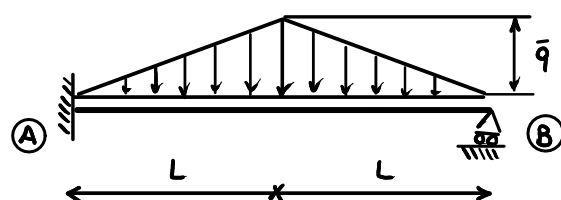
12

Trascurare la deformabilità tagliante ma non quella assiale

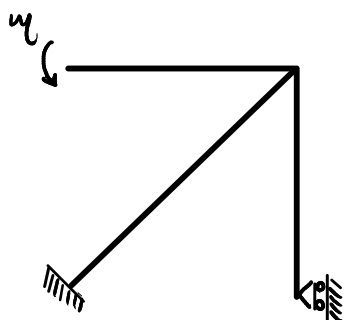


13

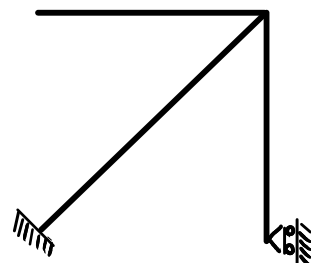
Trascurare la deformabilità assiale e quella tagliante



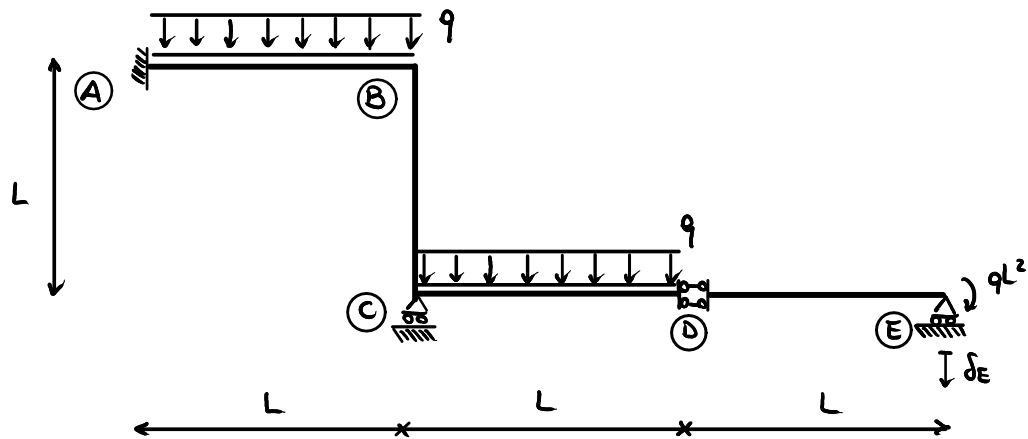
14



15



ESEMPIO SVOLTO 1



1) CALCOLO DEL GRADO DI LIBERTÀ

1.1) Metodo algebrico

INDIVIDUAZIONE DEI CORPI : $\textcircled{I} = ABCD$, $\textcircled{II} = DE$

EQUAZIONI DI VINCOLO

$\textcircled{A}$	$\textcircled{C}$	$\textcircled{D}$	$\textcircled{E}$
$\underline{u}_A = \underline{0}$	$\underline{u}_C \cdot \underline{e}_C = 0$	$\Delta \underline{u}_D \cdot \underline{e}_D = (\underline{u}_D^{\textcircled{II}} - \underline{u}_D^{\textcircled{I}}) \cdot \underline{e}_D = 0$	$\underline{u}_E \cdot \underline{e}_E = -\delta_E$
$\underline{\vartheta}_I = \underline{0}$		$\Delta \underline{\vartheta} = \underline{\vartheta}_{\textcircled{II}} - \underline{\vartheta}_{\textcircled{I}} = 0$	

POLI CINEMATICI : A per il corpo $\textcircled{I}$, D per $\textcircled{II}$

SISTEMI DI COORDINATE : $\{x, y, z\}$

FORMULA GENERALE DEL CAMPO DI SPOSTAMENTO RIGIDO INFINITESIMO : $\underline{u}_P = \underline{u}_O + \underline{\vartheta} \times (P - O)$

punti del corpo $\textcircled{I}$ $\underline{u}_P^{\textcircled{I}} = \underline{u}_A + \underline{\vartheta}_I \times (P - A)$, $P \in \textcircled{I}$

punti del corpo $\textcircled{II}$ $\underline{u}_P^{\textcircled{II}} = \underline{u}_D + \underline{\vartheta}_{\textcircled{II}} \times (P - D)$, $P \in \textcircled{II}$

Osservare che rispetto al riferimento adottato

$$A = (0, 0), \quad C = (L, -L), \quad D = (2L, -L), \quad E = (3L, -L), \quad P = (X, Y)$$

$$\underline{e}_C = \underline{e}_x, \quad \underline{e}_D = \underline{e}_x, \quad \underline{e}_E = \underline{e}_x$$

e quindi i vettori posizione di un punto generico rispetto ai poli cinematici sono

$$(P - A) = (X, Y) - (0, 0) = (X - 0, Y - 0) = (X, Y) = X \underline{e}_x + Y \underline{e}_y$$

$$(P - D) = (X, Y) - (2L, -L) = (X - 2L, Y - (-L)) = (X - 2L, Y + L) = (X - 2L) \underline{e}_x + (Y + L) \underline{e}_y$$

In particolare:

$$\underline{u}_P^I = u_{Ax} \underline{e}_x + u_{Ay} \underline{e}_y + \vartheta_I \underline{e}_z \times \underbrace{(X \underline{e}_x + Y \underline{e}_y)}_{P-A} = u_{Ax} \underline{e}_x + u_{Ay} \underline{e}_y + \vartheta_I X \underline{e}_y - \vartheta_I Y \underline{e}_x = (u_{Ax} - \vartheta_I Y) \underline{e}_x + (u_{Ay} + \vartheta_I X) \underline{e}_y$$

$$\begin{aligned} \underline{u}_P^{II} &= u_{Dx} \underline{e}_x + u_{Dy} \underline{e}_y + \vartheta_{II} \underline{e}_z \times \underbrace{[(X-2L) \underline{e}_x + (Y+L) \underline{e}_y]}_{P-D} = \\ &= u_{Dx} \underline{e}_x + u_{Dy} \underline{e}_y + \vartheta_{II} (X-2L) \underline{e}_y - \vartheta_{II} (Y+L) \underline{e}_x = [u_{Dx} - \vartheta_{II} (Y+L)] \underline{e}_x + [u_{Dy} + \vartheta_{II} (X-2L)] \underline{e}_y \end{aligned}$$

A questo punto è possibile scrivere le

EQUAZIONI SCALARI DI VINCOLO RISPETTO AI SISTEMI DI COORDINATE SCELTI

$$u_{Ax} = 0$$

$$u_{Ay} = 0$$

$$\vartheta_I = 0$$

$$\underline{u}_C \cdot \underline{e}_C^{\prime\prime} = [(u_{Ax} + \vartheta_{II} L) \underline{e}_x + (u_{Ay} + \vartheta_{II} L) \underline{e}_y] \cdot \underline{e}_y = u_{Ay} + \vartheta_{II} L = 0$$

$$(\underline{u}_D^I - \underline{u}_D^I) \cdot \underline{e}_D^{\prime\prime} = [(u_{Ax} + \vartheta_{II} L) \underline{e}_x + (u_{Ay} + \vartheta_{II} L) \underline{e}_y - u_{Dx} \underline{e}_x - u_{Dy} \underline{e}_y] \cdot \underline{e}_x = u_{Ax} + \vartheta_{II} L - u_{Dx} = 0$$

$$\vartheta_{II} - \vartheta_I = 0$$

$$\underline{u}_E \cdot \underline{e}_E^{\prime\prime} = [u_{Dx} \underline{e}_x + (u_{Dy} + \vartheta_{II} L) \underline{e}_y] \cdot \underline{e}_y = u_{Dy} + \vartheta_{II} L = -\delta_E$$

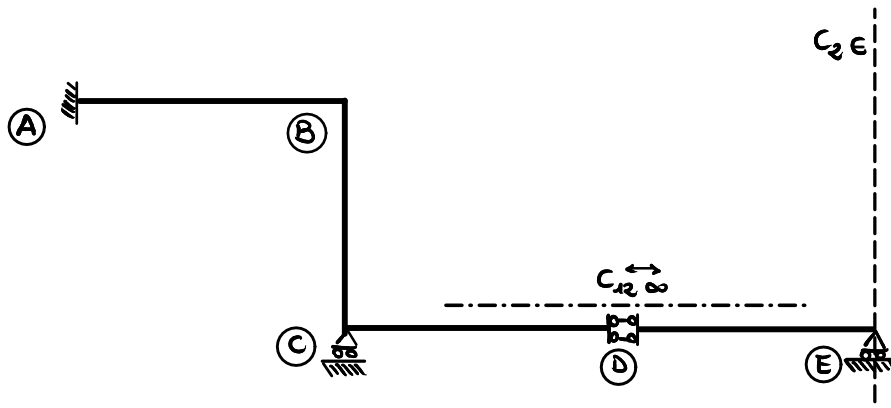
$$\begin{cases} u_{Ax} = 0 \\ u_{Ay} = 0 \\ \vartheta_I = 0 \\ u_{Ay} + \vartheta_{II} L = 0 \\ u_{Ax} + \vartheta_{II} L - u_{Dx} = 0 \\ \vartheta_{II} - \vartheta_I = 0 \\ u_{Dy} + \vartheta_{II} L = -\delta_E \end{cases} \rightsquigarrow \begin{vmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & L & 0 & 0 & 0 \\ 1 & 0 & L & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & 1 & L \end{vmatrix} \begin{vmatrix} u_{Ax} \\ u_{Ay} \\ \vartheta_I \\ u_{Dx} \\ u_{Dy} \\ \vartheta_{II} \end{vmatrix} = \begin{vmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ -\delta_E \end{vmatrix}$$

$$\ell =: \dim(\ker[A]) = n_A - \text{rg}(A) = 6 - 6 = 0$$

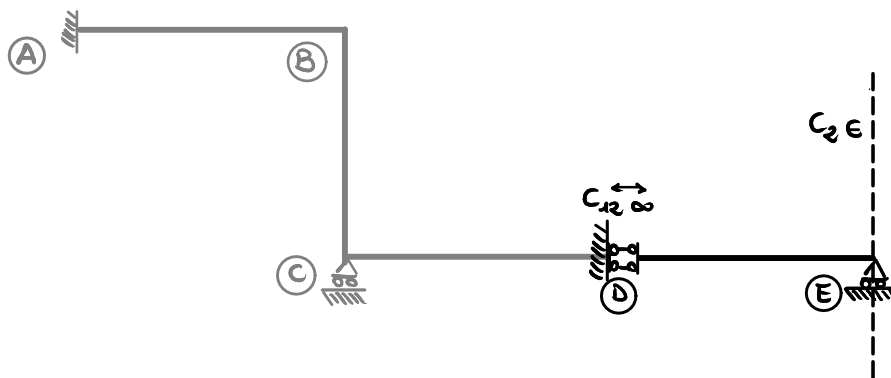
↓
rango di [A]

(n_A = numero di colonne di [A])

1.2) Centri di rotazione



L'incastro A prescrive che il centro C_1 del corpo I non esista ($\Rightarrow$ il corpo I non subisce moti rigidi). Quindi il vincolo D può essere pensato come esterno



Si deduce immediatamente che il corpo II non ammette l'esistenza di un centro della rotazione, non potendo questo essere contemporaneamente un punto proprio appartenente all'asse del carrello in E ed un punto improprio avente direzione orizzontale. Il corpo II è fermo

Perché non esistono i centri assoluti C_1 e C_2 non esisterà neanche il centro relativo C_{12} .

Il corpo non è labile, $\ell = 0$.

In generale se un corpo di una struttura può essere ricondotto a uno dei seguenti schemi allora quel corpo è singolarmente non labile e isostatico:

SCHEMI NOTEVOLI ISOSTATICI

CARRELLO - CERNIERA	INCASTRO	CARRELLO - GLIFO	PANTOGRAFO - CERNIERA

2) CALCOLO DEL GRADO DI IPERSTATICITA'

2.1) Metodo algebrico

EQUAZIONI DI EQUILIBRIO

$$R_{Ax} + R_{Dx} = 0$$

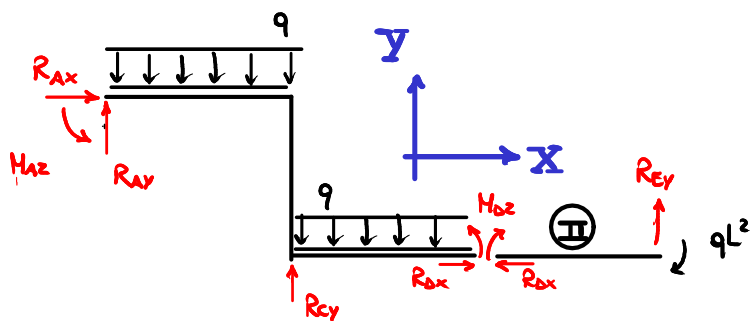
$$R_{Ay} + R_{Cy} + 2ql = 0$$

$$M_z(A) = M_{Az} - \frac{qL^2}{2} + R_{Cy}L - \frac{3}{2}qL^2 + R_{Dx}L + M_{Dz} = 0$$

$$-R_{Dx} = 0$$

$$R_{Ey} = 0$$

$$M_z(E) = -M_{Dz} - qL^2 = 0$$



$$\begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & L & L & -1 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} R_{Ax} \\ R_{Ay} \\ M_{Az} \\ R_{Cy} \\ R_{Dx} \\ M_{Dz} \\ R_{Ey} \end{bmatrix} = \begin{bmatrix} 0 \\ -2ql \\ 2ql^2 \\ 0 \\ 0 \\ ql^2 \\ 0 \end{bmatrix}$$

$$rg[B] = 6$$

$$i = \dim(\ker[B]) = n_B - rg(B) = 7 - 6 = 1$$

n_B = numero di colonne di $[B]$

↓
range di $[B]$

2.2) Formula di Grassmann

$$t = 2$$

$$s = 3 + 1 + 2 + 1 = 7$$

↓
incastro
A

↓
carrello
C

↓
ghjo
D

↓
carrello
E

$$\left. \begin{array}{l} t = 2 \\ s = 7 \end{array} \right\} 3t - s = -1 = \ell - i \quad \text{ma } \ell = 0 \quad \Rightarrow i = 1$$

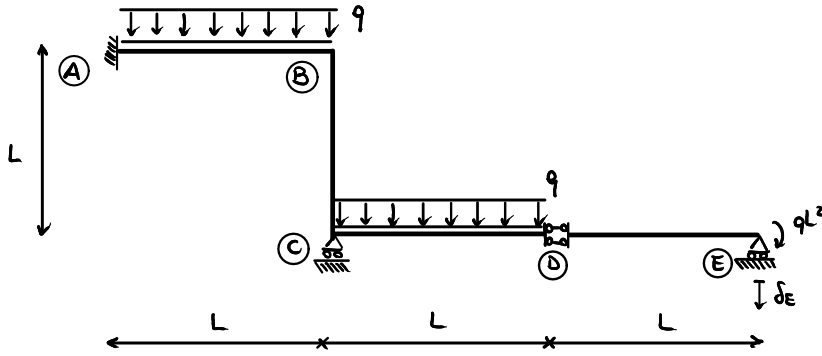
Si lascia come semplice calcolo la verifica che, prendendo come poli di calcolo dei momenti gli stessi punti usati come poli cinematici (A e D), risulta

$$[A]^T = [B] \quad \text{e} \quad [B]^T = [A]$$

Se le equazioni del problema statico e quelle del problema cinematico vengono scritte adottando gli stessi punti (poli) per il calcolo dei momenti (nel problema statico) e dei campi di spostamento rigido (nel problema cinematico), allora le matrici $[A]$ e $[B]$ associate a tali problemi sono una la trasposta dell'altra.

3) SOLUZIONE DELL'IPERSTATICITA' TRAMITE IL METODO DELLE FORZE

$$i = 1, \ell = 0$$



Occorre rimuovere una condizione semplice di vincolo senza alterare la labilità della struttura.

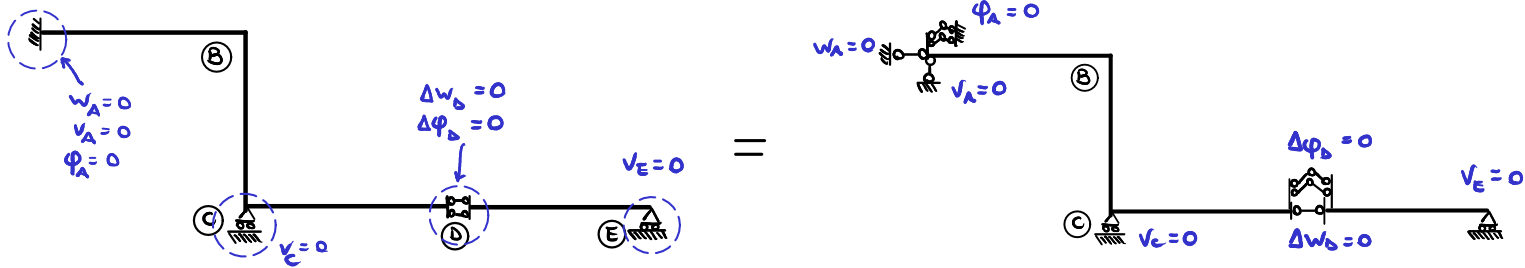
VINCOLO SEMPLICE: vincolo che ha molteplicità pari a uno. Sono due:

PENSOLO (o carrello)
INTERNO/ESTERNO

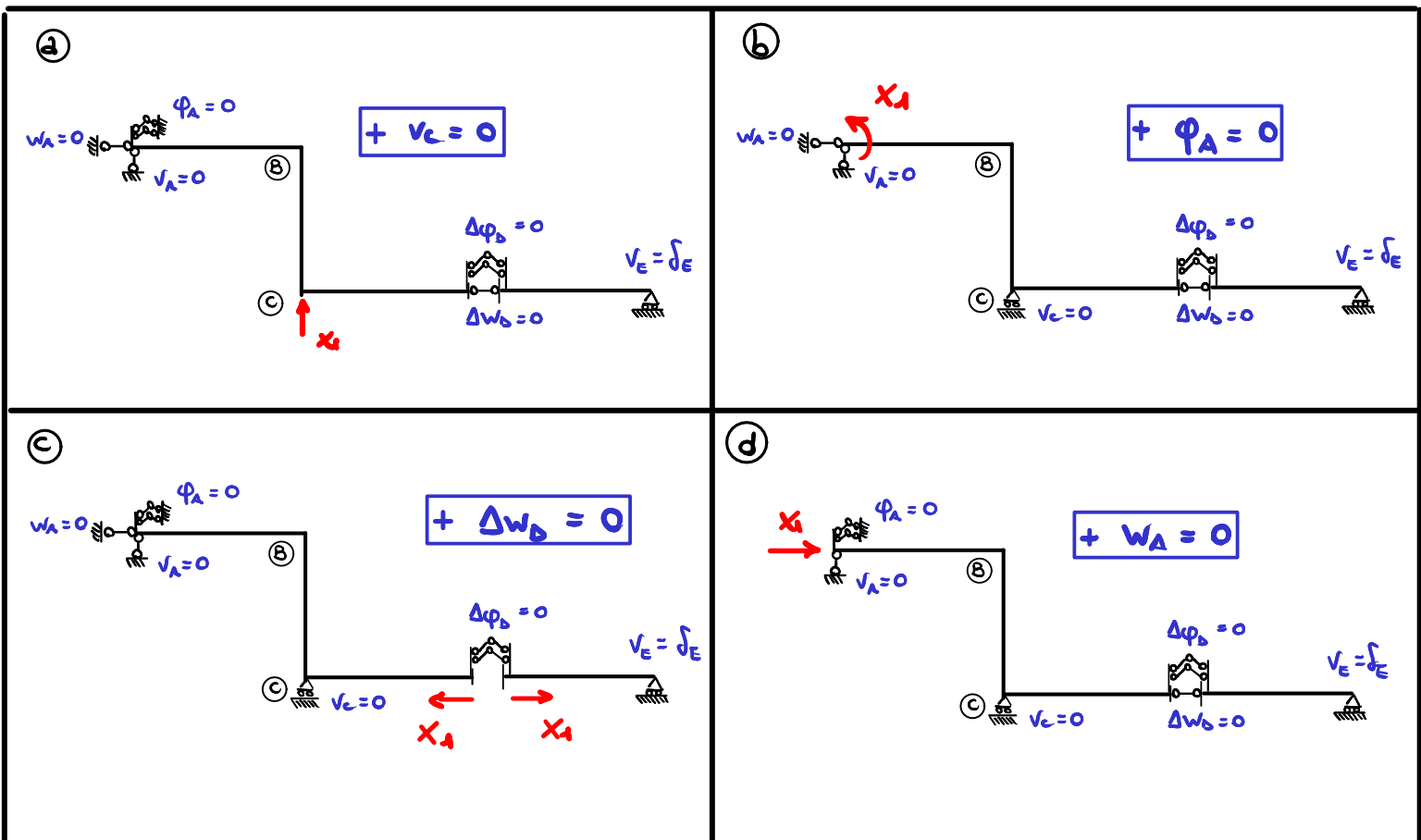
PANTOGRAFO
INTERNO/ESTERNO



Ogni altro vincolo può essere visto come sovrapposizione di vincoli semplici. Per la struttura in esame:

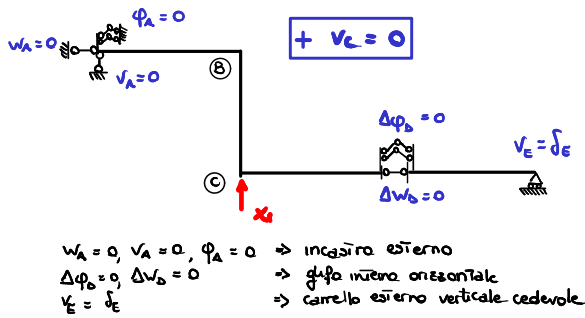


Si deve rimuovere uno di questi vincoli semplici. Non tutte le scelte sono possibili, perchè alcune rimozioni rendono la struttura labile:



Delle strutture precedenti la ③ e la ⑥ hanno $\ell = 0$, mentre ④ e ⑤ sono una volta labili ($\ell = 1$). Vediamo brevemente perché:

②

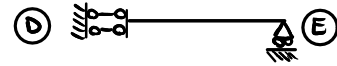


VERIFICA LABILITÀ

AD riconducibile a

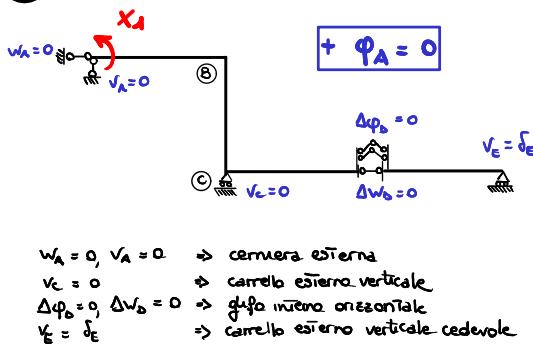


di conseguenza



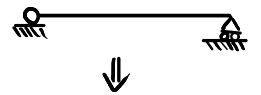
Con questa scelta di rimozione di vincolo i corpi ① e ② sono individualmente non labili perché riconducibili a schemi notevoli isostatici. Pertanto tutta la struttura non altera il grado di labilità della struttura assegnata: questa scelta di rimozione vincolare è una possibile scelta per l'applicazione del metodo delle forze.

⑥

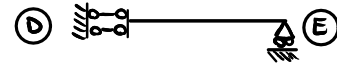


VERIFICA LABILITÀ

AD riconducibile a

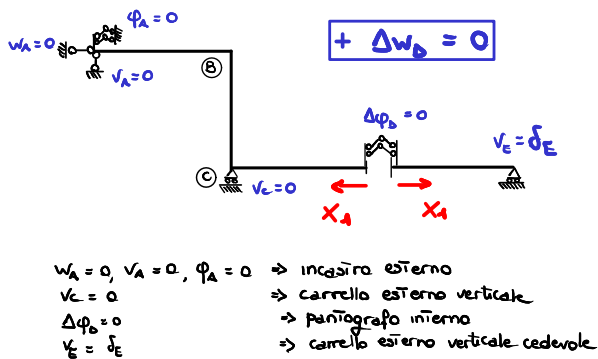


di conseguenza



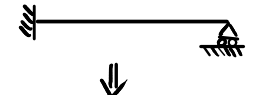
Con questa scelta di rimozione di vincolo i corpi ① e ② sono individualmente non labili perché riconducibili a schemi notevoli isostatici. Pertanto tutta la struttura non altera il grado di labilità della struttura assegnata: questa scelta di rimozione vincolare è una possibile scelta per l'applicazione del metodo delle forze.

④

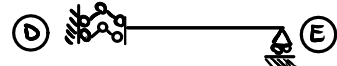


VERIFICA LABILITÀ

AD riconducibile a



di conseguenza



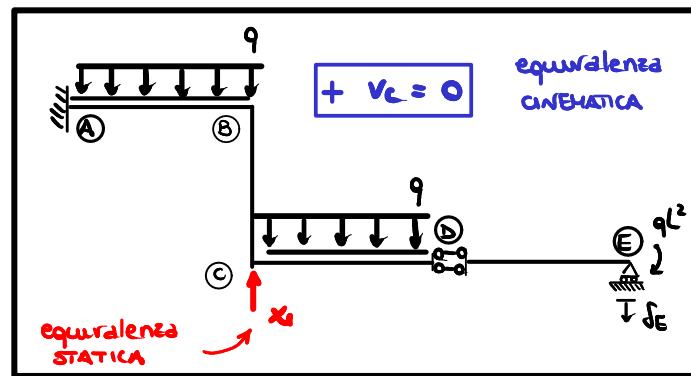
Con questa scelta di rimozione di vincolo non solo il corpo I rimane iperstatico (incastro + carrello), ma il corpo II diventa labile, non esistendo nessun vincolo che ne impedisca la traslazione orizzontale. La struttura ottenuta è una volta labile ed una volta iperstatica.

Ogni scelta di rimozione di vincolo che non alteri la labilità della struttura conduce alla stessa soluzione (per il teorema di Kirchhoff) - Scegliamo ad esempio la ②.

Il sistema equivalente è dunque

$S^{(eq)}$

=



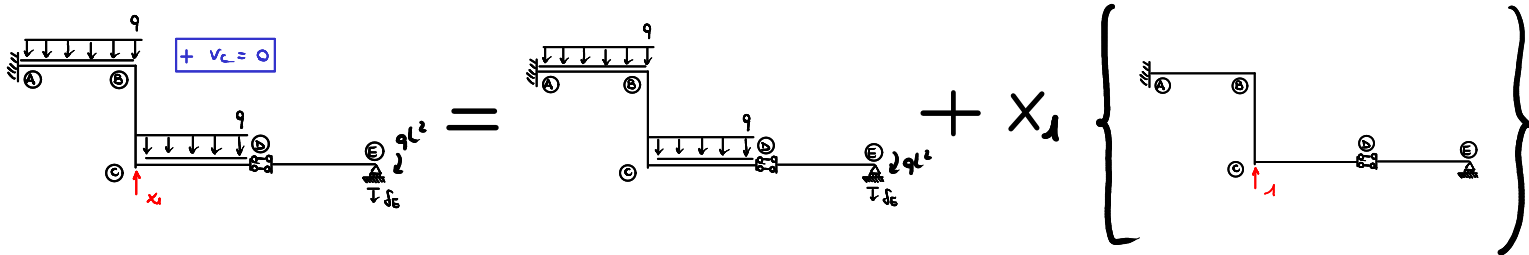
$S^{(eq)}$

=

$S^{(0)}$

+

$x_1 S^{(1)}$



$S^{(eq)}$: struttura di partenza resa isostatica + tutti i carichi esterni assegnati
 + tutti i cedimenti vincolari assegnati + reazioni dei vincoli soppressi (INCOGNITE)
 + condizioni cinematiche dei vincoli soppressi, scritte come equazioni a lato della
 struttura (CONDIZIONI DI CONGRUENZA con la struttura assegnata)

$S^{(0)}$: struttura di partenza resa isostatica + tutti i carichi esterni assegnati
 + tutti i cedimenti vincolari assegnati

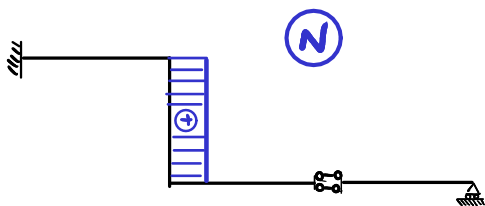
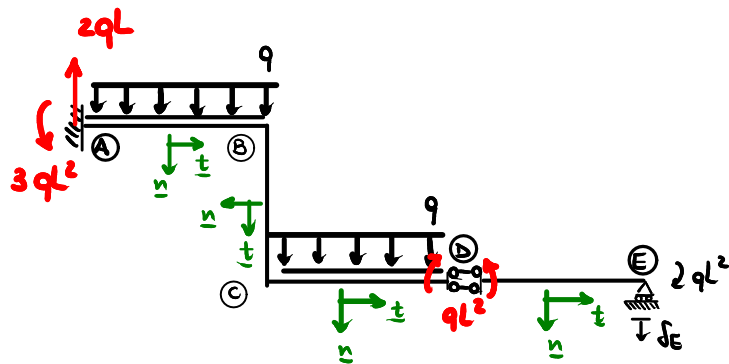
$S^{(1)}$: struttura di partenza resa isostatica + reazioni dei vincoli soppressi rese unitarie

Per risolvere l'iperstaticità della struttura occorre conoscere la reazione x_1 . A tal fine si sfrutta il teorema dei lavori virtuali, nel quale

SISTEMA DELLE FORZE $S^{(p)} = S^{(1)}$
 (sistema virtuale)

SISTEMA DEGLI SPOSTAMENTI $S^{(s)} = S^{(eq)}$
 (sistema effettivo)

$S^{(0)}$

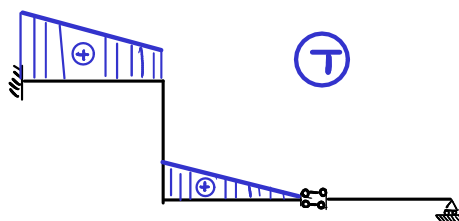


$$N_{AB}^{(0)} = 0$$

$$N_{CB}^{(0)} = 0$$

$$N_{BC}^{(0)} = qL$$

$$N_{DE}^{(0)} = 0$$

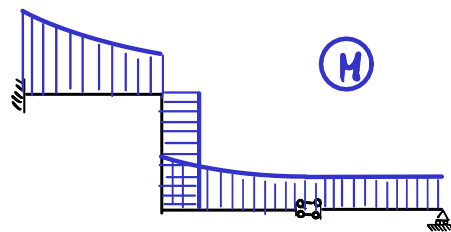


$$T_{AB}^{(0)} = 2qL - qz$$

$$T_{CB}^{(0)} = q(L-z)$$

$$T_{BC}^{(0)} = 0$$

$$T_{DE}^{(0)} = 0$$



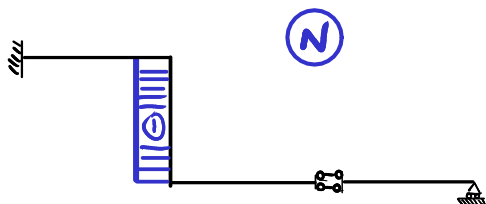
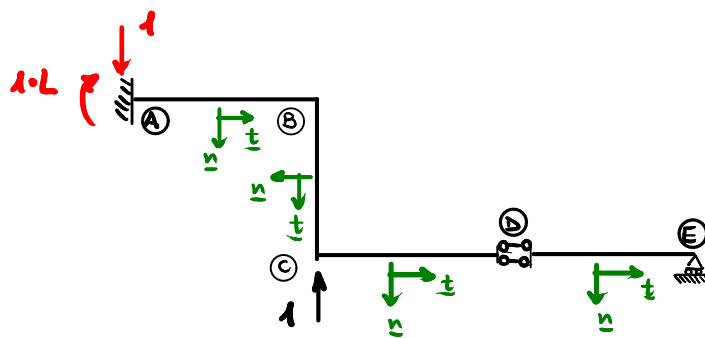
$$M_{AB}^{(0)} = -3qL^2 + 2qLz - \frac{qz^2}{2}$$

$$M_{BC}^{(0)} = -\frac{3}{2}qL^2$$

$$M_{CB}^{(0)} = -qL^2 - q\frac{(L-z)^2}{2}$$

$$M_{DE}^{(0)} = -qL^2$$

$S^{(1)}$

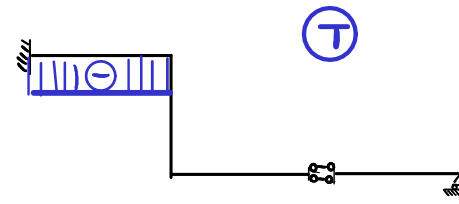


$$N_{AB}^{(1)} = 0$$

$$N_{CB}^{(1)} = 0$$

$$N_{BC}^{(1)} = -1$$

$$N_{DE}^{(1)} = 0$$

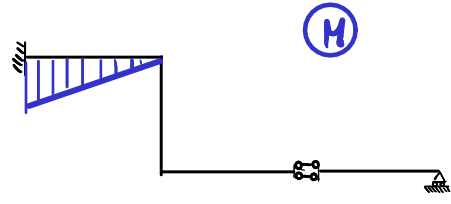


$$T_{AB}^{(1)} = -1$$

$$T_{CB}^{(1)} = 0$$

$$T_{BC}^{(1)} = 0$$

$$T_{DE}^{(1)} = 0$$



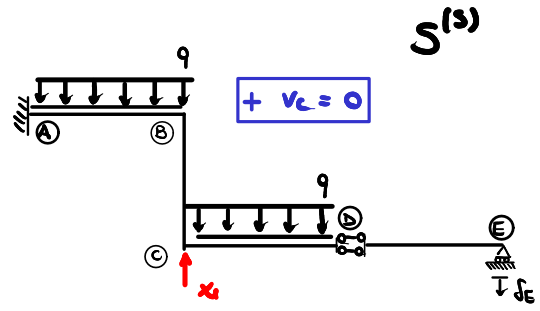
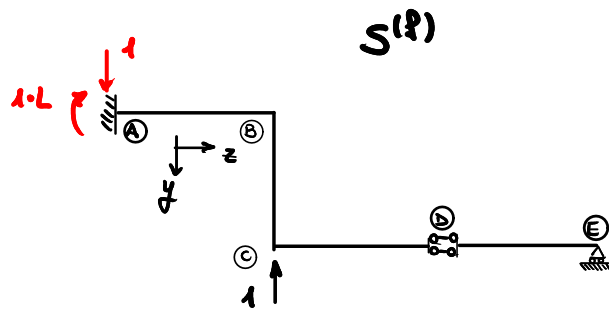
$$M_{AB}^{(1)} = L-z$$

$$M_{CB}^{(1)} = 0$$

$$M_{BC}^{(1)} = 0$$

$$M_{DE}^{(1)} = 0$$

Applicazione del TLV:



δ_{ve} = forze / momenti nel sistema delle forze $\cdot$ spostamenti / rotazioni nel sistema degli spostamenti

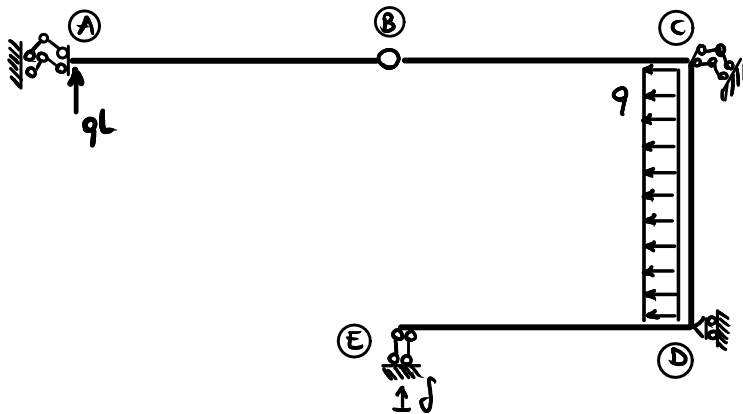
$$\delta_{ve} = M_{xA}^{(P)} \cdot \phi_A^{(S)} + R_{Ay}^{(P)} \cdot v_A^{(S)} + 1^{(P)} \cdot v_c^{(S)} = 0$$

$= 1 \cdot L$ $= 0$ (INCASTRATO) $= 1$ $= 0$ (INCASTRATO) $= 0$ (CONDIZIONE DI CONGRUENZA)

$$\begin{aligned} \delta_{vi} &= \int_{A \rightarrow E} \frac{M^{(P)} M^{(S)}}{EI} dz = \int_{A \rightarrow E} \frac{M^{(1)} M^{(eq)}}{EI} dz = \int_{A \rightarrow E} \frac{M^{(1)} [M^{(0)} + x_1 M^{(1)}]}{EI} dz = \\ &= \int_{AB} \frac{M_{AB}^{(1)} (M_{AB}^{(0)} + x_1 M_{AB}^{(1)})}{EI} dz + \int_{BC} \frac{M_{BC}^{(1)} (M_{BC}^{(0)} + x_1 M_{BC}^{(1)})}{EI} dz + \int_{CD} \frac{M_{CD}^{(1)} (M_{CD}^{(0)} + x_1 M_{CD}^{(1)})}{EI} dz + \\ &+ \int_{DE} \frac{M_{DE}^{(1)} (M_{DE}^{(0)} + x_1 M_{DE}^{(1)})}{EI} dz = \int_{AB} \frac{M_{AB}^{(1)} (M_{AB}^{(0)} + x_1 M_{AB}^{(1)})}{EI} dz = \\ &= \int_0^L \frac{(L-z) (-3qL^2 + 2qLz - \frac{qz^2}{3} + x_1 (L-z))}{EI} dz = \int_0^L \frac{-3qL^3 + 2qL^2z - \frac{qLz^2}{3} + 2qLz^2 - 2qLz^2 + \frac{qz^3}{3} + x_1 (L-z)^2}{EI} dz \\ &= \frac{1}{EI} \left[-3qL^3z + 2qL^2 \frac{z^2}{2} - \frac{qLz^3}{6} + qL^2z^2 - 2qL \frac{z^3}{3} + \frac{qz^4}{8} - x_1 \frac{(L-z)^3}{3} \right]_0^L = -\frac{29}{24} qL^4 + x_1 \frac{L^3}{3} \end{aligned}$$

$$\delta_{ve} = \delta_{vi} \quad \Leftrightarrow \quad x_1 = \frac{29}{8} qL$$

ESEMPIO SVOLTO 2



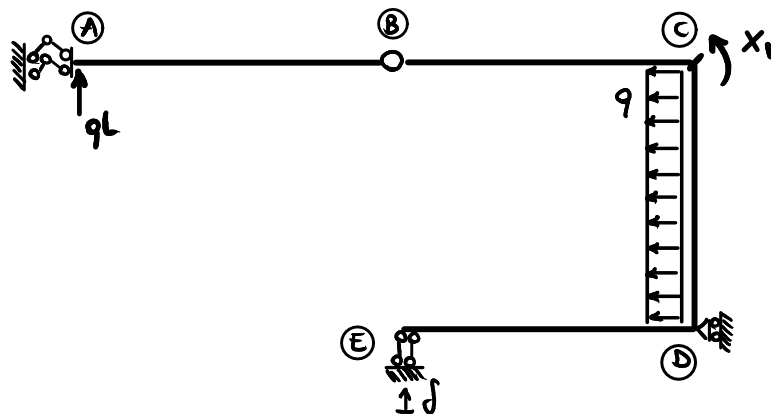
$$t=2$$

$$s=7 \quad 3t-s=l-i \Rightarrow i=1$$

$$l=0$$

Occorre rimuovere una condizione di vincolo semplice

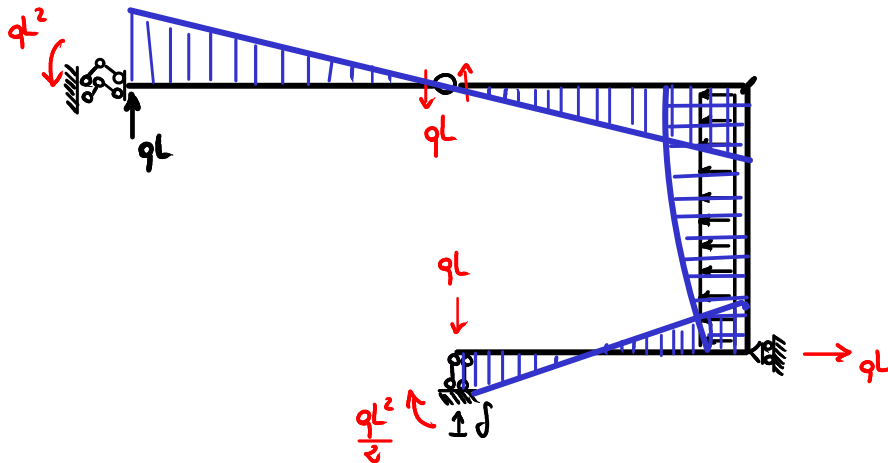
$S^{(4)}$ = struttura isostatica staticamente e cinematicamente equivalente alla struttura assegnata



EQUIVALENZA STATICA : va aggiunta la reazione x_1 del vincolo rimosso

EQUIVALENZA CINEMATICA : va aggiunta la condizione cinematica $\varphi_C = 0$ del vincolo rimosso

$S^{(0)}$



$$N_{AB}^{(0)} = 0$$

$$T_{AB}^{(0)} = qL$$

$$M_{AB}^{(0)} = -qL^2 + qLz$$

$$N_{BC}^{(0)} = 0$$

$$T_{BC}^{(0)} = qL$$

$$M_{BC}^{(0)} = qLz$$

$$N_{CD}^{(0)} = 0$$

$$T_{CD}^{(0)} = -qL$$

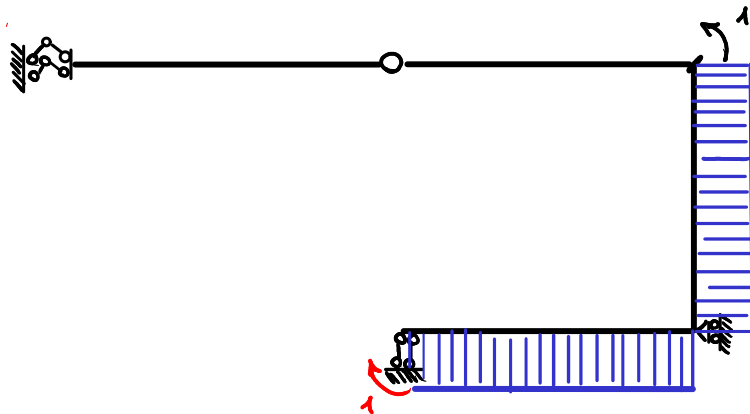
$$M_{CD}^{(0)} = qL^2 - \frac{qL^2}{2}$$

$$N_{DE}^{(0)} = 0$$

$$T_{DE}^{(0)} = -qL$$

$$M_{DE}^{(0)} = -\frac{qL^2}{2} + qL(L-z)$$

5⁽¹⁾



$$\delta_{ve} = 1 \cdot \varphi_E^{(eq)} = 0$$

$$\begin{aligned} \delta_{vi} &= \int_{CD} \frac{M_{CD}^{(1)} (M_{CD}^{(0)} + \chi_1 M_{CD}^{(1)})}{EI} dz + \int_{AB} \frac{M_{AB}^{(1)} (M_{AB}^{(0)} + \chi_1 M_{AB}^{(1)})}{EI} dz = \\ &= \int_0^L 1 \cdot \frac{qL^2 - \frac{qz^2}{2} + \chi_1 \cdot 1}{EI} dz + \int_0^L 1 \cdot \frac{-\frac{qL^2}{2} + qL(L-z) + \chi_1 \cdot 1}{EI} dz = \end{aligned}$$

$$= \frac{1}{EI} \left[qL^2 z - \frac{qz^3}{6} + \chi_1 z \right]_0^L + \frac{1}{EI} \left[-\frac{qL^2}{2} z + qL^2 z - \frac{qLz^2}{2} + \chi_1 z \right]_0^L =$$

$$= \frac{1}{EI} \left[qL^3 - \frac{qL^3}{6} + \chi_1 L \right] + \frac{1}{EI} \left[-\frac{qL^3}{2} + qL^3 - \frac{qL^3}{2} + \chi_1 L \right] = 2 \frac{\chi_1 L}{EI} + \frac{5}{6} \frac{qL^3}{EI}$$

$$\chi_1 = -\frac{5}{12} qL^2$$

Rispetto al riferimento X, Y, Z

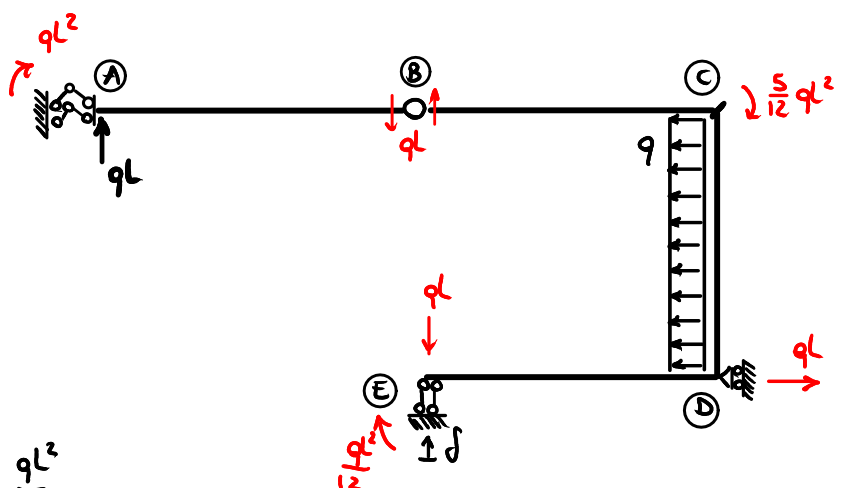
$$M_{ZA} = M_{ZA}^{(0)} + \chi_1 M_{ZA}^{(1)} = qL^2$$

$$R_{BY} = R_{BY}^{(0)} + \chi_1 R_{BY}^{(1)} = qL$$

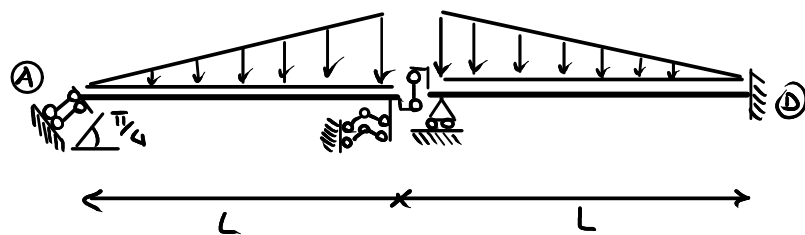
$$R_{DX} = R_{DX}^{(0)} + \chi_1 R_{DX}^{(1)} = qL$$

$$R_{EY} = R_{EY}^{(0)} + \chi_1 R_{EY}^{(1)} = -qL$$

$$M_{EZ} = M_{EZ}^{(0)} + \chi_1 M_{EZ}^{(1)} = -\frac{qL^2}{2} - \frac{5}{12} qL^2 (-1) = -\frac{qL^2}{12}$$

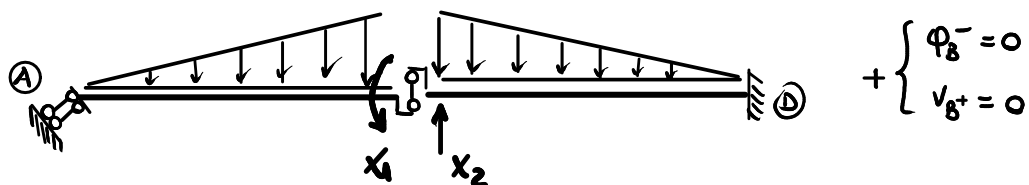


ESEMPIO SVOLTO 3



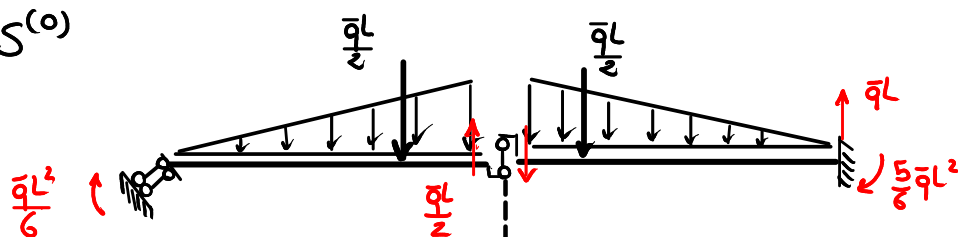
$$i = 2$$

$S^{(eq)}$



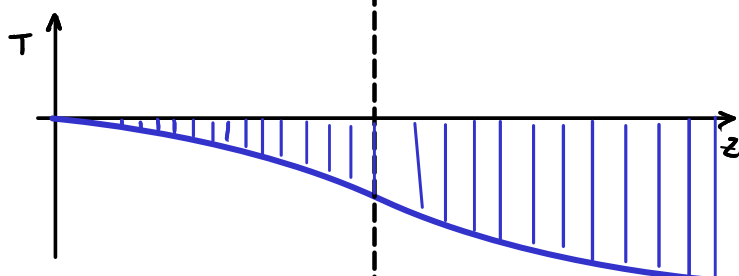
$$+ \begin{cases} \varphi_B^- = 0 \\ v_B^+ = 0 \end{cases}$$

$S^{(o)}$



$$q_{AB}(z) = \frac{\bar{q}z}{L}$$

$$q_{BC}(z) = \bar{q} \left(1 - \frac{z}{L}\right)$$

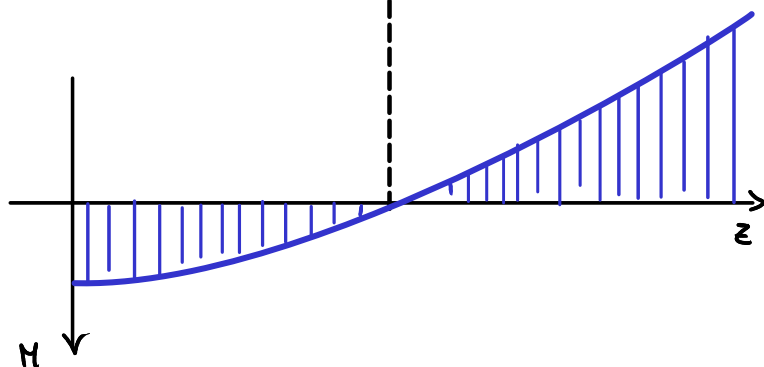


AB: guardando a precedere

$$T_{AB}^{(o)} = - \frac{q_{AB}(z) \cdot z}{2} = - \frac{\bar{q}z^2}{2L}$$

BC: guardando a seguire

$$T_{BC}^{(o)} = -\bar{q}L + \frac{q_{BC}(z)(L-z)}{2} = -\bar{q}L + \frac{\bar{q}(L-z)^2}{2L}$$



AB: guardando a precedere

$$M_{AB}^{(o)} = \frac{\bar{q}L^2}{6} - \frac{q_{AB}(z)z}{2} \cdot \frac{z}{3} = \frac{\bar{q}L^2}{6} - \frac{\bar{q}z^3}{6L}$$

BC: guardando a seguire

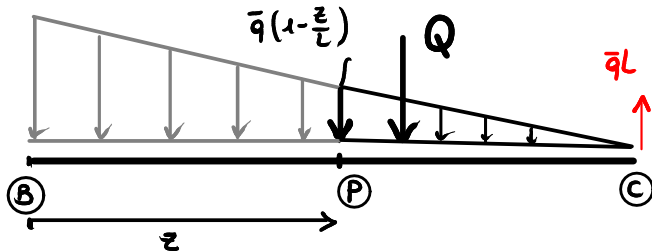
$$\begin{aligned} M_{BC}^{(o)} &= -\frac{5}{6}\bar{q}L^2 + \bar{q}L(L-z) - \frac{q_{BC}(z)(L-z)}{2} \cdot \frac{(L-z)}{3} \\ &= -\frac{5}{6}\bar{q}L^2 + \bar{q}L(L-z) - \frac{\bar{q}(L-z)^3}{6L} \end{aligned}$$

Nota sulle caratteristiche di $S^{(0)}$. Quando il carico distribuito è lineare conviene sempre scrivere le caratteristiche a precedere/ a seguire guardando il punto di nullo della distribuzione (che qui sono A sul tratto AB e C su BC)

Nulla vieta la scelta opposta, ma bisogna decomporre opportunamente il carico distribuito.

Si consideri ad esempio BC:

BC, guardando a seguire da un punto P



La porzione di carico distribuito vista guardando a seguire è triangolare, con risultante

$$Q = \frac{\bar{q}(1 - \frac{z}{L}) \cdot (L - z)}{2} = \bar{q} \frac{(L - z)^2}{2L}$$

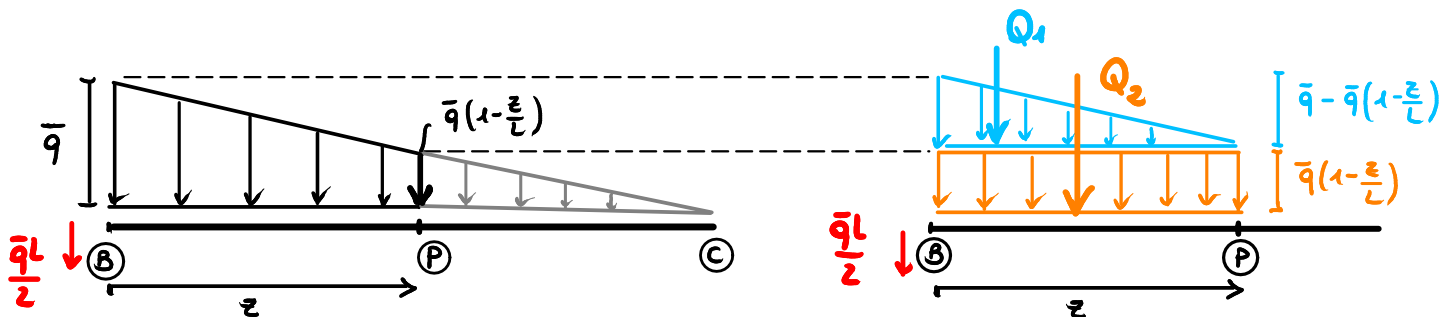
seguono quindi $T_{BC} = Q - |R_{yc}| = \frac{\bar{q}(L - z)^2}{2L} - \bar{q}L = \frac{\bar{q}z^2}{2L} - \bar{q}z - \frac{\bar{q}L}{2}$

$$M_{BC} = -Q \cdot \frac{L - z}{3} + |R_{yc}|(L - z)$$

BC, guardando a precedere da un punto P

La porzione di carico distribuito vista guardando a precedere è trapezoidale, e va decomposta in una componente triangolare $q_{BC}^{(1)}(z)$ ed una componente rettangolare $q_{BC}^{(2)}(z)$

Queste componenti sono determinabili una volta riconosciuta che nel punto generico P, posto all'ascissa z, l'intensità di tutta la distribuzione è banalmente $\bar{q}(1 - \frac{z}{L})$



$$q_{BC}^{(1)}(z) = \bar{q} - \bar{q}(1 - \frac{z}{L}) = \bar{q} \frac{z}{L}$$

con risultante $Q_1 = \frac{q_{BC}^{(1)}(z) \cdot z}{2} = \frac{\bar{q}z^2}{2L}$

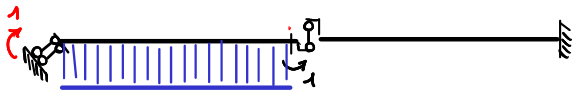
$$q_{BC}^{(2)}(z) = \bar{q}(1 - \frac{z}{L})$$

con risultante $Q_2 = q_{BC}^{(2)}(z) \cdot z = \bar{q}z(1 - \frac{z}{L})$

Dalle quali seguono le caratteristiche. Ad esempio il taglio

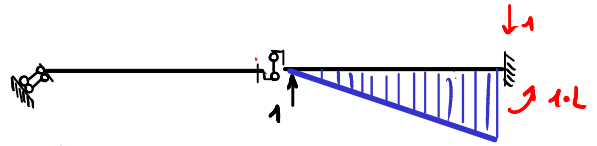
$$T_{BC}(z) = -\frac{\bar{q}L}{2} - Q_1 - Q_2 = -\frac{\bar{q}L}{2} - \frac{\bar{q}z^2}{2L} - \bar{q}z(1 - \frac{z}{L}) = -\frac{\bar{q}L}{2} + \frac{\bar{q}z^2}{2L} - \bar{q}z$$

(cfr quanto scritto guardando a seguire)

$S^{(1)}$ 

$$M_{AB}^{(1)} = 1$$

$$M_{BC}^{(1)} = 0$$

 $S^{(2)}$ 

$$M_{AB}^{(2)} = 0$$

$$M_{BC}^{(2)} = 1 \cdot z$$

SOLUZIONE IPERSTATICA

TLV 1

$$S^{(1)} = S^{(1)}$$

$$S^{(2)} = S^{(eq)}$$

$$\Delta v_c = 0$$

$$\Delta v_i = \int_{AB} M_{AB}^{(1)} \frac{M_{AB}^{(0)} + x_1 M_{AB}^{(1)} + x_2 M_{AB}^{(2)}}{EI} dz + \int_{BC} M_{BC}^{(1)} \frac{M_{BC}^{(0)} + x_1 M_{BC}^{(1)} + x_2 M_{BC}^{(2)}}{EI} dz$$

$$\Rightarrow \frac{1}{EI} \int_0^L 1 \cdot \left(\frac{\bar{q} L^2}{6} - \frac{\bar{q} z^3}{6L} + x_1 \cdot 1 \right) dz = 0$$

TLV 2

$$S^{(1)} = S^{(2)}$$

$$S^{(2)} = S^{(eq)}$$

$$\Delta v_c = 0$$

$$\Delta v_i = \int_{AB} M_{AB}^{(2)} \frac{M_{AB}^{(0)} + x_1 M_{AB}^{(1)} + x_2 M_{AB}^{(2)}}{EI} dz + \int_{BC} M_{BC}^{(2)} \frac{M_{BC}^{(0)} + x_1 M_{BC}^{(1)} + x_2 M_{BC}^{(2)}}{EI} dz$$

$$\Rightarrow \frac{1}{EI} \int_0^L z \cdot \left(-\frac{5}{6} q L^2 + q L (L-z) - \frac{q (L-z)^3}{6L} + x_2 z \right) dz = 0$$

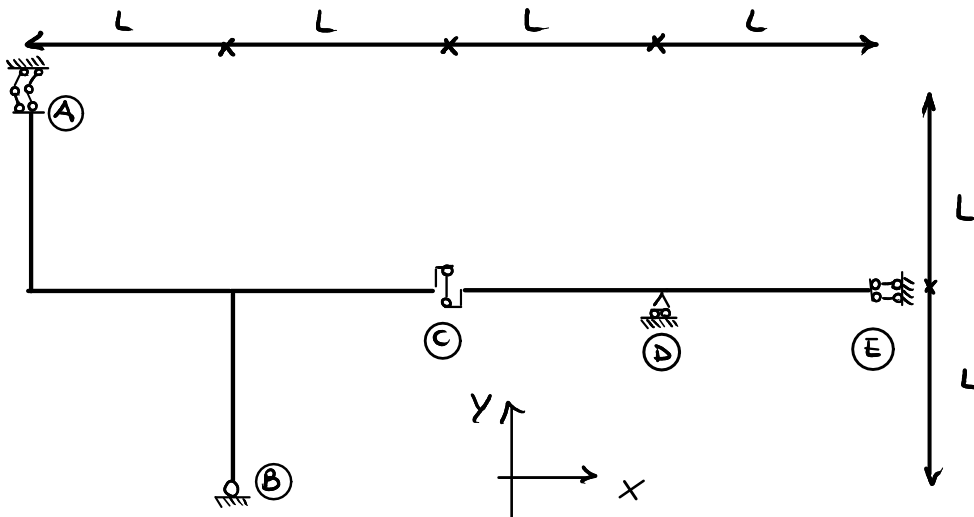
SOLUZIONE

$$x_1 = -\frac{q L^2}{8}$$

$$x_2 = \frac{31}{40} q L$$

Esercizio 1

Grado di libertà tramite la matrice cinematica



Si assumono i seguenti poli cinematici

Corpo I : B

Corpo II : E

$$\underline{u}^I = \underline{u}_B + \underline{\dot{\theta}}_1 \times (P - B)$$

$$\underline{u}^{II} = \underline{u}_E + \underline{\dot{\theta}}_2 \times (P - E)$$

$$\underline{\dot{\theta}}_1 = 0$$

$$\underline{u}_B = \underline{0}$$

$$\Delta \underline{u}_C \cdot \underline{e}_y = 0 \quad (*)$$

$$u_{By} = 0 \quad (**)$$

$$\dot{\theta}_2 = 0$$

$$u_{Ex} = 0$$

$$\dot{\theta}_1 = 0$$

$$u_{Bx} = 0$$

$$u_{By} = 0$$

$$u_{Ey} - 2\dot{\theta}_2 L - u_{By} - \dot{\theta}_1 L = 0$$

$$u_{Ey} - \dot{\theta}_2 L = 0$$

$$\dot{\theta}_2 = 0$$

$$u_{Ex} = 0$$

0	0	1	0	0	0	u_{Bx}	0
1	0	0	0	0	0	u_{By}	0
0	1	0	0	0	0	$\dot{\theta}_1$	0
0	-1	-L	0	1	-2L	u_{Ex}	0
0	0	0	0	1	-L	u_{Ey}	0
0	0	0	0	0	1	$\dot{\theta}_2$	0
0	0	0	1	0	0		

$$(*) \quad \Delta \underline{u}_C = \underline{u}_C^{II} - \underline{u}_C^I = [\underline{u}_E + \underline{\dot{\theta}}_2 \times (C - E)] - [\underline{u}_B + \underline{\dot{\theta}}_1 \times (C - B)] =$$

$$= [\underline{u}_E + \dot{\theta}_2 \underline{e}_z \times (-2L \underline{e}_x)] - [\underline{u}_B + \dot{\theta}_1 \underline{e}_z \times (L \underline{e}_x + L \underline{e}_y)] =$$

$$= [u_{Ex} \underline{e}_x + u_{Ey} \underline{e}_y - 2\dot{\theta}_2 L \underline{e}_y] - [u_{Bx} \underline{e}_x + u_{By} \underline{e}_y + \dot{\theta}_1 L (\underline{e}_y - \underline{e}_x)] =$$

$$= [u_{Ex} - u_{Bx} + \dot{\theta}_1 L] \underline{e}_x + [u_{Ey} - 2\dot{\theta}_2 L - u_{By} - \dot{\theta}_1 L] \underline{e}_y$$

$$(**) \quad \underline{u}_D = \underline{u}_E + \underline{\dot{\theta}}_2 \times (D - E) = \underline{u}_E + \dot{\theta}_2 \underline{e}_z \times (-L \underline{e}_x) = u_{Ex} \underline{e}_x + [u_{Ey} - \dot{\theta}_2 L] \underline{e}_y$$

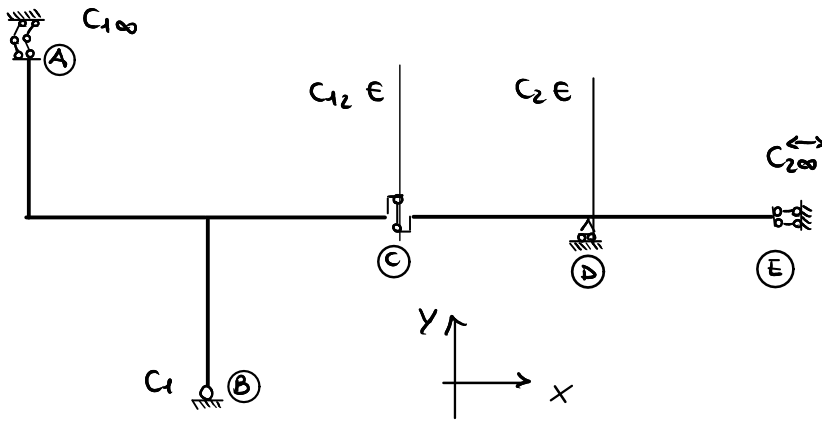
La matrice cinematica non ha rango massimo: nominando infatti le righe dall'alto verso il basso si può verificare che

$$R_4 = R_3 - L R_1 + R_5 - 2R_6$$

ovvero R_4 è linearmente dipendente da R_1, R_3, R_5, R_6 , ossia essa è "ridondante" al fine del calcolo del rango. Rimuovendo R_4 dalla matrice si ottiene una seconda matrice 6×6 di rango massimo, e allora

$$rg(A) = 6 \quad \Rightarrow \quad \ell = \#col(A) - rg(A) = 0$$

Grado di libertà tramite centri dello spostamento



Lo spostamento rigido del corpo I, se consentito, avviene attorno al centro C_1 la cui posizione è determinata dai vincoli esterni del corpo I:

- pantografo: C_1 è un punto improprio
- cerniera: C_1 è un punto proprio posto in B

Le due condizioni sono incompatibili tra loro: il centro C_1 non esiste

Se il centro C_1 non esiste il corpo I è fermo, il che consente di vedere il pendolo C come esterno. Si conclude rapidamente che anche il corpo II non ammette centro: esso infatti si degenera in un punto improprio in direzione verticale ($C_2 = C_{2\infty} \downarrow$) per via dei carrelli in C e D, o è un punto improprio orizzontale (vincolo E) -

Nessun centro esiste: $\ell = 0$

Grado di iperstaticità tramite la matrice di equilibrio

Assumendo come poli statici di riduzione dei momenti gli stessi poli usati nella descrizione della cinematica, B e E, si può sfruttare la proprietà

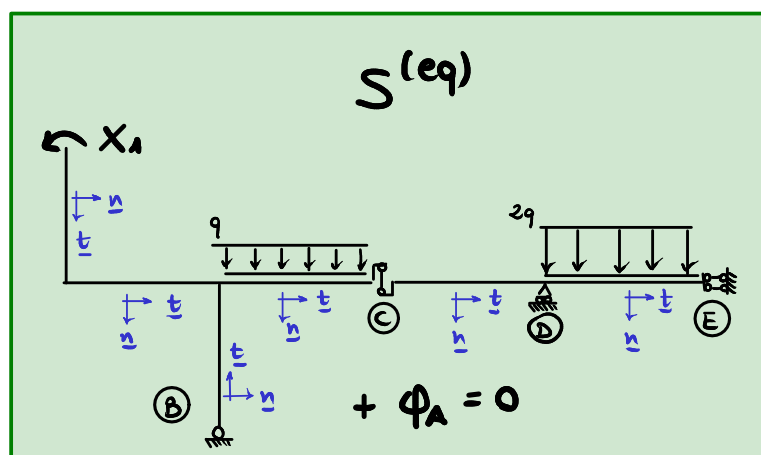
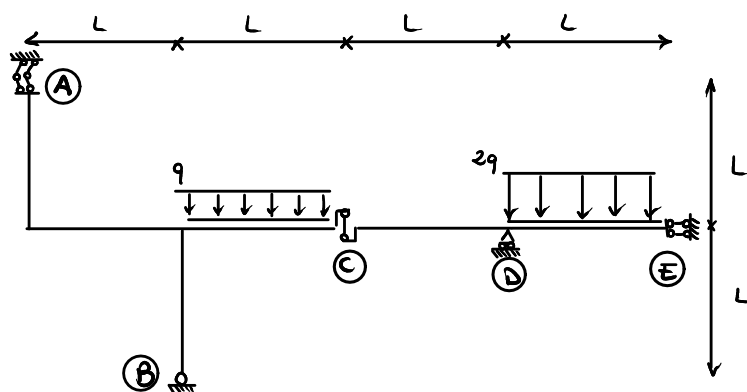
$$[A] = [B]^T \Leftrightarrow [A]^T = [B] \Rightarrow i = \# \text{col}[A] - \text{rg}[A] = 7 - 6 = 1$$

dove, si ricordi, $\text{rg}[M] = \text{rg}[M]^T$

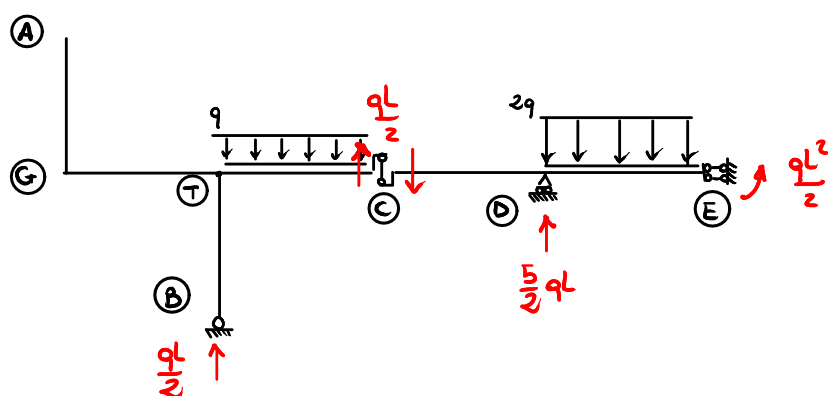
Grado di iperstaticità tramite la formula di Grassmann

Nota ℓ si conclude immediatamente che $3t - s = 3 \cdot 2 - 7 = 0 - i \Rightarrow i = 1$

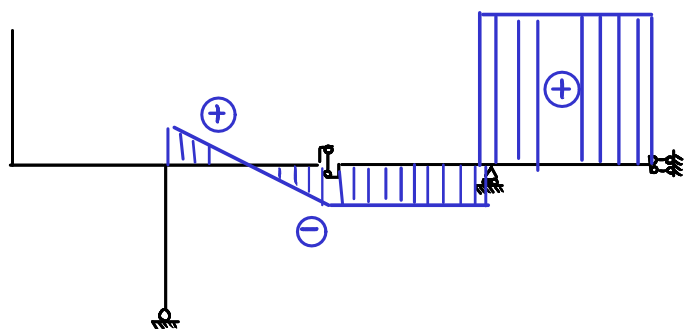
Soluzione tramite il metodo delle forze



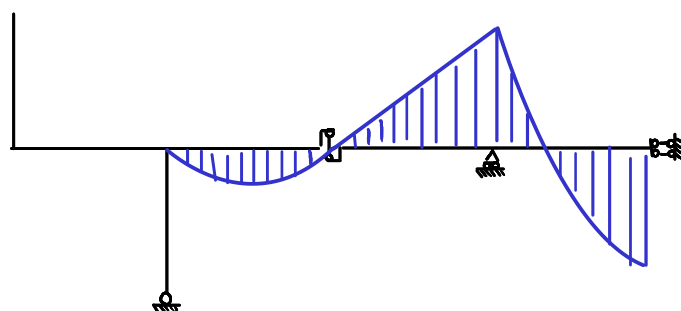
$S^{(0)}$



$T^{(0)}$



$M^{(0)}$



$$T_{AG}^{(0)} = 0$$

$$T_{TC}^{(0)} = -qz + \frac{qL}{2}$$

$$T_{GT}^{(0)} = 0$$

$$T_{CD}^{(0)} = -qL/2$$

$$T_{BT}^{(0)} = 0$$

$$T_{DE}^{(0)} = 2q(L-z)$$

$$M_{AG}^{(0)} = 0$$

$$M_{TC}^{(0)} = -\frac{qz^2}{2} + \frac{qLz}{2}$$

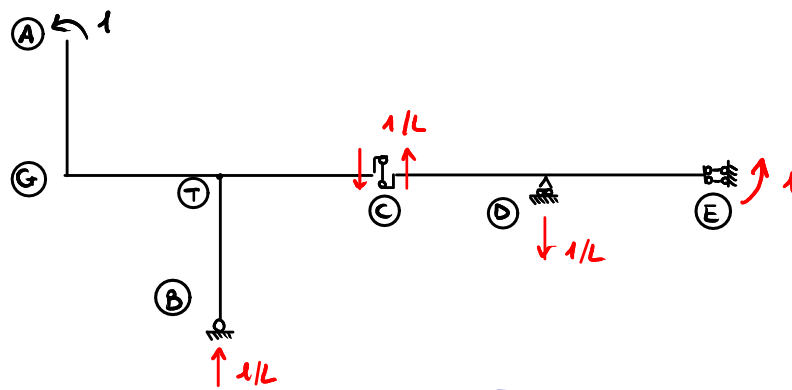
$$M_{GT}^{(0)} = 0$$

$$M_{CD}^{(0)} = -\frac{qLz}{2}$$

$$M_{BT}^{(0)} = 0$$

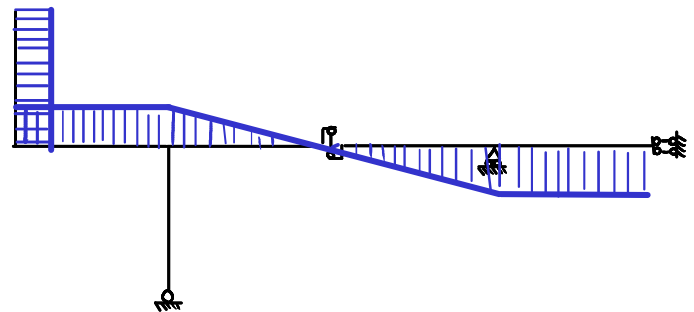
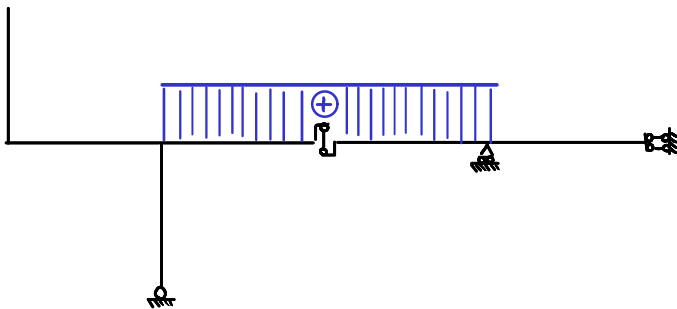
$$M_{DE}^{(0)} = \frac{qL^2}{2} - 2q\frac{(L-z)^2}{2}$$

$S^{(1)}$



$T^{(1)}$

$M^{(1)}$



$$\begin{aligned} T_{AG}^{(1)} &= 0 & T_{TC}^{(1)} &= \frac{1}{L} \\ T_{GT}^{(1)} &= 0 & T_{CD}^{(1)} &= \frac{1}{L} \\ T_{BT}^{(1)} &= 0 & T_{DE}^{(1)} &= 0 \end{aligned}$$

$$\begin{aligned} M_{AG}^{(1)} &= -1 & M_{TC}^{(1)} &= -1 + \frac{z}{L} \\ M_{GT}^{(1)} &= -1 & M_{CD}^{(1)} &= \frac{z}{L} \\ M_{BT}^{(1)} &= 0 & M_{DE}^{(1)} &= 1 \end{aligned}$$

$$S^{(1)} = S^{(1)}, \quad S^{(3)} = S^{(eq)}$$

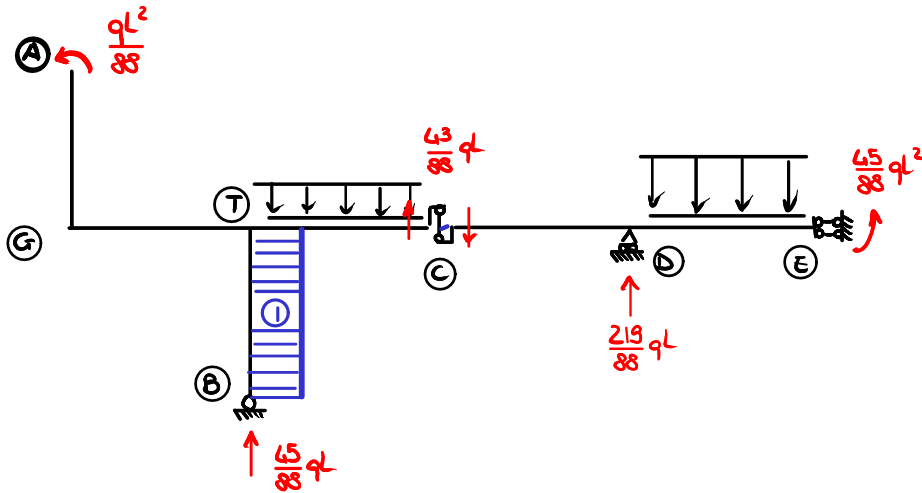
$$\delta v_c = 0$$

$$\begin{aligned} \delta v_c = & \int_{AG} M_{AG}^{(1)} \frac{M_{AG}^{(0)} + x_1 M_{AG}^{(1)}}{EI} dz + \int_{GT} M_{GT}^{(1)} \frac{M_{GT}^{(0)} + x_1 M_{GT}^{(1)}}{EI} dz + \int_{BT} M_{BT}^{(1)} \frac{M_{BT}^{(0)} + x_1 M_{BT}^{(1)}}{EI} dz + \\ & \int_{TC} M_{TC}^{(1)} \frac{M_{TC}^{(0)} + x_1 M_{TC}^{(1)}}{EI} dz + \int_{CD} M_{CD}^{(1)} \frac{M_{CD}^{(0)} + x_1 M_{CD}^{(1)}}{EI} dz + \int_{DE} M_{DE}^{(1)} \frac{M_{DE}^{(0)} + x_1 M_{DE}^{(1)}}{EI} dz \end{aligned}$$

$$\delta v_c = \delta v_{c1} \quad \Leftrightarrow \quad x_1 = \frac{2L^2}{88}$$

Struttura iperstatica risolta: Sovrapposizione degli effetti

(N)



$$N_{AG} = 0$$

$$N_{TC} = 0$$

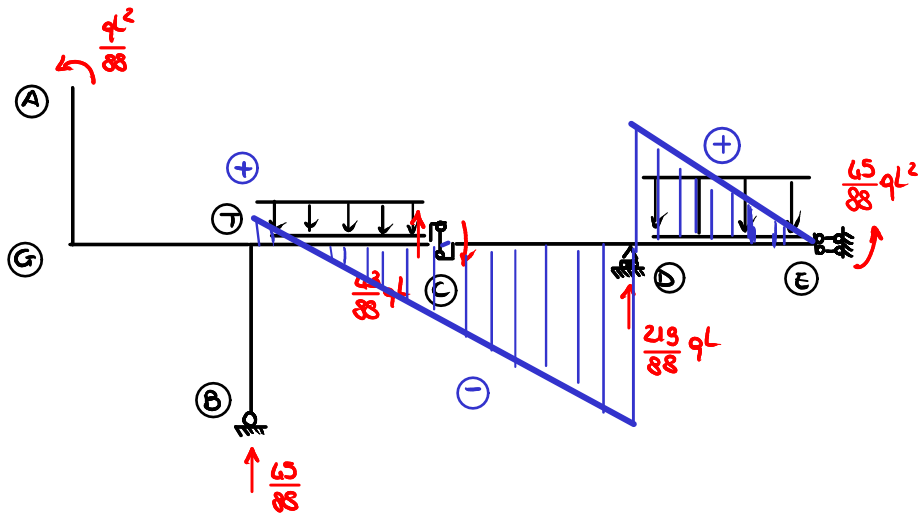
$$N_{GT} = 0$$

$$N_{CD} = 0$$

$$N_{TB} = -\frac{45}{88} qL$$

$$N_{DE} = 0$$

(T)



$$T_{AG} = 0$$

$$T_{TC} = \frac{45}{88} qL - qz$$

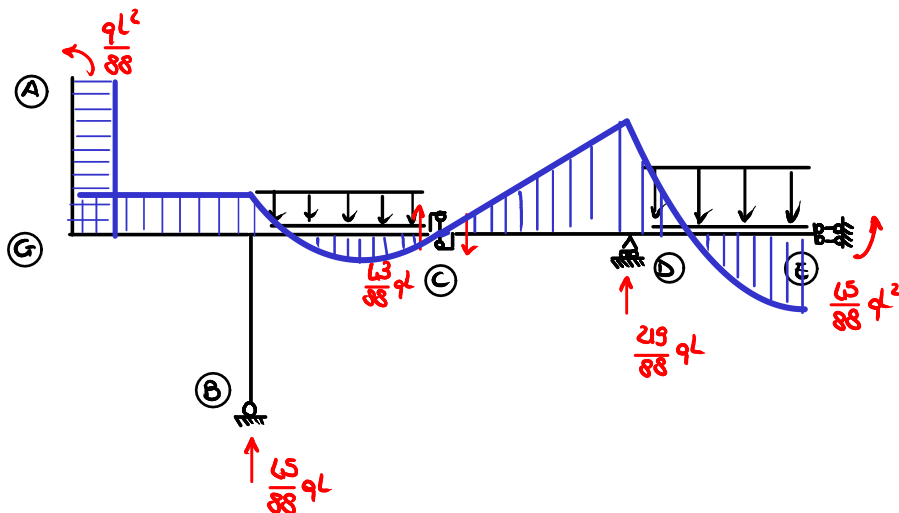
$$T_{GT} = 0$$

$$T_{CD} = -\frac{43}{88} qL$$

$$T_{TB} = 0$$

$$T_{DE} = 2q(L-z)$$

(M)



$$M_{AG} = -\frac{qL^2}{88}$$

$$M_{GT} = -\frac{qL^2}{88}$$

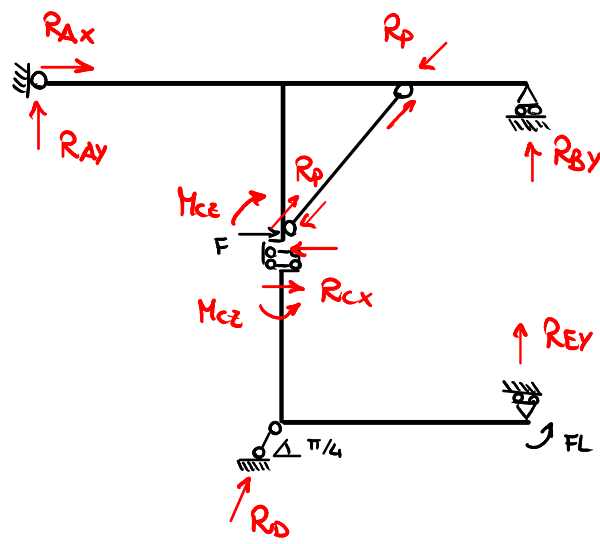
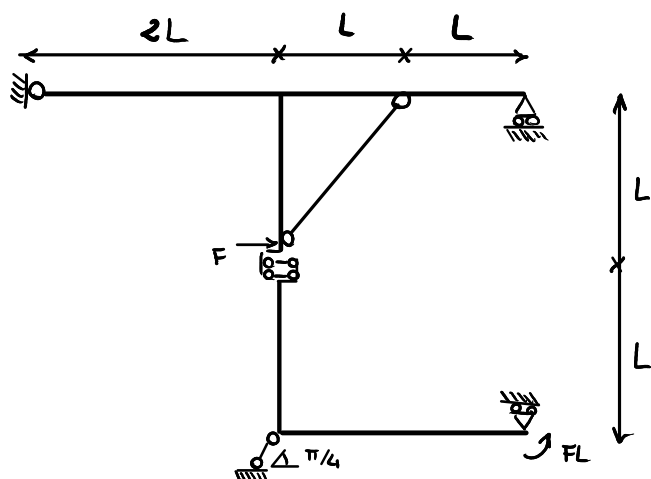
$$M_{TB} = 0$$

$$M_{TC} = -\frac{qz^2}{2} + \frac{45}{88} qLz - \frac{qL^2}{88}$$

$$M_{CD} = -\frac{43}{88} qLz$$

$$M_{DE} = \frac{45}{88} qL^2 - q(L-z)^2$$

Esercizio 2



Sfruttando la proprietà $[A]=[B]^T$ si calcolano ℓ ed i a partire dalla matrice di equilibrio

$$R_{Ax} - R_p \frac{\sqrt{2}}{2} + R_p \frac{\sqrt{2}}{2} - R_{Cx} + F = 0$$

$$R_{Ay} - R_p \frac{\sqrt{2}}{2} + R_p \frac{\sqrt{2}}{2} + R_{By} = 0$$

$$M_z(C) = -M_{Cx} - R_{Ax}L - R_{Ay}L + R_{By}L = 0$$

$$R_{Cx} + R_D \frac{\sqrt{2}}{2} = 0$$

$$R_D \frac{\sqrt{2}}{2} + R_{Ey} = 0$$

$$M_z(E) = M_{Cx} - R_{Cx}L - R_D \frac{\sqrt{2}}{2}L + FL = 0$$

$\begin{vmatrix} 1 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ -L & -L & 0 & L & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & \frac{\sqrt{2}}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{2} & 1 \\ 0 & 0 & 0 & 0 & -L & 1 & -\frac{\sqrt{2}}{2} & 0 \end{vmatrix}$	$\begin{matrix} R_{Ax} \\ R_{Ay} \\ R_p \\ R_{By} \\ R_{Cx} \\ M_{Cx} \\ R_D \\ R_{Ey} \end{matrix}$	$\begin{matrix} = \\ \\ \\ \\ \\ -FL \\ \\ \end{matrix}$
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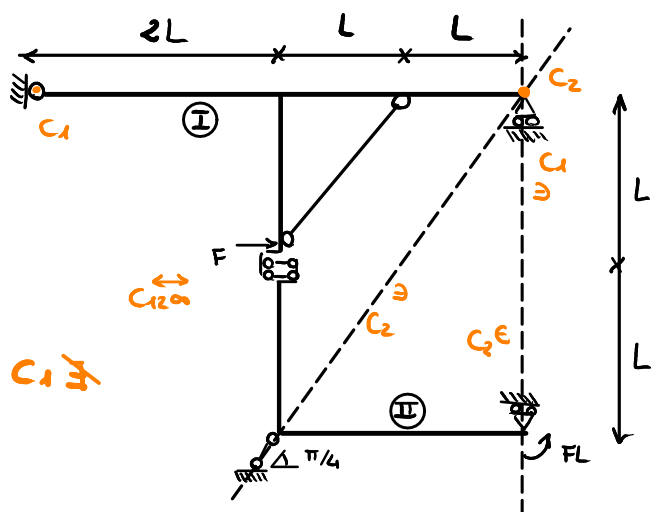
$$[B]_{n \times m}, \quad n = 6, \quad m = 8 \quad \text{rk}[B] = 6$$

$$[A]_{m \times n} \quad \text{rk}[A] = \text{rk}[A]^T = \text{rk}[B] = 6$$

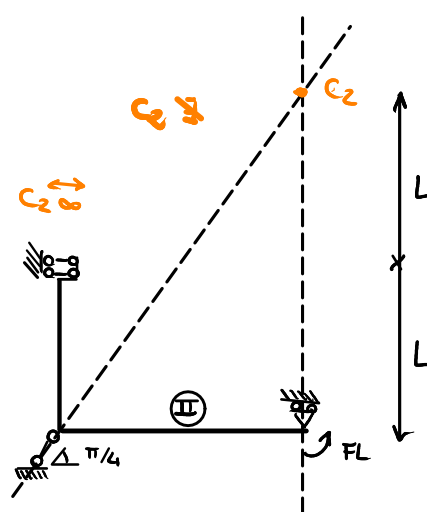
$$i = \dim(\ker[B]) = \text{col}[B] - \text{rk}[B] = m - \text{rk}[B] = 2$$

$$\ell = \dim(\ker[A]) = \text{col}[A] - \text{rk}[B] = n - \text{rk}[A] = n - \text{rk}[B] = 0$$

Labilità via centri dello spostamento

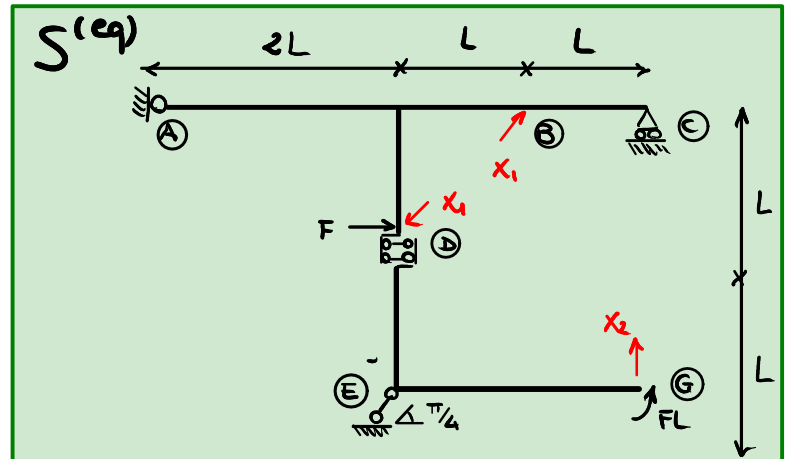
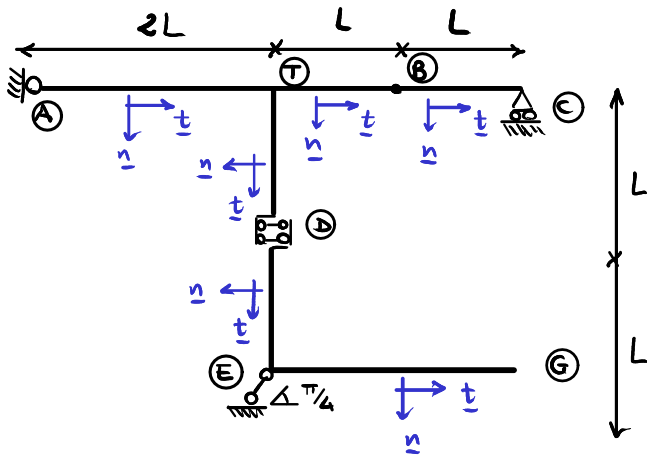
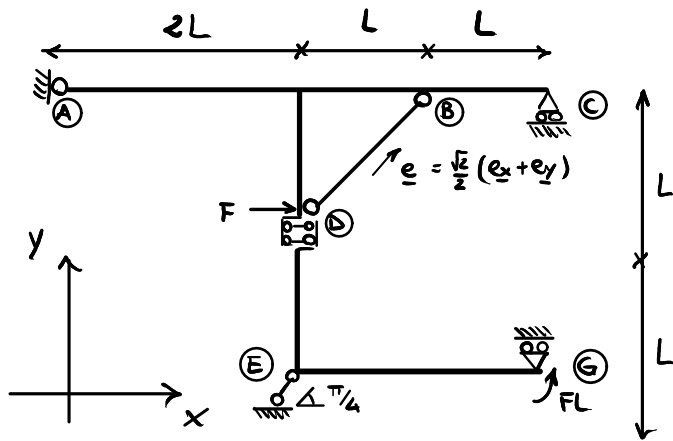


III



Soluzione iperstatica tramite metodo delle forze

$$i = 2$$



Condizioni di congruenza

$$(\underline{s}_B - \underline{s}_D) \cdot \underline{e} = 0 \quad \underline{e} = \frac{\sqrt{2}}{2} \underline{e}_x + \frac{\sqrt{2}}{2} \underline{e}_y$$

$$v_G = 0$$

$$\underline{s}_B = w_B \underline{t}_{BC} + v_B \underline{n}_{BC} = w_B \underline{e}_x - v_B \underline{e}_y$$

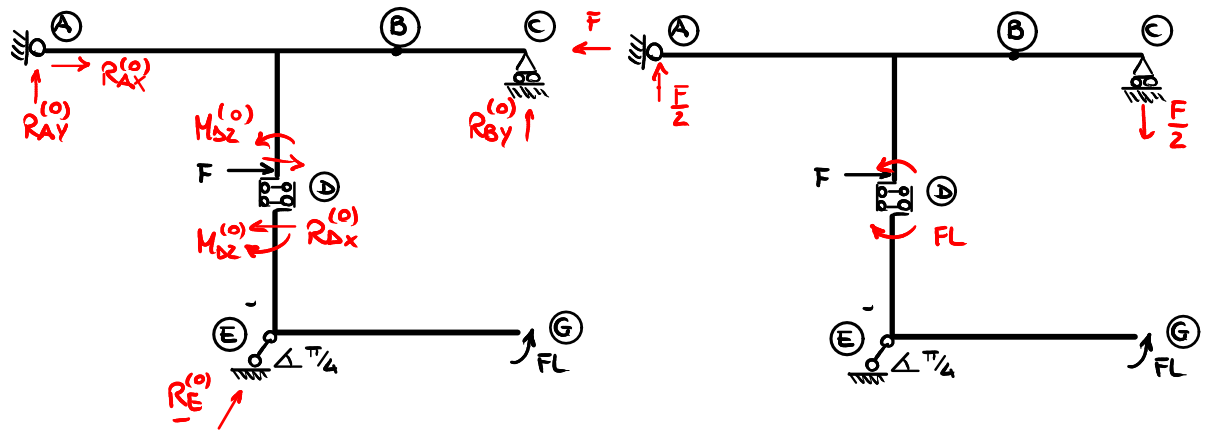
$$\underline{s}_D = w_D \underline{t}_{ED} + v_D \underline{n}_{ED} = -w_D \underline{e}_y - v_D \underline{e}_x$$

$$\underline{s}_B - \underline{s}_D = (w_B + v_D) \underline{e}_x + (w_D - v_B) \underline{e}_y$$

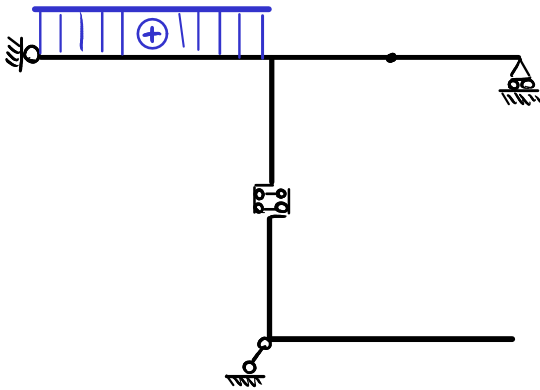
$$(\underline{s}_B - \underline{s}_D) \cdot \underline{e} = \begin{Bmatrix} w_B + v_D \\ w_D - v_B \end{Bmatrix} \cdot \begin{Bmatrix} \sqrt{2}/2 \\ \sqrt{2}/2 \end{Bmatrix} = \frac{\sqrt{2}}{2} (w_B + v_D) + \frac{\sqrt{2}}{2} (w_D - v_B) = 0$$

OSSERVAZIONE: la scelta dei versi dei vettori $\underline{t}$ e $\underline{n}$ su ciascun tratto è arbitraria (purché rispetti $\underline{n} \times \underline{t} = \underline{e}_z$) ma, una volta fissata, deve essere mantenuta per tutti i sottoschemi $S^{(0)}$, $S^{(1)}$, $S^{(2)}$, etc.

$S^{(0)}$



N



$$N_{AT}^{(0)} = F$$

$$N_{TB}^{(0)} = 0$$

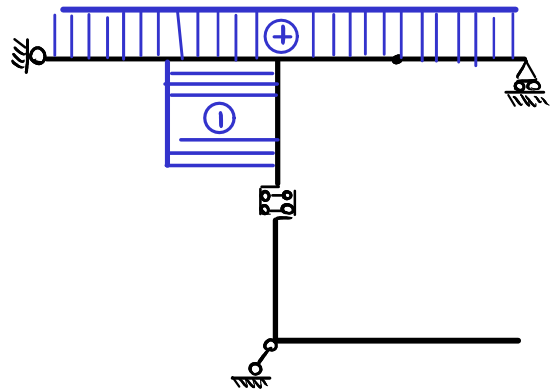
$$N_{TB}^{(0)} = 0$$

$$N_{DE}^{(0)} = 0$$

$$N_{BC}^{(0)} = 0$$

$$N_{EG}^{(0)} = 0$$

T



$$T_{AT}^{(0)} = F/2$$

$$T_{TB}^{(0)} = -F$$

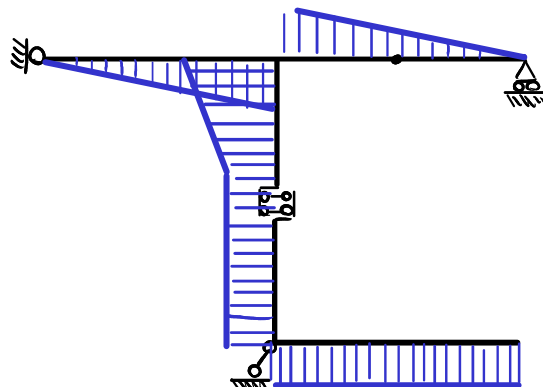
$$T_{TB}^{(0)} = F/2$$

$$T_{DE}^{(0)} = 0$$

$$T_{BC}^{(0)} = F/2$$

$$T_{EG}^{(0)} = 0$$

M



$$M_{AT}^{(0)} = \frac{Fz}{2}$$

$$M_{TB}^{(0)} = FL + F(L-z)$$

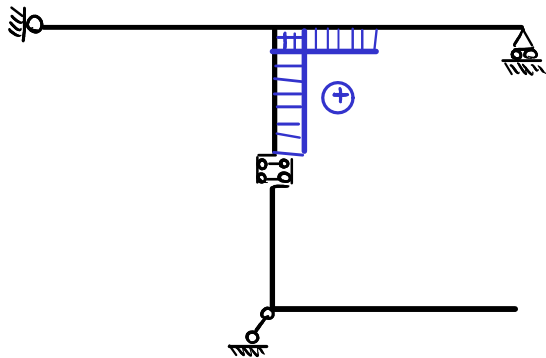
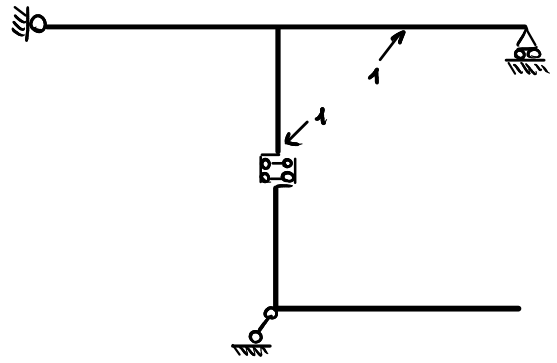
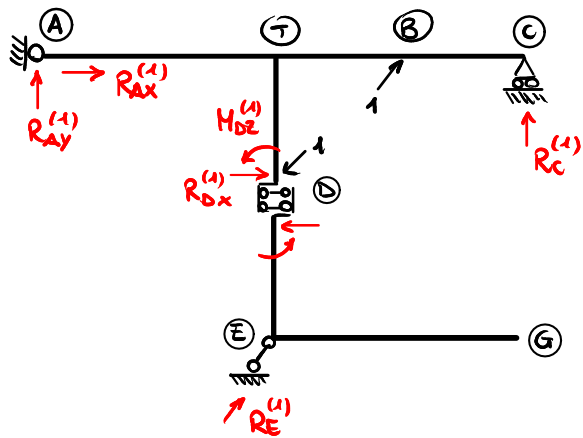
$$M_{TB}^{(0)} = -\frac{F(2L-z)}{2}$$

$$M_{DE}^{(0)} = FL$$

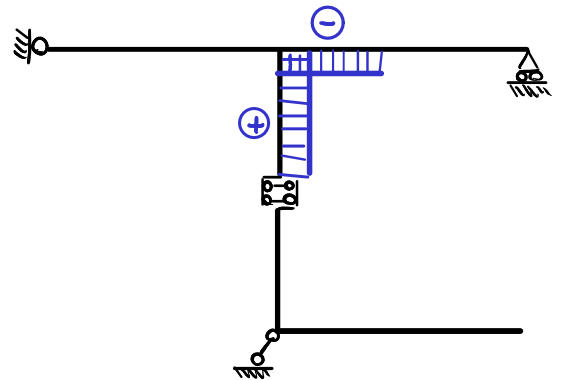
$$M_{BC}^{(0)} = -\frac{F}{2}(L-z)$$

$$M_{EG}^{(0)} = FL$$

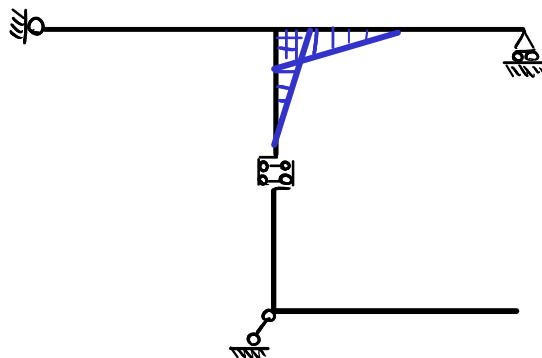
$S^{(1)}$



$$\begin{aligned} N_{AT}^{(1)} &= 0 & N_{TD}^{(1)} &= \frac{\sqrt{2}}{2} \\ N_{TB}^{(1)} &= \frac{\sqrt{2}}{2} & N_{DE}^{(1)} &= 0 \\ N_{BC}^{(1)} &= 0 & N_{EG}^{(1)} &= 0 \end{aligned}$$

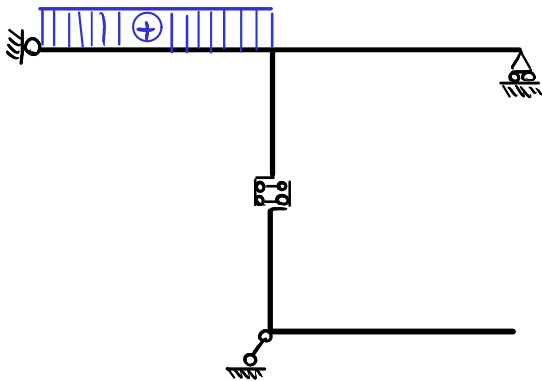
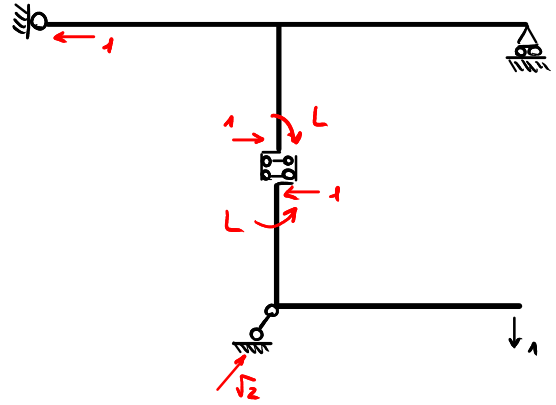
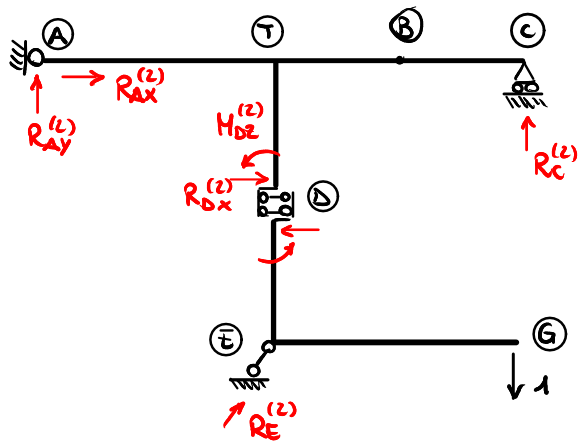


$$\begin{aligned} T_{AT}^{(1)} &= 0 & T_{TD}^{(1)} &= \frac{\sqrt{2}}{2} \\ T_{TB}^{(1)} &= -\frac{\sqrt{2}}{2} & T_{DE}^{(1)} &= 0 \\ T_{BC}^{(1)} &= 0 & T_{EG}^{(1)} &= 0 \end{aligned}$$

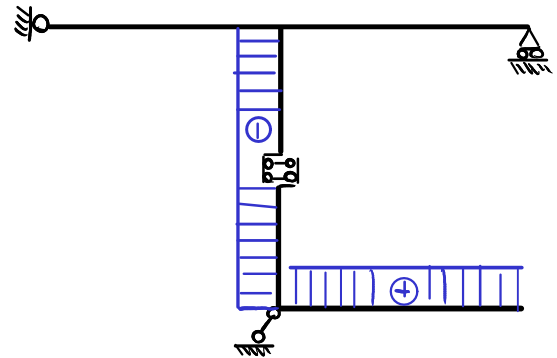


$$\begin{aligned} M_{AT}^{(1)} &= 0 & M_{TD}^{(1)} &= \frac{\sqrt{2}}{2} (L-z) \\ M_{TB}^{(1)} &= \frac{\sqrt{2}}{2} (L-z) & M_{DE}^{(1)} &= 0 \\ M_{BC}^{(1)} &= 0 & M_{EG}^{(1)} &= 0 \end{aligned}$$

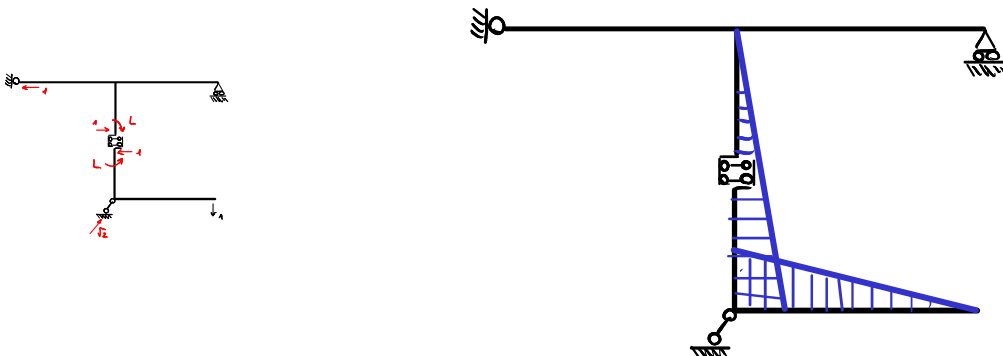
$S^{(2)}$



$$\begin{aligned} N_{AT}^{(2)} &= & N_{TD}^{(2)} &= \\ N_{TB}^{(2)} &= & N_{DE}^{(2)} &= \\ N_{BC}^{(2)} &= & N_{EG}^{(2)} &= \end{aligned}$$



$$\begin{aligned} T_{AT}^{(2)} &= & T_{TD}^{(2)} &= \\ T_{TB}^{(2)} &= & T_{DE}^{(2)} &= \\ T_{BC}^{(2)} &= & T_{EG}^{(2)} &= \end{aligned}$$



$$M_{AT}^{(2)} = 0$$

$$M_{TD}^{(2)} = -z$$

$$M_{TB}^{(2)} = 0$$

$$M_{DE}^{(2)} = -L - z$$

$$M_{BC}^{(2)} = 0$$

$$M_{EG}^{(2)} = -(2L - z)$$

Soluzione iperstatica

$$1 \cdot (S_B - S_D) \cdot e = 0 = \int_0^{2L} \frac{M_{AT}^{(1)} H_{AT}^{(eq)}}{EI} dz + \int_0^L \frac{M_{TB}^{(1)} H_{TB}^{(eq)}}{EI} dz + \int_0^L \frac{M_{BC}^{(1)} H_{BC}^{(eq)}}{EI} dz + \int_0^L \frac{M_{TD}^{(1)} H_{TD}^{(eq)}}{EI} dz +$$

$$+ \int_0^L \frac{M_{DE}^{(1)} H_{DE}^{(eq)}}{EI} dz + \int_0^{2L} \frac{M_{EG}^{(1)} H_{EG}^{(eq)}}{EI} dz = \frac{(4x_1 - \sqrt{2}5F)L^3}{24EI} + \frac{(2x_1 - 5\sqrt{2}F + \sqrt{2}x_2)L^3}{12EI}$$

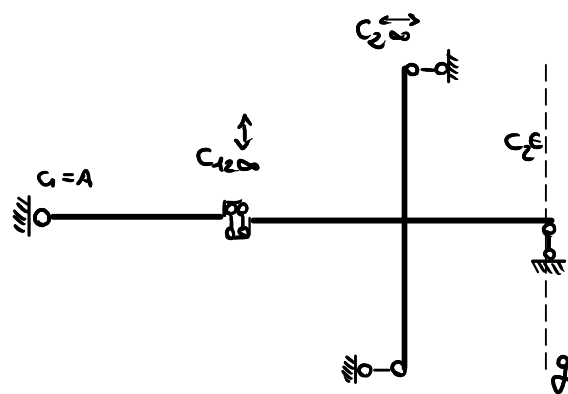
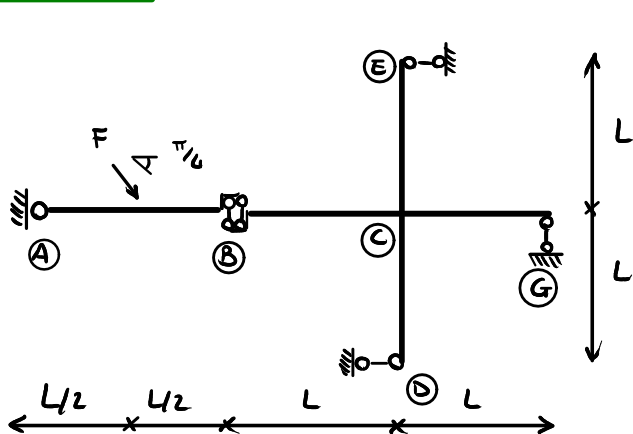
$$1 \cdot V_G = 0 = \int_0^{2L} \frac{M_{AT}^{(2)} H_{AT}^{(eq)}}{EI} dz + \int_0^L \frac{M_{TB}^{(2)} H_{TB}^{(eq)}}{EI} dz + \int_0^L \frac{M_{BC}^{(2)} H_{BC}^{(eq)}}{EI} dz + \int_0^L \frac{M_{TD}^{(2)} H_{TD}^{(eq)}}{EI} dz +$$

$$+ \int_0^L \frac{M_{DE}^{(2)} H_{DE}^{(eq)}}{EI} dz + \int_0^{2L} \frac{M_{EG}^{(2)} H_{EG}^{(eq)}}{EI} dz = - \frac{2L^3(3F - 4x_2)}{3EI} + \frac{L^3(4x_2 - 8F + \sqrt{2}x_1)}{12EI} - \frac{L^3(9F - 14x_2)}{6EI}$$

$$x_1 = \frac{215}{122} \sqrt{2} F$$

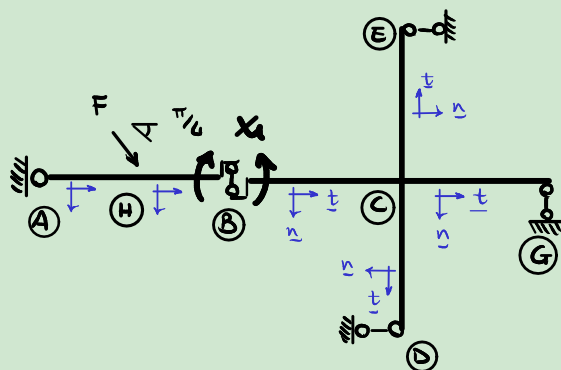
$$x_2 = \frac{185}{254} F$$

Esercizio 3



Il centro del corpo (II) non esiste, dovendo essere contemporaneamente un punto proprio sulla retta g passante per G e un punto improprio di direzione orizzontale (condizione data dai due pendoli in E e D). Pertanto il corpo (I) è una trave cerniera - doppio pendolo (esterni), sicuramente non labile. Tutta la struttura ha $l=0$ e quindi $i = -(3t-s) = 1$

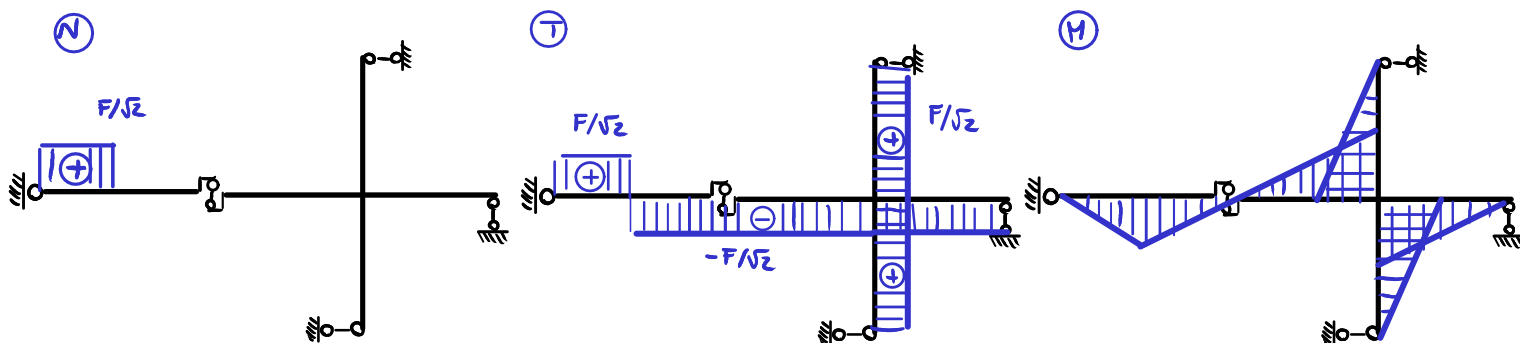
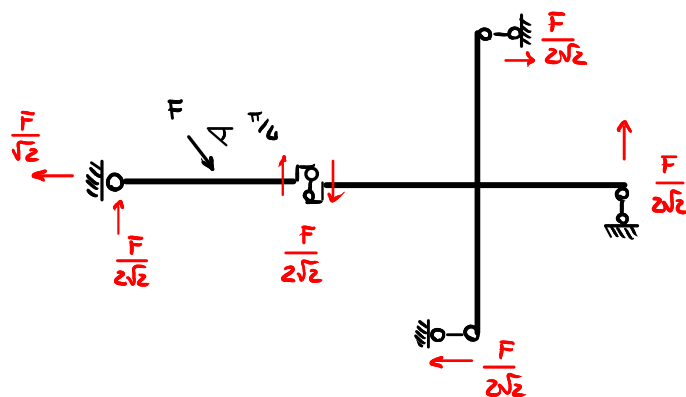
$S^{(eq)}$



CONDIZIONE DI CONGRUENZA

$$\Delta\varphi_B = 0$$

$S^{(0)}$



$$M_{AH}^{(0)} = \frac{Fz}{2\sqrt{2}}$$

$$M_{HB}^{(0)} = \frac{F}{2\sqrt{2}} \left(\frac{L}{2} - z \right)$$

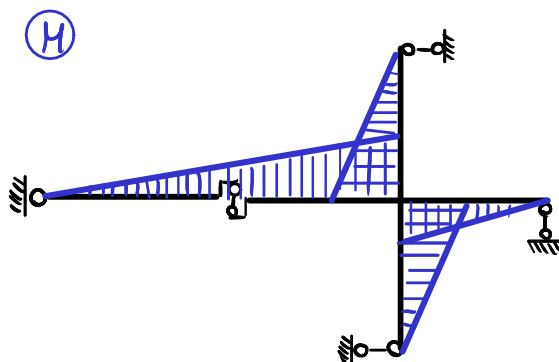
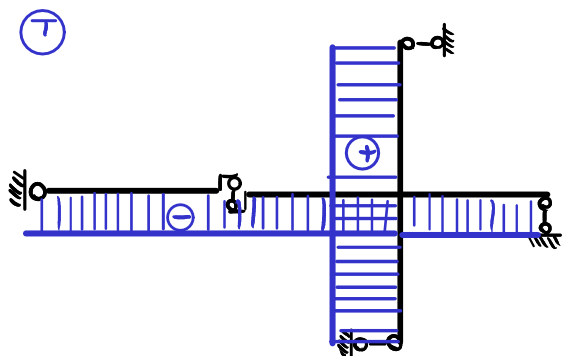
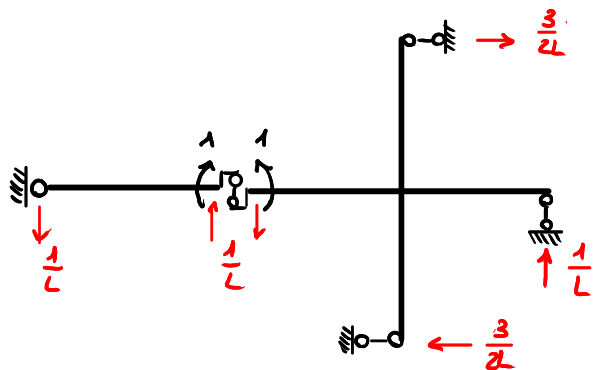
$$M_{BC}^{(0)} = -\frac{Fz}{2\sqrt{2}}$$

$$M_{CB}^{(0)} = -\frac{F(L-z)}{2\sqrt{2}}$$

$$M_{CE}^{(0)} = -\frac{F(L-z)}{2\sqrt{2}}$$

$$M_{CG}^{(0)} = \frac{F(L-z)}{2\sqrt{2}}$$

$S^{(1)}$



$$\begin{aligned} H_{AH}^{(1)} &= -\frac{z}{L} & H_{CD}^{(1)} &= -\frac{3}{2L}(L-z) \\ H_{HB}^{(1)} &= -\frac{1}{L}\left(\frac{L}{2}+z\right) & H_{CE}^{(1)} &= -\frac{3}{2L}(L-z) \\ H_{BC}^{(1)} &= -\frac{L+z}{L} & H_{CG}^{(1)} &= \frac{L-z}{L} \end{aligned}$$

Soluzione iperstatica

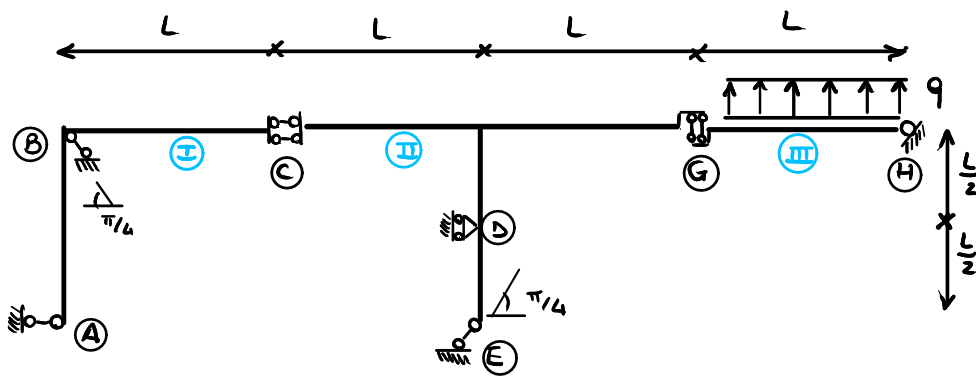
$$\delta w_e = -1(B^-) \cdot \varphi_B^{(eq)} + 1(B^+) \varphi_B^{(eq)} = 1(\varphi_B^{(eq)} - \varphi_B^{(eq)}) = 0$$

$$\begin{aligned} \delta w_i &= \int_0^{L/2} \frac{M_{AH}^{(1)} M_{AH}^{(eq)}}{EI} dz + \int_0^L \frac{M_{HB}^{(1)} M_{HB}^{(eq)}}{EI} dz + \int_0^L \frac{M_{BC}^{(1)} M_{BC}^{(eq)}}{EI} dz + \int_0^L \frac{M_{CD}^{(1)} M_{CD}^{(eq)}}{EI} dz + \\ &\quad \int_0^L \frac{M_{CE}^{(1)} M_{CE}^{(eq)}}{EI} dz + \int_0^L \frac{M_{EG}^{(1)} M_{EG}^{(eq)}}{EI} dz \end{aligned}$$

$$\delta w_e = \delta w_i \Rightarrow$$

$$X_1 = -\frac{49}{432} \sqrt{2} FL$$

Esercizio 4

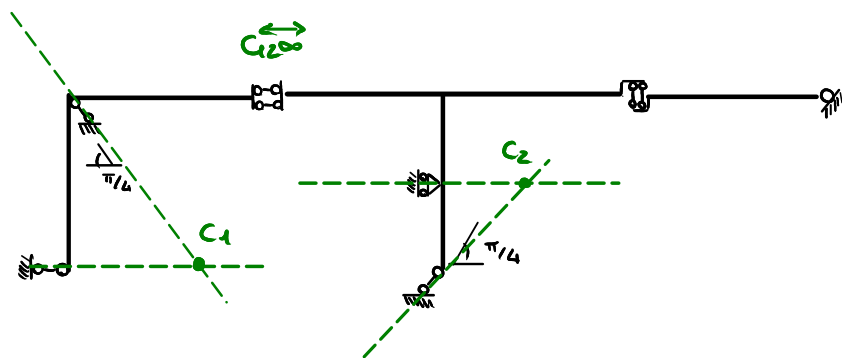


$$t = 3$$

$$s = 10$$

GRADO DI LABILITA'

- I corpi $\textcircled{\text{I}}$ e $\textcircled{\text{II}}$ costituiscono un insieme di corpi fermo:



C_1 e C_2 , se esistono, debbono trovarsi alla intersezione tra le rette d'azione dei pendoli

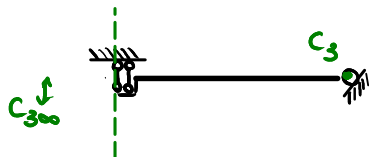
C_2 tuttavia, improprio nella direzione orizzontale, non può allineare C_1 e C_2



L'allineamento dei centri C_1, C_2, C_3 , necessario perché la struttura sia labile (1 teorema di allineamento) è impossibile: nessuno dei centri esiste, ovvero

- $C_1 \nexists$: lo spostamento assoluto del corpo $\textcircled{\text{I}}$ è nullo ($\textcircled{\text{I}}$ è fermo)
- $C_2 \nexists$: lo spostamento assoluto del corpo $\textcircled{\text{II}}$ è nullo ($\textcircled{\text{II}}$ è fermo)
- $C_3 \nexists$: lo spostamento relativo dei corpi $\textcircled{\text{I}}$ e $\textcircled{\text{II}}$ è nullo ($\textcircled{\text{I}}$ e $\textcircled{\text{II}}$ si comportano come se fossero un unico corpo)

- Il corpo $\textcircled{\text{III}}$ è pertanto riguardabile come segue:



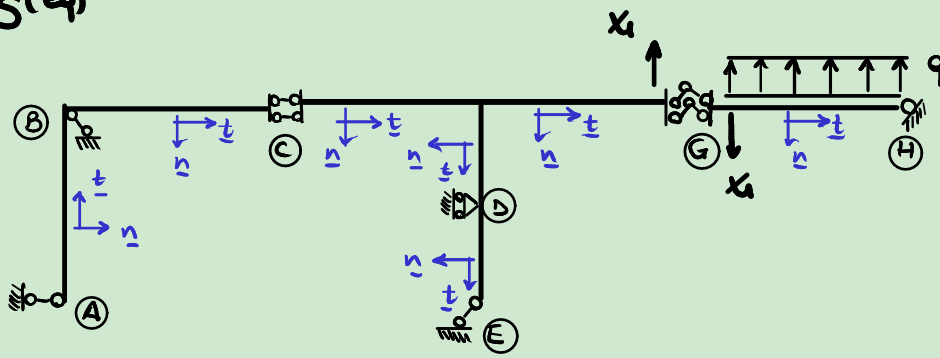
$\Rightarrow C_3 \nexists$ non potendo essere contemporaneamente un punto proprio a posizione fissata (H) e un punto improprio (G)

il corpo $\textcircled{\text{III}}$ è fermo

- I centri relativi C_{12}, C_{13}, C_{23} non possono essere allineati su una stessa direzione

Pertanto l'intera struttura è ferma ($\ell = 0$) e $i = -(3t - s) = 1$

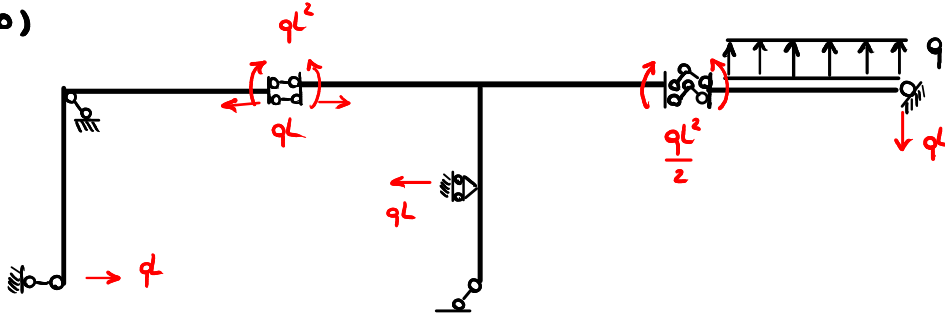
$S^{(q)}$



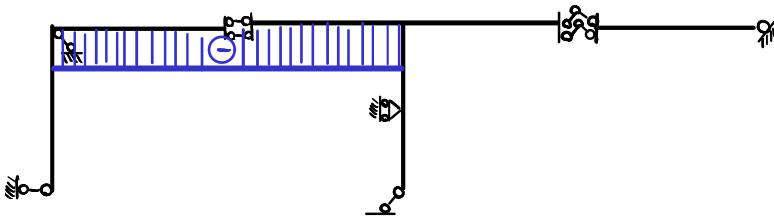
CONDIZIONE DI CONGRUENZA

$$\Delta V_G = 0$$

$S^{(0)}$



(N)



$$N_{AB}^{(0)} = 0$$

$$N_{DE}^{(0)} = 0$$

$$N_{BC}^{(0)} = -qL$$

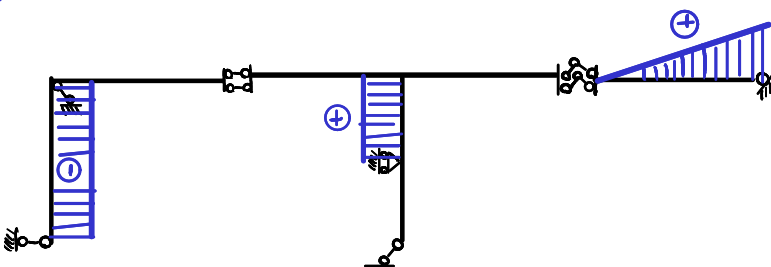
$$N_{TG}^{(0)} = 0$$

$$N_{CT}^{(0)} = -qL$$

$$N_{GH}^{(0)} = 0$$

$$N_{TD}^{(0)} = 0$$

(T)



$$T_{AB}^{(0)} = -qL$$

$$T_{DE}^{(0)} = 0$$

$$T_{BC}^{(0)} = 0$$

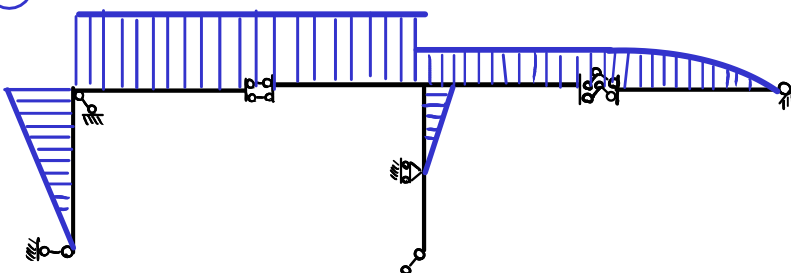
$$T_{TG}^{(0)} = 0$$

$$T_{CT}^{(0)} = 0$$

$$T_{GH}^{(0)} = qz$$

$$T_{TD}^{(0)} = qL$$

(M)



$$M_{AB}^{(0)} = -qL^2$$

$$M_{DE}^{(0)} = 0$$

$$M_{BC}^{(0)} = -qL^2$$

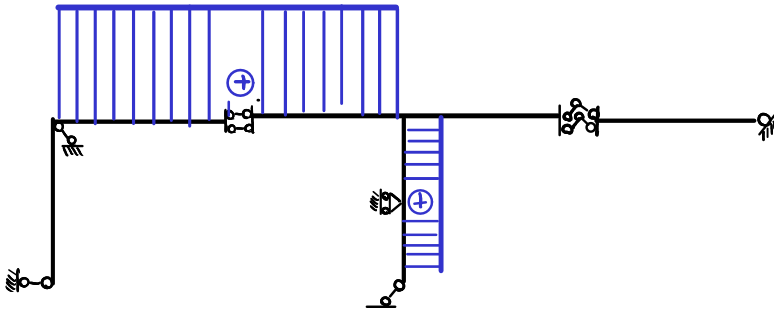
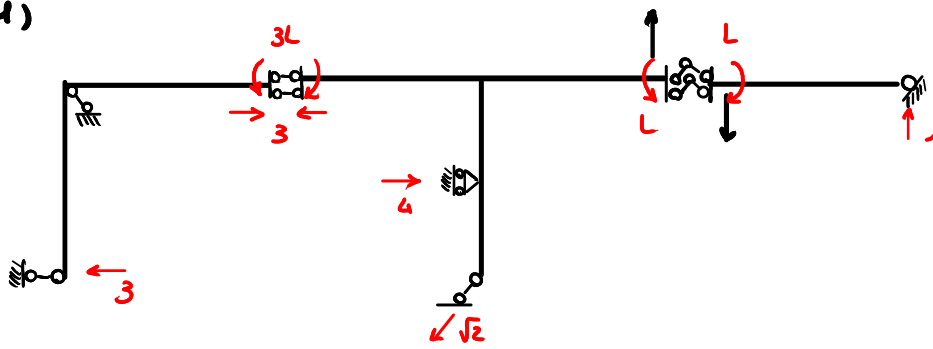
$$M_{TG}^{(0)} = -\frac{qL^2}{2}$$

$$M_{CT}^{(0)} = -qL^2$$

$$M_{GH}^{(0)} = -\frac{qL^2}{2} + \frac{qz^2}{2}$$

$$M_{TD}^{(0)} = qL\left(\frac{L}{2} - z\right)$$

S⁽¹⁾

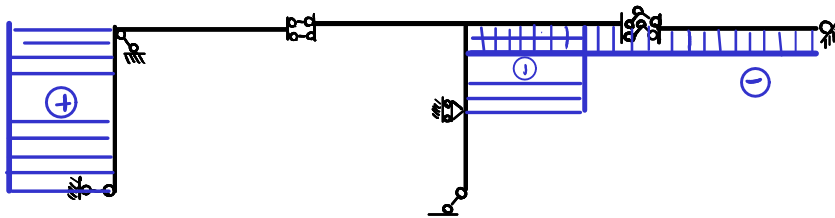


$$N_{AB}^{(1)} = 0 \quad N_{DE}^{(1)} = 1$$

$$N_{BC}^{(1)} = 3 \quad N_{TG}^{(1)} = 0$$

$$N_{CT}^{(1)} = 3 \quad N_{GH}^{(1)} = 0$$

$$N_{TD}^{(1)} = 1$$

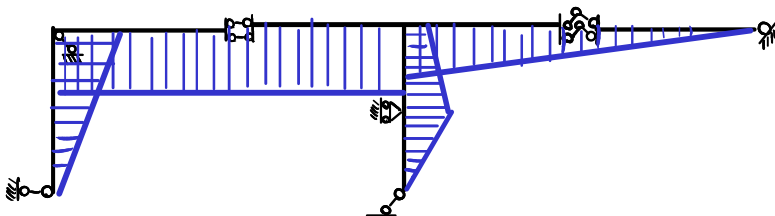


$$T_{AB}^{(1)} = 3 \quad T_{DE}^{(1)} = 0$$

$$T_{BC}^{(1)} = 0 \quad T_{TG}^{(1)} = -1$$

$$T_{CT}^{(1)} = 0 \quad T_{GH}^{(1)} = -1$$

$$T_{TD}^{(1)} = -1$$



$$M_{AB}^{(1)} = 3z \quad M_{DE}^{(1)} = -\left(\frac{L}{2} - z\right)$$

$$M_{BC}^{(1)} = 3L \quad M_{TG}^{(1)} = L + (L - z)$$

$$M_{CT}^{(1)} = 3L \quad M_{GH}^{(1)} = (L - z)$$

$$M_{TD}^{(1)} = 6\left(\frac{L}{2} - z\right) - (L - z)$$

Soluzione iperstatica

$$dve = 1 \cdot \Delta V_G^{(eq)} = 0$$

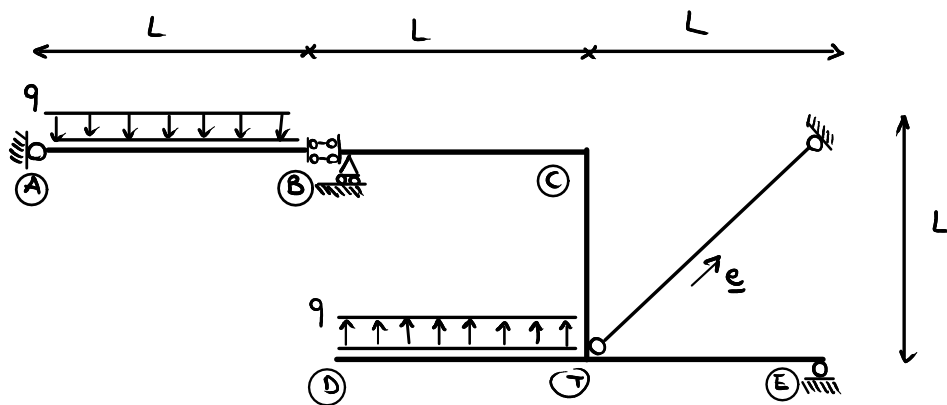
$$dvi = \int_0^L \frac{M_{AB}^{(1)} M_{AB}^{(eq)}}{EI} dz + \int_0^L \frac{M_{BC}^{(1)} M_{BC}^{(eq)}}{EI} dz + \int_0^L \frac{M_{CT}^{(1)} M_{CT}^{(eq)}}{EI} dz + \int_0^{L/2} \frac{M_{TD}^{(1)} M_{TD}^{(eq)}}{EI} dz +$$

$$\int_0^{L/2} \frac{M_{DE}^{(1)} M_{DE}^{(eq)}}{EI} dz + \int_0^L \frac{M_{EG}^{(1)} M_{EG}^{(eq)}}{EI} dz + \int_0^L \frac{M_{GH}^{(1)} M_{GH}^{(eq)}}{EI} dz$$

$$dve = dvi \Rightarrow$$

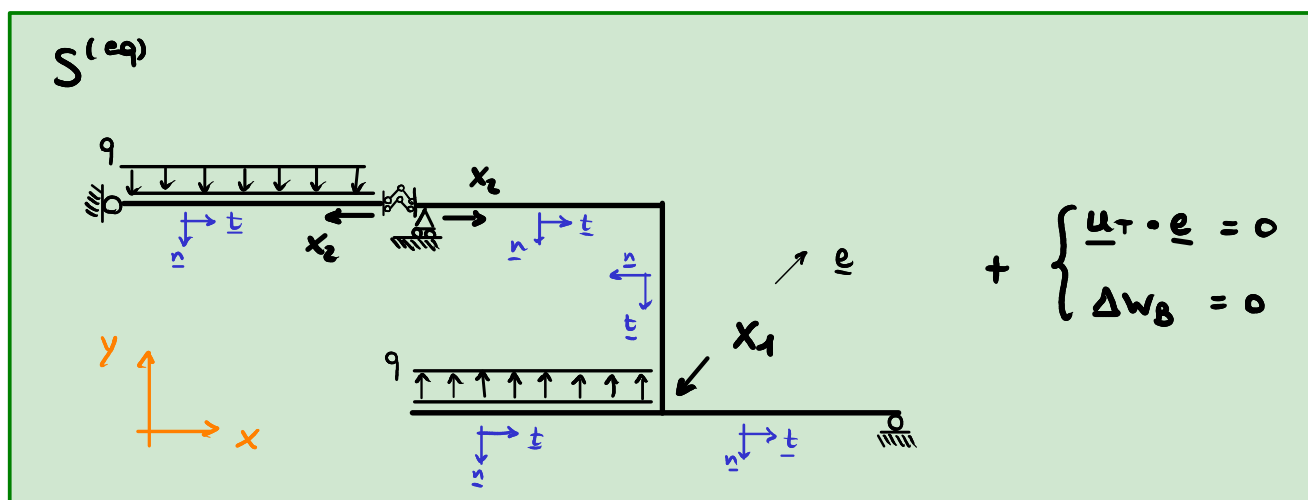
$$X_1 = \frac{379}{1644} qL$$

Esercizio 5



$$l = 0$$

$$i = 2$$

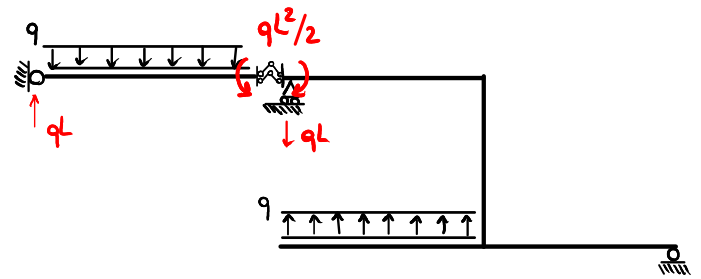
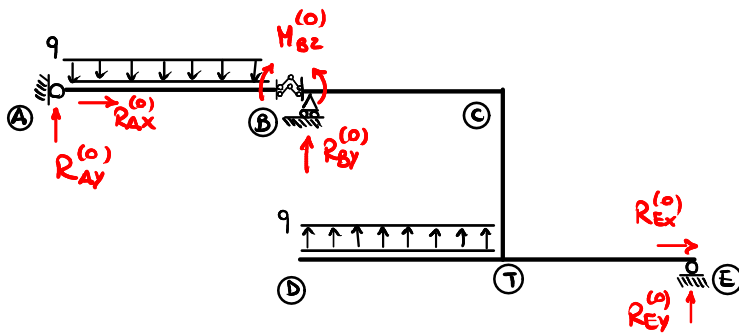


Pensando T appartenente al tratto TE e decomponendo il vettore e nel riferimento $\{\underline{t}_{TE}, \underline{n}_{TE}\}$ la prima equazione di congruenza diventa:

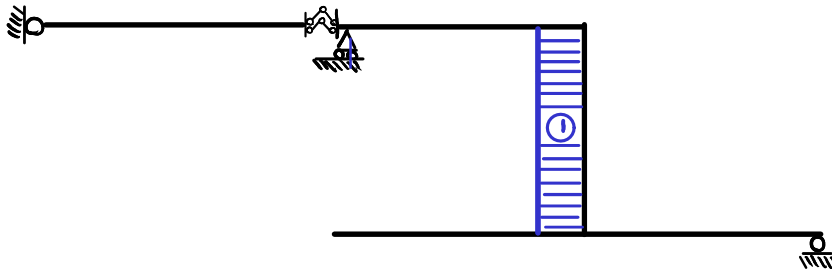
$$\underline{e} = \frac{\sqrt{2}}{2} (-\underline{t}_{TE} + \underline{n}_{TE})$$

$$\underline{u}_T \cdot \underline{e} = (w_T \underline{t}_{TE} + v_T \underline{n}_{TE}) \cdot \frac{\sqrt{2}}{2} (-\underline{t}_{TE} + \underline{n}_{TE}) = -\frac{\sqrt{2}}{2} w_T + \frac{\sqrt{2}}{2} v_T$$

$S^{(0)}$



N



$$N_{AB}^{(0)} = 0$$

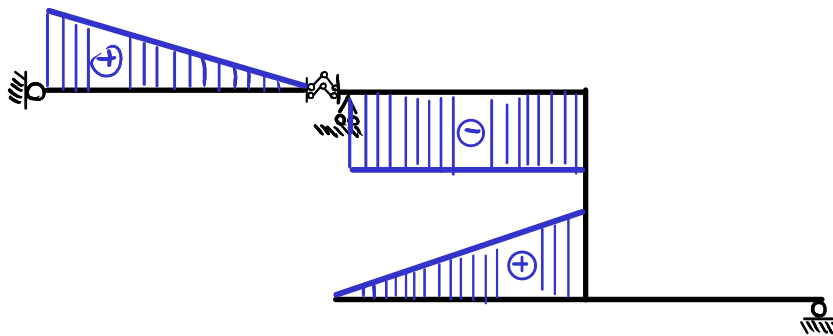
$$N_{BC}^{(0)} = 0$$

$$N_{CD}^{(0)} = -qL$$

$$N_{DE}^{(0)} = 0$$

$$N_{\Sigma E}^{(0)} = 0$$

T



$$T_{AB}^{(0)} = qL - qz$$

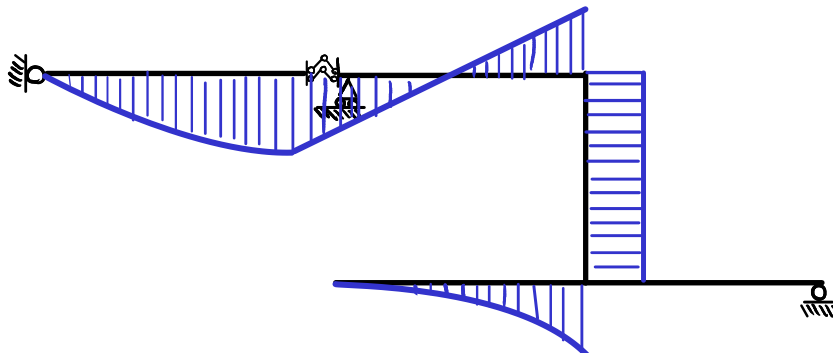
$$T_{BC}^{(0)} = -qL$$

$$T_{CD}^{(0)} = 0$$

$$T_{DE}^{(0)} = qz$$

$$T_{\Sigma E}^{(0)} = 0$$

M



$$M_{AB}^{(0)} = qLz - \frac{qz^2}{2}$$

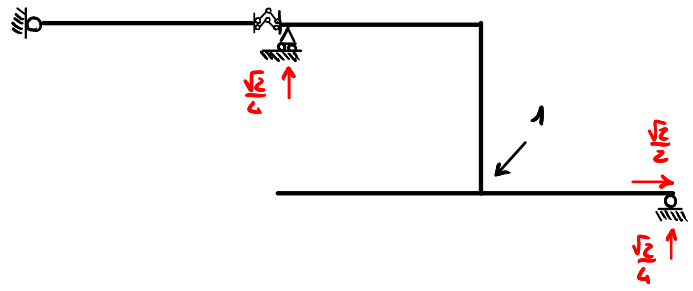
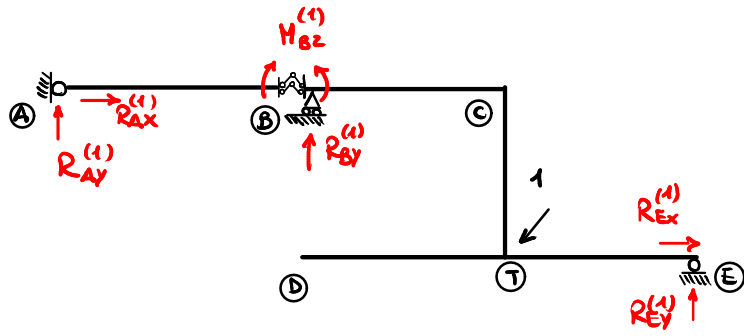
$$M_{BC}^{(0)} = -qLz + \frac{qL^2}{2}$$

$$M_{CD}^{(0)} = -\frac{qL^2}{2}$$

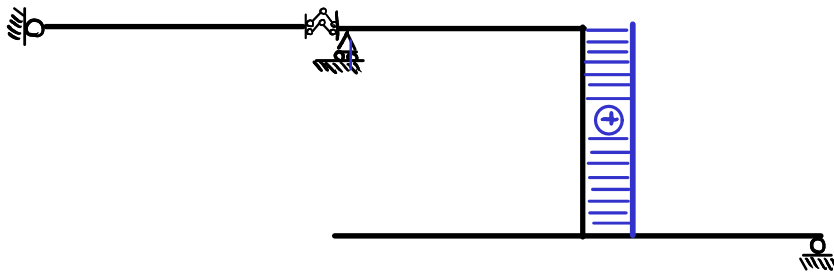
$$M_{DE}^{(0)} = +\frac{qL^2}{2}$$

$$M_{\Sigma E}^{(0)} = 0$$

$S^{(1)}$



N



$$N_{AB}^{(1)} = 0$$

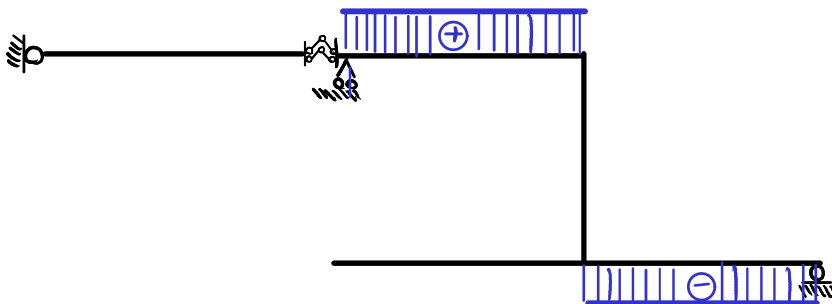
$$N_{BC}^{(1)} = 0$$

$$N_{CD}^{(1)} = \frac{\sqrt{2}}{4}$$

$$N_{TD}^{(1)} = 0$$

$$N_{DE}^{(1)} = 0$$

T



$$T_{AB}^{(1)} = 0$$

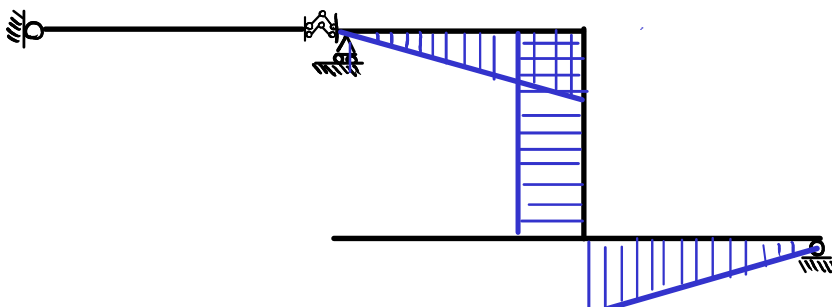
$$T_{BC}^{(1)} = \frac{\sqrt{2}}{4}$$

$$T_{CD}^{(1)} = 0$$

$$T_{TD}^{(1)} = 0$$

$$T_{DE}^{(1)} = -\frac{\sqrt{2}}{4}$$

M



$$M_{AB}^{(1)} = 0$$

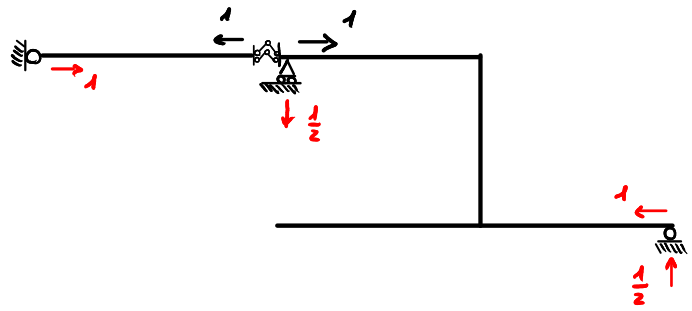
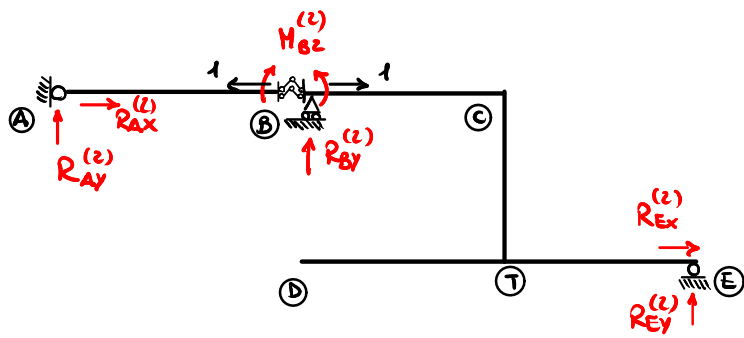
$$M_{BC}^{(1)} = \frac{\sqrt{2}}{4} z$$

$$M_{CD}^{(1)} = \frac{\sqrt{2}}{4} L$$

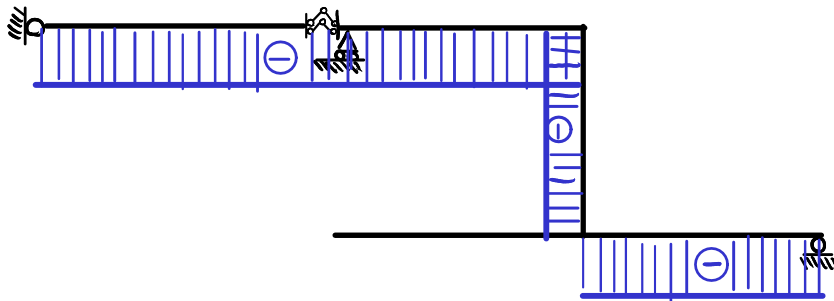
$$M_{TD}^{(1)} = 0$$

$$M_{DE}^{(1)} = \frac{\sqrt{2}}{4} (L - z)$$

$S^{(2)}$



N



$$N_{AB}^{(2)} = -1$$

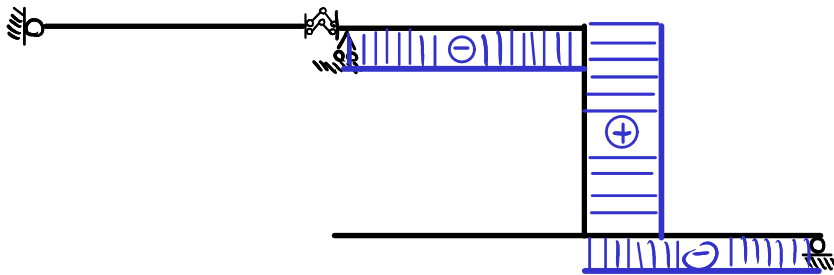
$$N_{BC}^{(2)} = -1$$

$$N_{CT}^{(2)} = -1/2$$

$$N_{TD}^{(2)} = 0$$

$$N_{DE}^{(2)} = -1$$

T



$$T_{AB}^{(2)} = 0$$

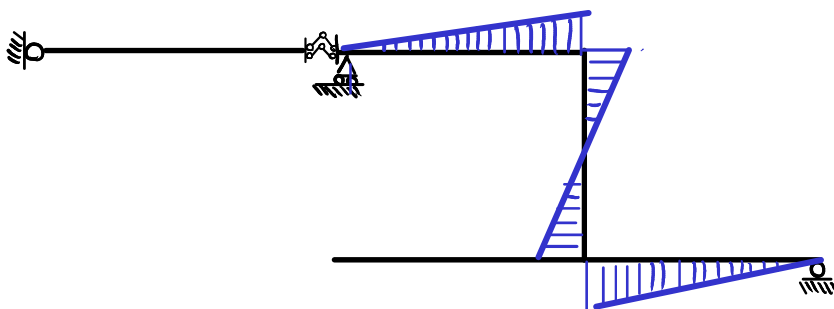
$$T_{BC}^{(2)} = -1/2$$

$$T_{CT}^{(2)} = 1$$

$$T_{TD}^{(2)} = 0$$

$$T_{DE}^{(2)} = -1/2$$

M



$$M_{AB}^{(2)} = 0$$

$$M_{BC}^{(2)} = -\frac{x}{2}$$

$$M_{CT}^{(2)} = -\frac{L}{2} + x$$

$$M_{TD}^{(2)} = 0$$

$$M_{DE}^{(2)} = \frac{L-x}{2}$$

Soluzione iperstatica

TLV 1

$$\delta v_e = 1 \cdot [\underline{u}_T^{(eq)} \cdot \underline{e}] = 0$$

$$\delta v_i = \int_0^L \frac{M_{AB}^{(1)} M_{AB}^{(eq)}}{EI} dz + \int_0^L \frac{M_{BC}^{(1)} M_{BC}^{(eq)}}{EI} dz + \int_0^L \frac{M_{CT}^{(1)} M_{CT}^{(eq)}}{EI} dz + \int_0^L \frac{M_{DT}^{(1)} M_{DT}^{(eq)}}{EI} dz + \int_0^L \frac{M_{TE}^{(1)} M_{TE}^{(eq)}}{EI} dz$$

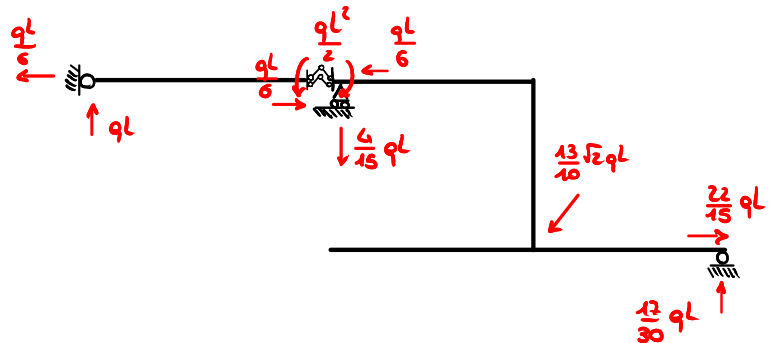
TLV 2

$$\delta v_e = 1 \cdot \Delta W_B^{(eq)} = 0$$

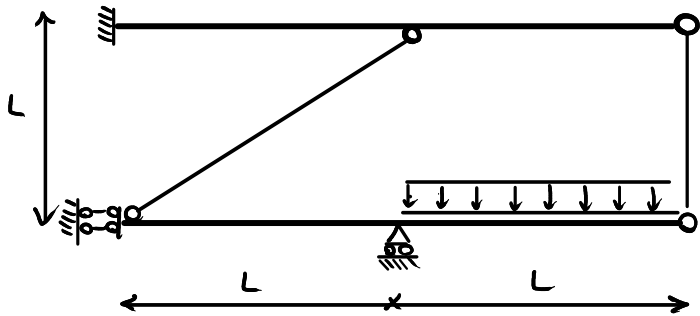
$$\delta v_i = \int_0^L \frac{M_{AB}^{(2)} M_{AB}^{(eq)}}{EI} dz + \int_0^L \frac{M_{BC}^{(2)} M_{BC}^{(eq)}}{EI} dz + \int_0^L \frac{M_{CT}^{(2)} M_{CT}^{(eq)}}{EI} dz + \int_0^L \frac{M_{DT}^{(2)} M_{DT}^{(eq)}}{EI} dz + \int_0^L \frac{M_{TE}^{(2)} M_{TE}^{(eq)}}{EI} dz$$

$$\delta v_e = \delta v_i \quad \Leftrightarrow$$

$$\begin{aligned} x_1 &= \frac{13}{10} \sqrt{2} qL \\ x_2 &= -\frac{qL}{6} \end{aligned}$$



Esercizio 6

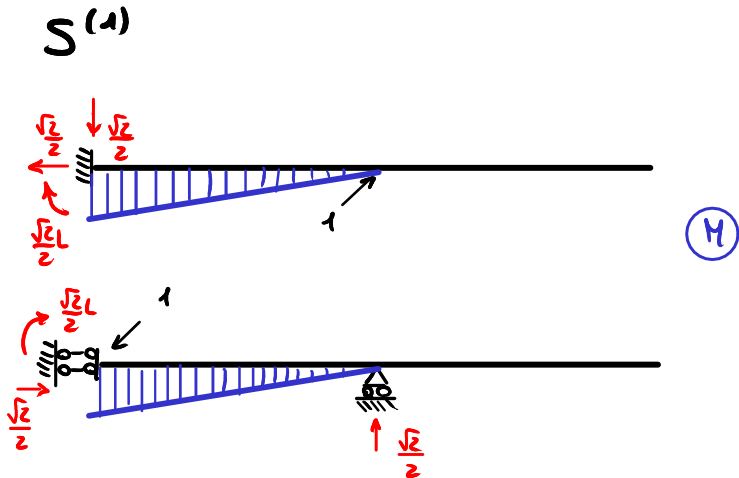
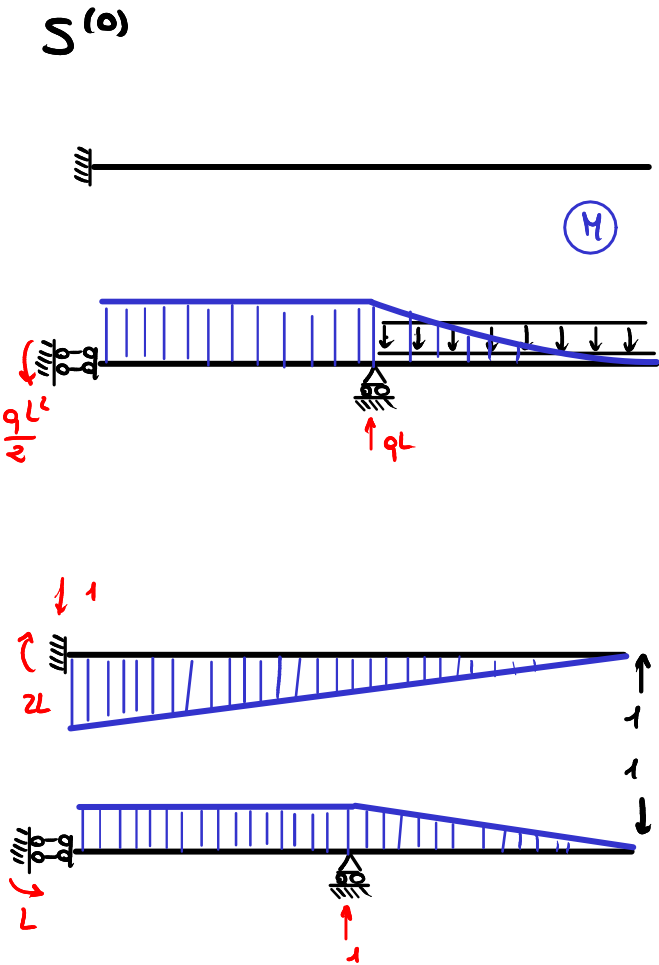


$$l = 0$$

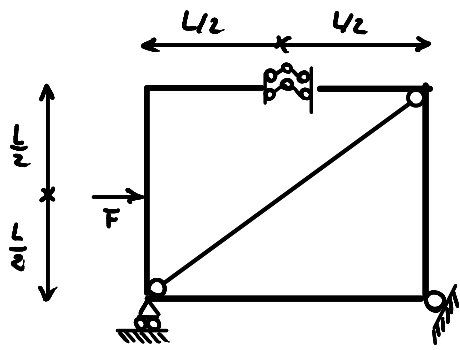
$$i = 2$$

$S^{(eq)}$

$(\underline{u}_B - \underline{u}_A) \cdot \underline{e}_1 = 0$
 $+ (\underline{u}_C - \underline{u}_D) \cdot \underline{e}_2 = 0$



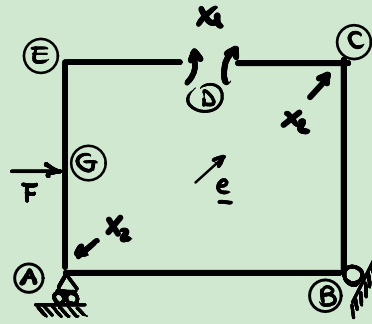
Esercizio 7



$$l = 0$$

$$i = 2$$

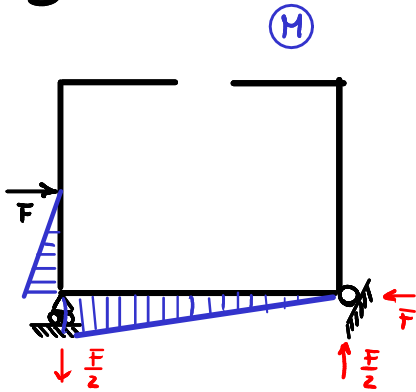
$S^{(eq)}$



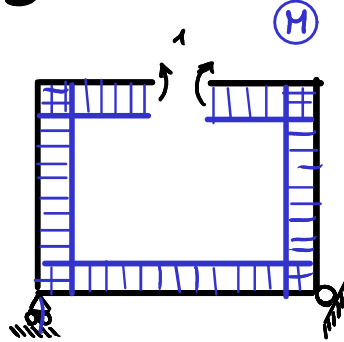
$$\Delta\varphi_D = 0$$

$$(\underline{u}_C - \underline{u}_A) \cdot \underline{e} = 0$$

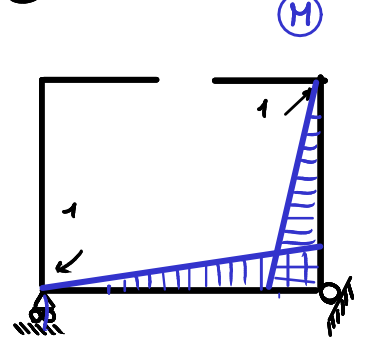
$S^{(0)}$



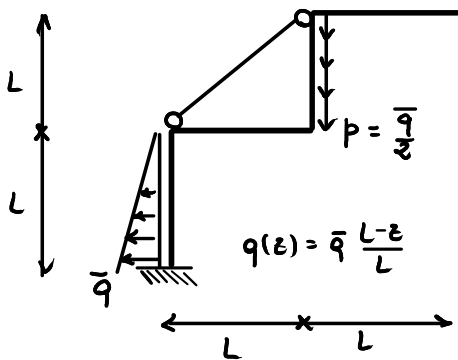
$S^{(1)}$



$S^{(2)}$



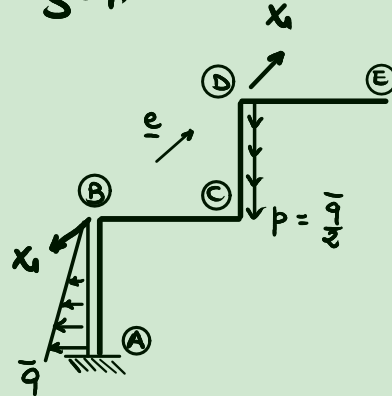
Esercizio 8



$$l = 0$$

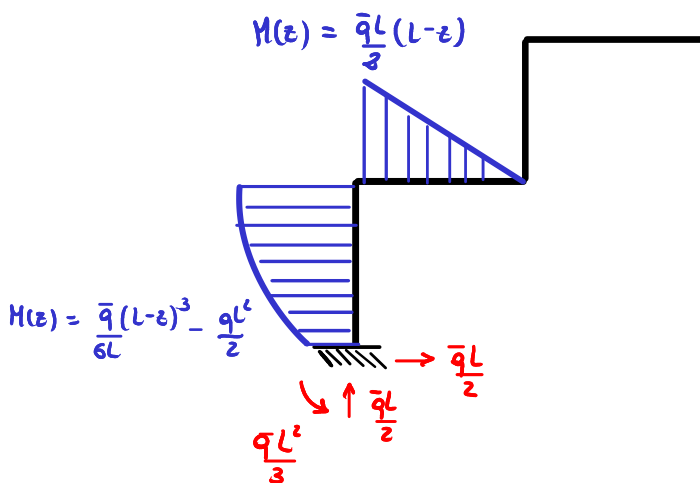
$$i = 1$$

$S^{(eq)}$

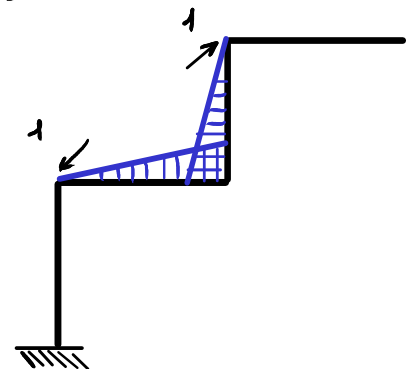


$$+ (\underline{u}_D - \underline{u}_B) \cdot \underline{e} = 0$$

$S^{(0)}$

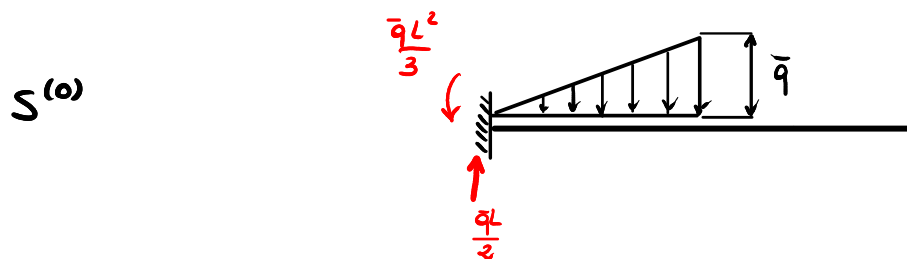
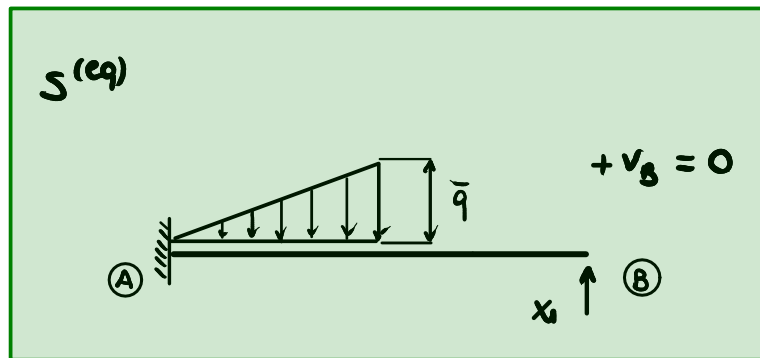
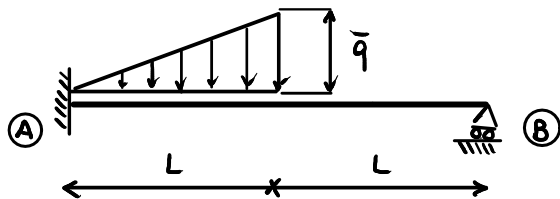


$S^{(1)}$



Esercizio 10

Trascurare la deformabilità assiale
e quella tagliante



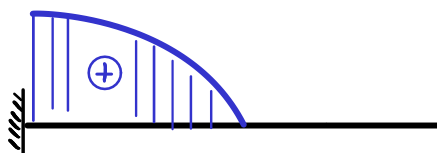
(N)



$$N_{AB}^{(0)} = 0$$

$$N_{BC}^{(0)} = 0$$

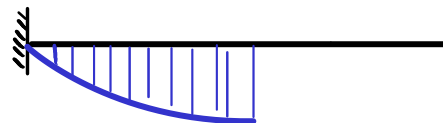
(T)



$$T_{AB}^{(0)} = \frac{\bar{q}L}{2} - q(z) \frac{z}{2}$$

$$T_{BC}^{(0)} = 0$$

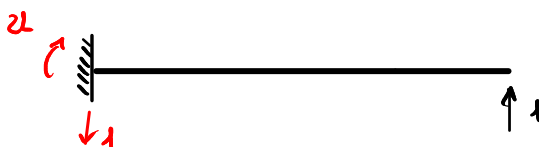
(M)



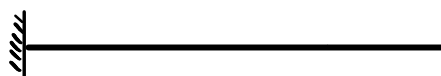
$$M_{AB}^{(0)} = \frac{\bar{q}Lz}{2} - q(z) \frac{z^2}{6} - \frac{\bar{q}L^2}{3}$$

$$M_{BC}^{(0)} = 0$$

$S^{(1)}$



(N)



$$N_{AB}^{(1)} = 0$$

$$N_{BC}^{(1)} = 0$$

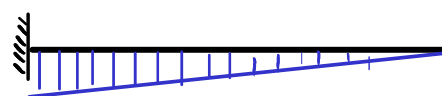
(T)



$$T_{AB}^{(1)} = -1$$

$$T_{BC}^{(1)} = -1$$

(M)



$$M_{AB}^{(1)} = 2L - z$$

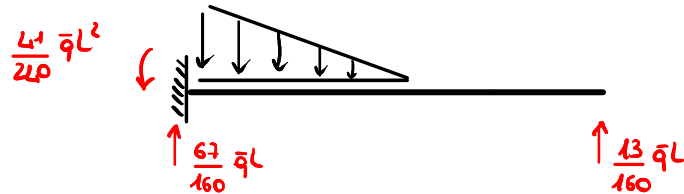
$$M_{BC}^{(1)} = L - z$$

Soluzione iperstatica

$$\delta v_e = 1 \cdot v_B^{(eq)} = 0$$

$$\delta v_i = \int_0^L \frac{M_{AB}^{(1)} M_{AB}^{(eq)}}{EI} dz + \int_0^L \frac{M_{BC}^{(1)} M_{BC}^{(eq)}}{EI} dz$$

$$\delta v_e = \delta v_i \Rightarrow x_1 = \frac{13}{160} \bar{q} L$$



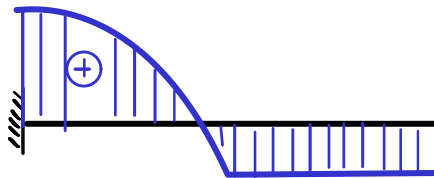
(N)



$$N_{AB} = 0$$

$$N_{BC} = 0$$

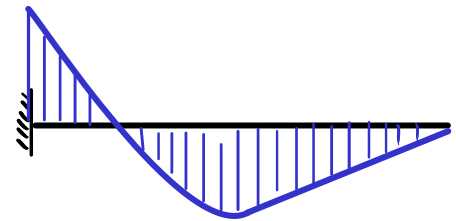
(T)



$$T_{AB} = \frac{67}{160} \bar{q} L - \frac{\bar{q} z^2}{2L}$$

$$T_{BC} = -\frac{13}{160} \bar{q} L$$

(M)

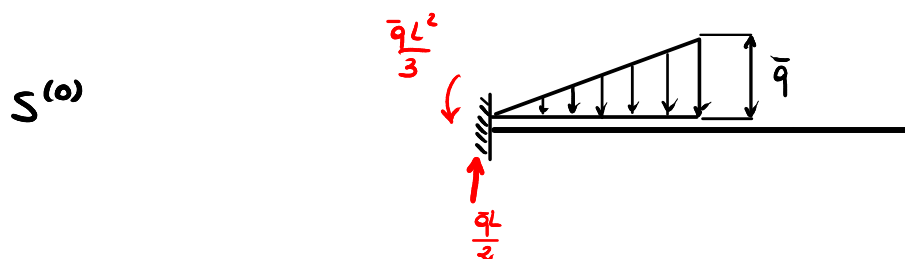
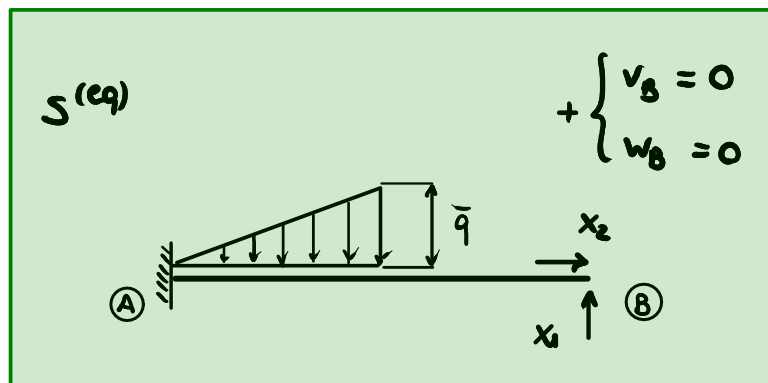
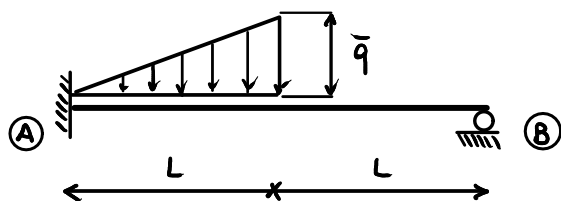


$$M_{AB} = \frac{13}{160} \bar{q} L (2L - z) - \bar{q} \left(\frac{L^2}{3} + \frac{z^3}{6L} - \frac{Lz}{2} \right)$$

$$M_{BC} = \frac{13}{160} \bar{q} L (L - z)$$

Esercizio 11

Trascurare la deformabilità assiale
e quella tagliante



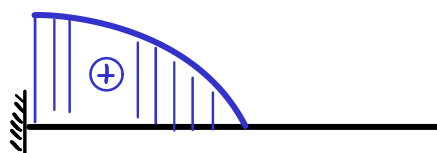
(N)



$$N_{AB}^{(0)} = 0$$

$$N_{BC}^{(0)} = 0$$

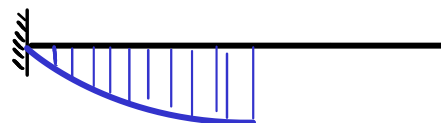
(T)



$$T_{AB}^{(0)} = \frac{\bar{q}L}{2} - q(z) \frac{z}{2}$$

$$T_{BC}^{(0)} = 0$$

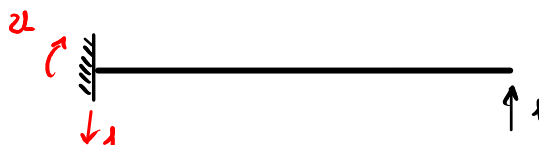
(M)



$$M_{AB}^{(0)} = \frac{\bar{q}Lz}{2} - q(z) \frac{z^2}{6} - \frac{\bar{q}L^2}{3}$$

$$M_{BC}^{(0)} = 0$$

$S^{(1)}$



(N)



$$N_{AB}^{(1)} = 0$$

$$N_{BC}^{(1)} = 0$$

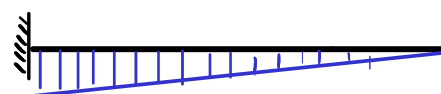
(T)



$$T_{AB}^{(1)} = -1$$

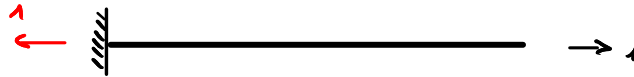
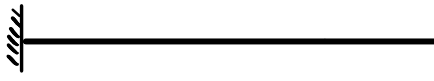
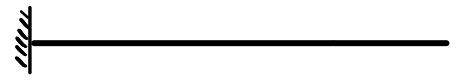
$$T_{BC}^{(1)} = -1$$

(M)



$$M_{AB}^{(1)} = 2L - z$$

$$M_{BC}^{(1)} = L - z$$

$S^{(2)}$  (N)  (T)  (M) 

$$N_{AB}^{(2)} = 1$$

$$N_{BC}^{(2)} = 1$$

$$T_{AB}^{(2)} = 0$$

$$T_{BC}^{(2)} = 0$$

$$M_{AB}^{(2)} = 0$$

$$M_{BC}^{(2)} = 0$$

Soluzione iperstatica

$$\begin{cases} dvc = 1 \cdot v_B^{(eq)} = 0 \\ dvi = \int_0^L \frac{M_{AB}^{(1)} M_{AB}^{(eq)}}{EI} dz + \int_0^L \frac{M_{BC}^{(1)} M_{BC}^{(eq)}}{EI} dz \end{cases}$$

$$dvc = dvi \Rightarrow$$

$$X_1 = \frac{13}{160} \bar{q}L$$

$$X_2 \text{ qualsiasi}$$

$$\begin{cases} dvc = 1 \cdot w_B^{(eq)} = 0 \\ dvi = \int_0^L \frac{\cancel{M_{AB}^{(2)}} M_{AB}^{(eq)}}{EI} dz + \int_0^L \frac{\cancel{M_{BC}^{(2)}} M_{BC}^{(eq)}}{EI} dz \end{cases}$$

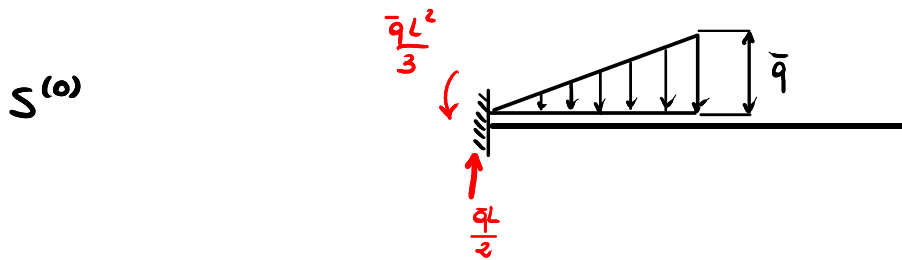
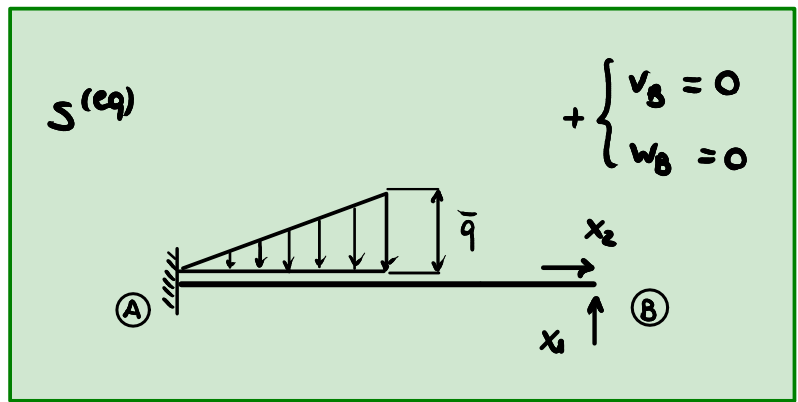
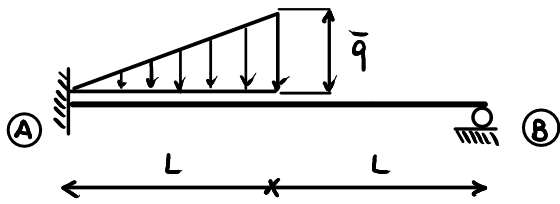
In questo caso X_2 resta indeterminata perché si è trascurata la deformabilità assiale: la iperstatica associata a X_2 non può essere risolta fintanto che si assume $EA \rightarrow \infty$, a riprova del fatto che

Per risolvere una struttura iperstatica bisogna abbandonare il modello di corpo rigido ($EI, EA \rightarrow \infty$) ed assumere l'elasticità (lineare) dei corpi che la costituiscono

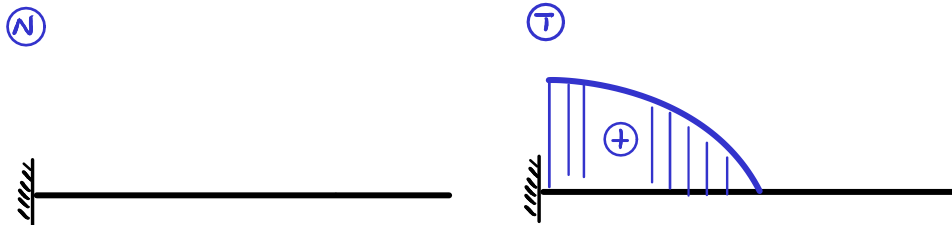
La soluzione qui ottenuta, indeterminata sul problema assiale, è tuttavia identica alla precedente per quanto riguarda il problema flessionale (stessa X_1 dell'esempio 10).

Esercizio 12

Trascurare la deformabilità tagliante
ma non quella assiale



$$q(z) = \bar{q} \frac{z}{L}$$

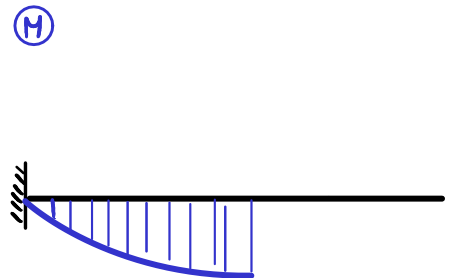


$$N_{AB}^{(0)} = 0$$

$$N_{BC}^{(0)} = 0$$

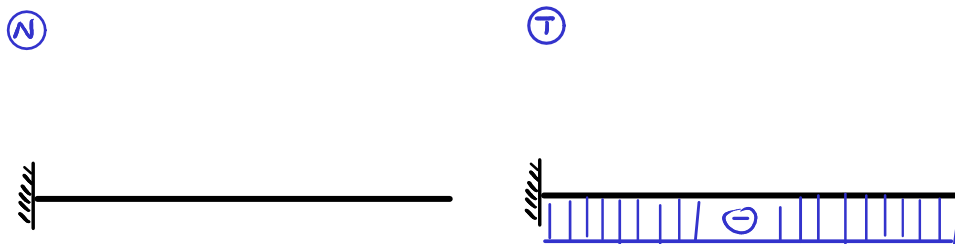
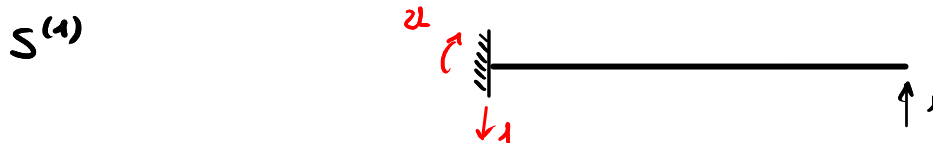
$$T_{AB}^{(0)} = \frac{\bar{q}L}{2} - q(z) \frac{z}{2}$$

$$T_{BC}^{(0)} = 0$$



$$M_{AB}^{(0)} = \frac{\bar{q}Lz}{2} - q(z) \frac{z^2}{6} - \frac{\bar{q}L^2}{3}$$

$$M_{BC}^{(0)} = 0$$

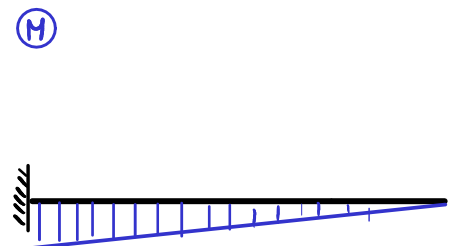


$$N_{AB}^{(1)} = 0$$

$$N_{BC}^{(1)} = 0$$

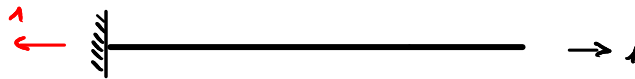
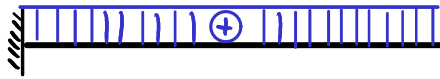
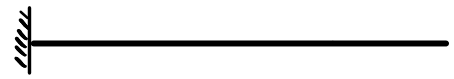
$$T_{AB}^{(1)} = -1$$

$$T_{BC}^{(1)} = -1$$



$$M_{AB}^{(1)} = 2L - z$$

$$M_{BC}^{(1)} = L - z$$

$S^{(2)}$  (N)  (T)  (M) 

$$N_{AB}^{(2)} = 1$$

$$N_{BC}^{(2)} = 1$$

$$T_{AB}^{(2)} = 0$$

$$T_{BC}^{(2)} = 0$$

$$M_{AB}^{(2)} = 0$$

$$M_{BC}^{(2)} = 0$$

Soluzione iperstatica

$$TLV1 \quad \begin{cases} \delta v_e = 1 \cdot v_B^{(eq)} = 0 \\ \delta v_i = \int_0^L \frac{N_{AB}^{(1)} N_{AB}^{(eq)}}{EA} dz + \int_0^L \frac{N_{BC}^{(1)} N_{BC}^{(eq)}}{EA} dz + \int_0^L \frac{M_{AB}^{(1)} M_{AB}^{(eq)}}{EI} dz + \int_0^L \frac{M_{BC}^{(1)} M_{BC}^{(eq)}}{EI} dz \end{cases}$$

$$TLV2 \quad \begin{cases} \delta v_e = 1 \cdot w_B^{(eq)} = 0 \\ \delta v_i = \int_0^L \frac{N_{AB}^{(2)} N_{AB}^{(eq)}}{EI} dz + \int_0^L \frac{N_{BC}^{(2)} N_{BC}^{(eq)}}{EI} dz + \int_0^L \frac{\overset{=0}{\cancel{M_{AB}^{(2)}}} M_{AB}^{(eq)}}{EI} dz + \int_0^L \frac{\overset{=0}{\cancel{M_{BC}^{(2)}}} M_{BC}^{(eq)}}{EI} dz \end{cases}$$

$$x_1 = \frac{13}{160} \bar{q} L$$

$$x_2 = 0$$

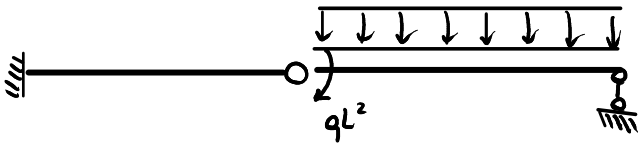
Soluzione di strutture iperstatiche: linea elastica (metodo degli spostamenti)

Per le strutture assegnate risolvere le equazioni di linea elastica al IV ordine ($EI v'''' = q$, $EA w'' = -p$) scrivendone gli integrali generali e le rispettive condizioni al contorno. Si trascuri la deformabilità tagliante (modello di Eulero-Bernoulli).

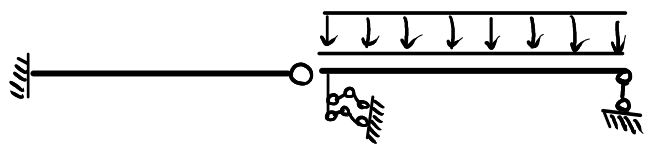
Tracciare una deformata qualitativa della struttura, basandosi sui diagrammi di M , N .

Se non diversamente specificato, sia la lunghezza di ciascun tratto pari a L .

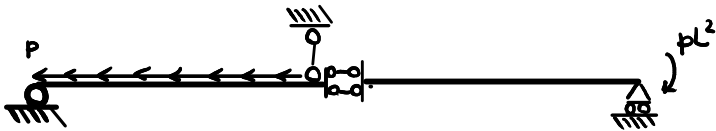
①



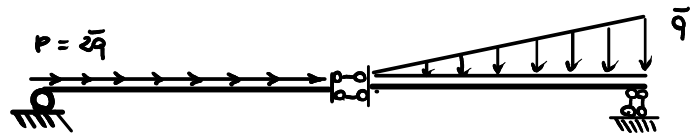
②



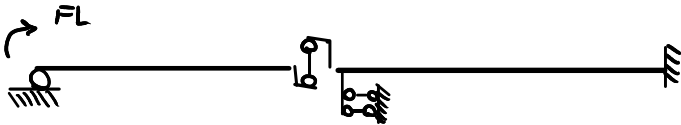
③



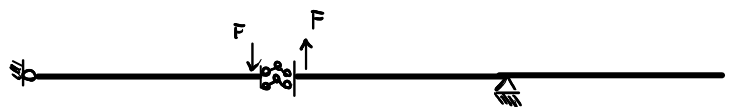
④



⑤

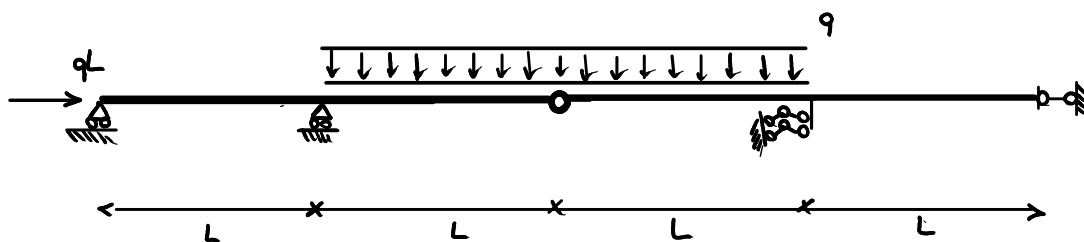


⑥

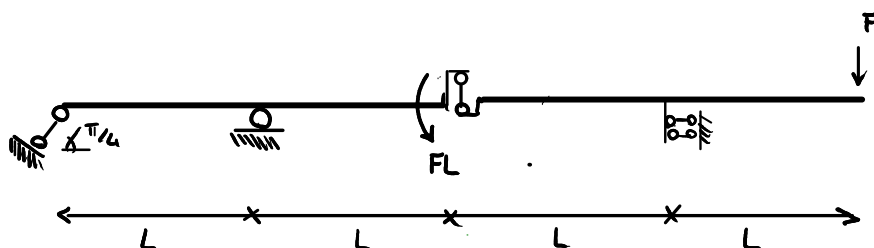


Scrivere le condizioni al contorno per le seguenti strutture, valutando di volta in volta se è necessario includere quelle del problema associato ($EA w'' = -p$)

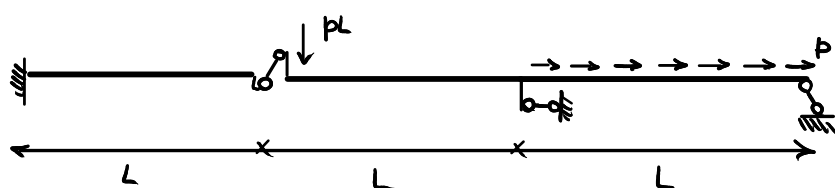
7



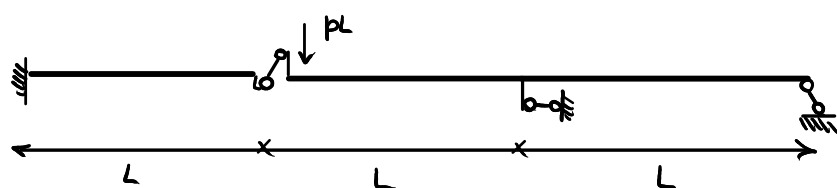
8



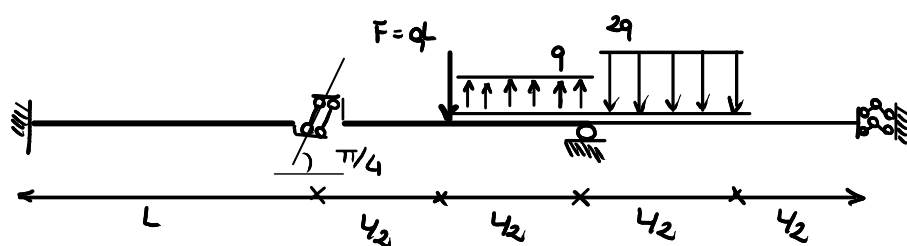
9



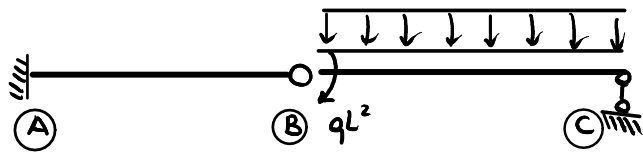
10



11



ESERCIZIO 1



N° corpi : 2

N° tratti : 2

$$EA w_{AB}'' = -p(z) = 0$$

$$EA w_{BC}'' = -p(z) = 0$$

$$EI v_{AB}'''' = q(z) = 0$$

$$EI v_{BC}'''' = q(z) = q$$

INTEGRALI GENERALI

$$EI v_{AB}'''' = c_1$$

$$EI v_{BC}'''' = qz + c_5$$

$$EI v_{AB}''' = c_2 z + c_2$$

$$EI v_{BC}''' = \frac{qz^2}{2} + c_5 z + c_6$$

$$EI v_{AB}'' = \frac{c_2 z^2}{2} + c_2 z + c_3$$

$$EI v_{BC}'' = \frac{qz^3}{6} + \frac{c_5 z^2}{2} + c_6 z + c_7$$

$$EI v_{AB}' = \frac{c_1 z^3}{6} + \frac{c_2 z^2}{2} + c_3 z + c_4$$

$$EI v_{BC}' = \frac{qz^4}{24} + \frac{c_5 z^3}{6} + \frac{c_6 z^2}{2} + c_7 z + c_8$$

$$EA w_{AB}' = c_9$$

$$EA w_{BC}' = c_{11}$$

$$EA w_{AB} = c_9 z + c_{10}$$

$$EA w_{BC} = c_{11} z + c_{12}$$

CONDIZIONI AL CONFINO

$$w_A = 0 \rightarrow w_{AB}(0) = 0$$

$$v_A = 0 \rightarrow v_{AB}(0) = 0$$

$$\phi_A = 0 \rightarrow -v_{AB}'(0) = 0$$

$$\Delta w_B = 0 \rightarrow w_{BC}(0) - w_{AB}(L) = 0$$

$$\Delta v_B = 0 \rightarrow v_{BC}(0) - v_{AB}(L) = 0$$

$$v_C = 0 \rightarrow v_{BC}(L) = 0$$

$$\Delta N_B = 0 \rightarrow EA w_{BC}'(0) - EA w_{AB}'(L) = 0$$

$$\Delta T_B = 0 \rightarrow -EI v_{BC}''(0) - [-EI v_{AB}''(L)] = 0$$

$$M_B^- = 0 \rightarrow -EI v_{AB}''(L) = 0$$

$$M_B^+ = qL^2 \rightarrow -EI v_{BC}''(0) = qL^2$$

$$N_C = 0 \rightarrow EA w_{BC}'(L) = 0$$

$$M_C = 0 \rightarrow -EI v_{BC}''(L) = 0$$

SOLUZIONE

$$c_1 = qL/2$$

$$c_2 = -qL^2/2$$

$$c_3 = 0$$

$$c_4 = 0$$

$$c_5 = qL/2$$

$$c_6 = -qL^2$$

$$c_7 = 13 qL^3/24$$

$$c_8 = -qL^4/6$$

$$c_9 = 0$$

$$c_{10} = 0$$

$$c_{11} = 0$$

$$c_{12} = 0$$

CARATTERISTICHE DELLA SOLLECITAZIONE

$$N_{AB}(z) = EA w'_{AB}(z) = 0$$

$$T_{AB}(z) = -EI v'''_{AB}(z) = -\frac{qL}{2}$$

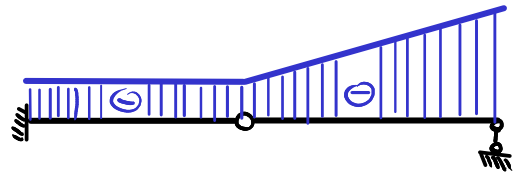
$$M_{AB}(z) = -EI v''_{AB}(z) = -\frac{qL}{2}z + \frac{qL^2}{2}$$

$$N_{BC}(z) = EA w'_{BC}(z) = 0$$

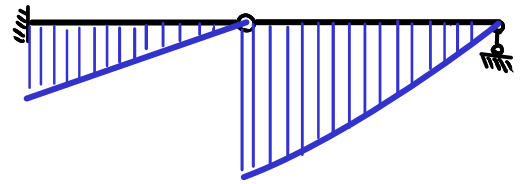
$$T_{BC}(z) = -EI v'''_{BC}(z) = -qz - \frac{qL}{2}$$

$$M_{BC}(z) = -EI v''_{BC}(z) = qL^2 - \frac{qLz}{2} - \frac{qz^2}{2}$$

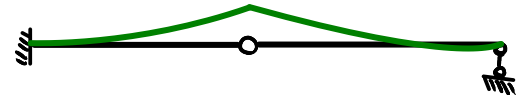
(T)



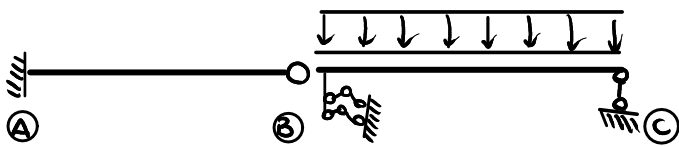
(M)



Deformazione flessionale



ESERCIZIO 2



N° corpi : 2

N° tratti : 2

$$EA w''_{AB} = -p(z) = 0$$

$$EA w''_{BC} = -p(z) = 0$$

$$EI v'''_{AB} = q(z) = 0$$

$$EI v'''_{BC} = q(z) = q$$

Integrali generali scritti come nell'esercizio precedente

CONDIZIONI AL CONFINO

$$w_A = 0 \rightarrow v_{AB}(0) = 0$$

$$v_A = 0 \rightarrow v_{AB}(0) = 0$$

$$\phi_A = 0 \rightarrow -v'_{AB}(0) = 0$$

$$\Delta w_B = 0 \rightarrow w_{BC}(0) - w_{AB}(L) = 0$$

$$\Delta v_B = 0 \rightarrow v_{BC}(0) - v_{AB}(L) = 0$$

$$\phi_B^+ = 0 \rightarrow -v'_{BC}(0) = 0$$

$$v_C = 0 \rightarrow v_{BC}(L) = 0$$

$$\Delta N_B = 0 \rightarrow EA w'_{BC}(0) - EA w'_{AB}(L) = 0$$

$$\Delta T_B = 0 \rightarrow -EI v'''_{BC}(0) - [-EI v'''_{AB}(L)] = 0$$

$$M_B^- = 0 \rightarrow -EI v''_{AB}(L) = 0$$

$$N_C = 0 \rightarrow EA w'_{BC}(L) = 0$$

$$M_C = 0 \rightarrow -EI v''_{BC}(L) = 0$$

SOLUZIONE

$$C_1 = -5qL/16$$

$$C_6 = -3qL^2/16$$

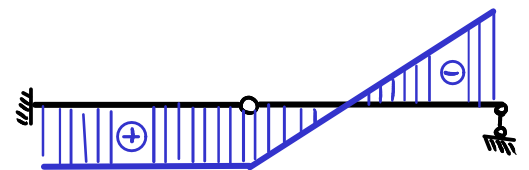
$$C_2 = 5qL^2/16$$

$$C_8 = 5qL^4/48$$

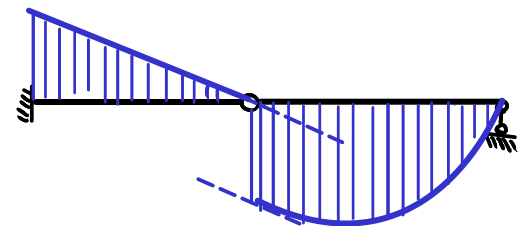
$$C_5 = -5qL/16$$

$$C_3 = C_4 = C_7 = C_9 = C_{10} = C_{11} = C_{12} = 0$$

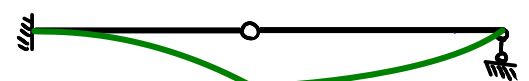
(T)



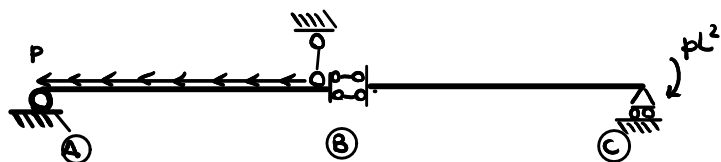
(M)



Deformazione flessionale



ESERCIZIO 3



N° corpi : 2

N° tratti : 2

$$EA w_{AB}'' = -p(\xi) = p$$

$$EA w_{BC}'' = -p(\xi) = 0$$

$$EI v_{AB}''' = q(\xi) = 0$$

$$EI v_{BC}''' = q(\xi) = 0$$

Integrali generali scritti come nell'esercizio 1

CONDIZIONI AL CONFINO

SOLUZIONE

$$w_A = 0 \rightarrow v_{AB}(0) = 0$$

$$v_A = 0 \rightarrow v_{AB}(0) = 0$$

$$v_B^- = 0 \rightarrow v_{AB}(L) = 0$$

$$\Delta w_B = 0 \rightarrow w_{BC}(0) - w_{AB}(L) = 0$$

$$\Delta \phi_B = 0 \rightarrow -v'_{BC}(0) - (-v'_{AB}(L)) = 0$$

$$v_C = 0 \rightarrow v_{BC}(L) = 0$$

$$M_A = 0 \rightarrow -EI v_{AB}''(0) = 0$$

$$\Delta N_B = 0 \rightarrow -EA w'_{BC}(0) - (-EA w'_{AB}(L)) = 0$$

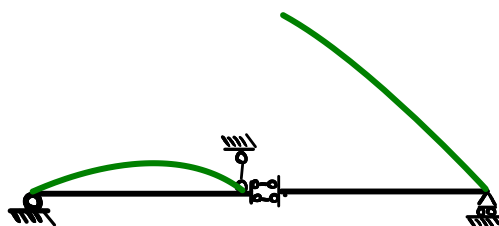
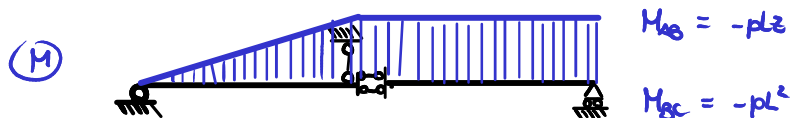
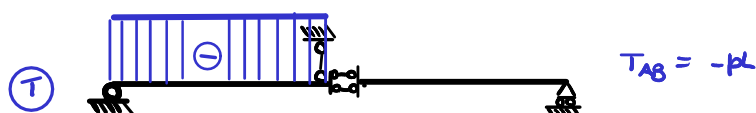
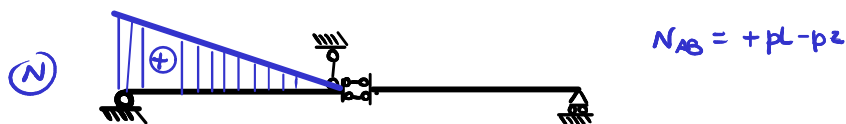
$$T_B^+ = 0 \rightarrow -EI v_{BC}'''(0) = 0$$

$$\Delta H_B = 0 \rightarrow -EI v_{BC}''(0) - (-EI v_{AB}''(L)) = 0$$

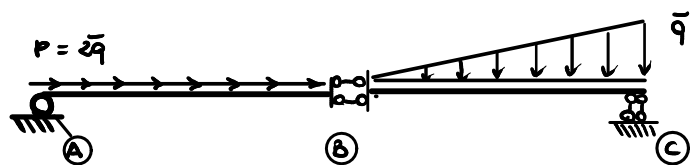
$$N_C = 0 \rightarrow -EA w_{BC}'(L) = 0$$

$$M_C = -pL^2 \rightarrow -EI v_{BC}''(L) = -pL^2$$

$$\begin{aligned} c_1 &= pL & c_7 &= pL^3/3 \\ c_3 &= -pL^3/6 & c_8 &= -5pL^4/6 \\ c_6 &= pL^2 & c_9 &= pL \\ c_{12} &= pL^2/2 & c_4 &= c_3 = c_{10} = c_{11} = 0 \end{aligned}$$



ESERCIZIO 4



N° corpi : 2

N° tratti : 2

$$EA w''_{AB} = -p(x) = -2\bar{q}$$

$$EA w''_{BC} = -p(x) = 0$$

$$EI v'''_{AB} = q(x) = 0$$

$$EI v'''_{BC} = q(x) = \frac{\bar{q}x}{L}$$

INTEGRALI GENERALI

$$EI v_{AB} = \frac{c_1 x^3}{6} + \frac{c_2 x^2}{2} + c_3 x + c_4$$

$$EI v_{BC} = \frac{\bar{q} x^5}{24L} + \frac{c_5 x^3}{6} + \frac{c_6 x^2}{2} + c_7 x + c_8$$

$$EA w_{AB} = \bar{q} x^2 + c_9 x + c_{10}$$

$$EA w_{BC} = c_{11} x + c_{12}$$

CONDIZIONI AL CONFINO

$$w_A = 0$$

$$M_A = 0$$

$$v_A = 0$$

$$\Delta N_B = 0$$

$$\Delta w_B = 0$$

$$T_B^- = 0$$

$$\Delta \phi_B = 0$$

$$T_B^+ = 0$$

$$v_C = 0$$

$$\Delta M_B = 0$$

$$\phi_C = 0$$

$$N_C = 0$$

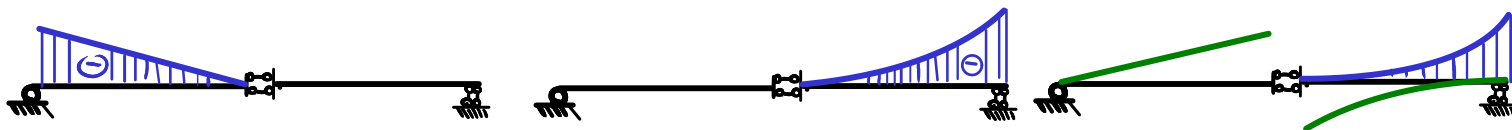
SOLUZIONE

$$c_3 = c_4 = -5\bar{q}L^3/24$$

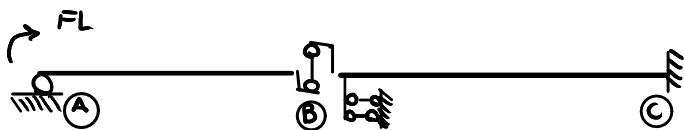
$$c_8 = \bar{q}L^4/6$$

$$c_9 = c_{12} = -2\bar{q}L$$

Le altre costanti sono nulle



ESERCIZIO 5



CONDIZIONI AL CONFINO

$$w_A = 0$$

$$M_A = FL$$

$$v_A = 0$$

$$\Delta T_B = 0$$

$$\Delta v_B = 0$$

$$N_B^- = 0$$

$$w_B^+ = 0$$

$$M_B^- = 0$$

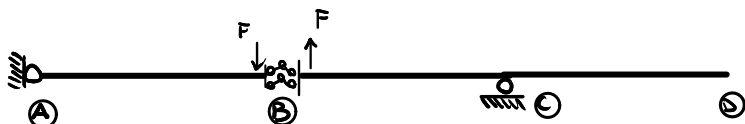
$$\phi_B^+ = 0$$

$$w_C = 0$$

$$v_C = 0$$

$$\phi_C = 0$$

ESERCIZIO 6



$$w_A = 0$$

$$\Delta w_C = 0$$

$$M_A = 0$$

$$\Delta M_C = 0$$

$$v_A = 0$$

$$\Delta v_C = 0$$

$$N_B^- = 0$$

$$N_B = 0$$

$$\Delta \phi_B = 0$$

$$\Delta \phi_C = 0$$

$$N_B^+ = 0$$

$$T_B = 0$$

$$w_C^+ = 0$$

$$T_B^- = F$$

$$M_B = 0$$

$$v_C^+ = 0$$

$$T_B^+ = 0$$

$$\Delta M_B = 0$$